Templates

Integrations & External Services

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CRMERP-09 Security & Compliance (PII, least privilege, audit)
CRMERP-10 Observability & Runbooks (dashboards, alerts, operator steps)

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WHCP-04 Delivery Semantics (ordering, replay, backoff)
WHCP-05 Security Rules (signatures, secrets rotation, allowlists)
WHCP-06 Endpoint Registration & Management (subscriptions, secrets)
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PAY-04 Webhook Handling for Payments (events, idempotency, retries)
PAY-05 Tax/VAT Rules & Compliance (if applicable)
PAY-06 Refunds, Chargebacks & Disputes Policy
PAY-07 Ledger & Reconciliation Rules (source of truth, audits)
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Notifications & Comms (NOTIF)
NOTIF-01 Channel Inventory (email/sms/push/in-app)
NOTIF-02 Provider Inventory (by provider_id)
NOTIF-03 Template & Localization Mapping (template_id, variables)
NOTIF-04 Send Policy (throttles, quiet hours, batching)
NOTIF-05 Deliverability Rules (bounces, spam, suppression lists)
NOTIF-06 Preference Center & Opt-Out Rules (global/per-type)
NOTIF-07 Security & Compliance (PII, CAN-SPAM, consent)
NOTIF-08 Event-to-Notification Mapping (events → messages)

NOTIF-09 Failure Handling (retries, fallback channels, DLQ)
NOTIF-10 Observability & Support Runbooks (delivery metrics, alerts)

Files/Media & Storage (FMS)
FMS-01 Storage Provider Inventory (by provider_id/bucket_id)
FMS-02 Upload/Download Integration Spec (signed URLs, auth, limits)
FMS-03 Media Processing Integration (transcode/resize/scan vendors)
FMS-04 CDN & Delivery Rules (cache keys, variants, purge)
FMS-05 Retention & Lifecycle Rules (TTL, archival, deletion)
FMS-06 Security & Compliance (PII, malware scanning, encryption)
FMS-07 Access Control Model (ACLs, signed access, tenancy)
FMS-08 Metadata & Indexing (search tags, hashes, dedupe)
FMS-09 Disaster Recovery & Backup (restore tests, RPO/RTO)
FMS-10 Observability & Cost Controls (egress, storage growth, alerts)

Integration & External Systems (INT)

IXS-01

IXS-01 — Integration Inventory (by integration_id)

Header Block

template_id: IXS-01
title: Integration Inventory (by integration_id)
type: integration_inventory
template_version: 1.0.0
output_path: 10_app/integrations/IXS-01_Integration_Inventory.md
compliance_gate_id: TMP-05.PRIMARY.IXS
upstream_dependencies: ["SPEC_INDEX", "DOMAIN_MAP"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Create the single, canonical inventory of all external integrations used by the application, indexed by integration_id. This inventory defines what systems exist, why they exist, what data moves, and which other integration templates apply. This document must not invent integrations not present in upstream inputs.

Inputs Required

SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields

Integration registry (integration_id list)
integration_id (stable identifier)
name (human-readable)
vendor/system name
integration category (SSO/CRMERP/WHCP/PAY/NOTIF/FMS/other)
direction (inbound/outbound/bidirectional)
purpose (why it exists)
data domains touched (domain ids or entity ids)
auth method summary (api key/oauth/saml/oidc/signed webhook/etc.)
criticality (low/med/high)
environments enabled (dev/stage/prod)
owner (team/role)
primary risks (PII, financial, availability)
references to detailed specs (template refs)

Optional Fields

Status (planned/active/deprecated) | OPTIONAL
Cost center / billing owner | OPTIONAL

Vendor contract/SLA notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new integration_ids; use only {{spec.integrations_by_id}} if present, else mark UNKNOWN and flag.
Each integration MUST have at least one “references to detailed specs” entry (or approved UNKNOWN).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Inventory Summary
   total_integrations: {{inv.total}}
   categories_present: {{inv.categories_present}} | OPTIONAL
   notes: {{inv.notes}} | OPTIONAL

2. Integration Entries (by integration_id)
   Integration
   integration_id: {{items[0].integration_id}}
   name: {{items[0].name}}
   vendor: {{items[0].vendor}}
   category: {{items[0].category}}
   direction: {{items[0].direction}}
   purpose: {{items[0].purpose}}
   data_domains: {{items[0].data_domains}}
   auth_method: {{items[0].auth_method}}
   criticality: {{items[0].criticality}}
   env_enabled: {{items[0].env_enabled}}
   owner: {{items[0].owner}}
   primary_risks: {{items[0].primary_risks}}
   spec_refs:

- {{items[0].spec_refs[0]}} (e.g., {{xref:IXS-02}} / {{xref:SSO-02}} / {{xref:WHCP-02}})

- {{items[0].spec_refs[1]}} | OPTIONAL
  status: {{items[0].status}} | OPTIONAL
  cost_center: {{items[0].cost_center}} | OPTIONAL
  sla_notes: {{items[0].sla_notes}} | OPTIONAL
  open_questions:
  {{items[0].open_questions[0]}} | OPTIONAL

(Repeat per integration_id.)

3. References
   Integration specs: {{xref:IXS-02}} | OPTIONAL
   Connectivity policy: {{xref:IXS-03}} | OPTIONAL
   Secrets policy: {{xref:IXS-04}} | OPTIONAL
   Data mapping: {{xref:IXS-05}} | OPTIONAL
   Error handling: {{xref:IXS-06}} | OPTIONAL
   Observability: {{xref:IXS-07}} | OPTIONAL
   Security/compliance: {{xref:IXS-08}} | OPTIONAL
   Sandbox/testing: {{xref:IXS-09}} | OPTIONAL
   Change management: {{xref:IXS-10}} | OPTIONAL

Cross-References
Upstream: {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:DOMAIN_MAP}} | OPTIONAL
Downstream: {{xref:SSO-01}}, {{xref:CRMERP-01}}, {{xref:WHCP-01}}, {{xref:PAY-01}}, {{xref:NOTIF-01}}, {{xref:FMS-01}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Populate registry and core fields; use UNKNOWN where inputs are missing; do not invent.
intermediate: Required. Add correct category/direction/auth summaries and spec_refs.
advanced: Required. Add risk notes, ownership clarity, and traceability pointers to domain/entity IDs.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, inv notes, status, cost_center, sla_notes, secondary spec refs, open_questions
If any Required Field is UNKNOWN, allow only if: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If items[*].integration_id is UNKNOWN → block Completeness Gate.*
*If items[*].spec_refs is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.IXS
Pass conditions:
required_fields_present == true
integration_ids_unique == true
no new integration_ids introduced
each integration has spec_refs (or approved UNKNOWN)
placeholder_resolution == true
no_unapproved_unknowns == true

IXS-02

IXS-02 — Integration Spec (per integration: purpose, data flow, auth)
Header Block
template_id: IXS-02
title: Integration Spec (per integration: purpose, data flow, auth)
type: integration_spec
template_version: 1.0.0
output_path: 10_app/integrations/IXS-02_Integration_Spec.md
compliance_gate_id: TMP-05.PRIMARY.IXS
upstream_dependencies: ["IXS-01", "API-01", "DATA-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "API-01", "API-02", "DATA-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical per-integration specification: what the integration does, how data moves, the exact interfaces (APIs/webhooks/files), authentication method, and operational behavior. This template must not invent endpoints, entities, or integration IDs beyond what exists in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}}
API-01 Endpoint Catalog: {{api.endpoint_catalog}} | OPTIONAL
API-02 Endpoint Specs: {{api.endpoint_specs}} | OPTIONAL
DATA-06 Canonical Data Schemas: {{data.schemas}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
integration_id (must exist in IXS-01)
integration name/vendor
purpose statement
direction (inbound/outbound/bidirectional)
interfaces used (REST/GraphQL/WS/Webhooks/SFTP/files/SDK)
data objects exchanged (entity_id/schema_ref list)
data flow diagram (text form is OK: source → transform → destination)
auth method (oauth/api key/signed webhook/saml/oidc/etc.)
credential location policy ref (IXS-04)
rate limit/quotas summary (or ref)
error handling policy ref (IXS-06)
observability requirements ref (IXS-07)

security/compliance notes ref (IXS-08)
environments enabled (dev/stage/prod)

Optional Fields
SLAs/latency expectations | OPTIONAL
Batch vs realtime notes | OPTIONAL
Data retention constraints | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
integration_id MUST exist in {{xref:IXS-01}}.
Do not introduce new endpoint_ids or schema_refs; reference existing IDs from {{xref:API-01}}/{{xref:DATA-06}}.
Auth must be specific and implementable; do not leave “secure” as a description.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Integration Identity
   integration_id: {{integration.id}}
   name: {{integration.name}}
   vendor: {{integration.vendor}}
   category: {{integration.category}} | OPTIONAL
   direction: {{integration.direction}}
   env_enabled: {{integration.env_enabled}}
2. Purpose
   purpose: {{purpose.statement}}
   success_definition: {{purpose.success_definition}} | OPTIONAL
3. Interfaces
   interfaces: {{interfaces.list}}
   endpoints_used: {{interfaces.endpoint_ids}} | OPTIONAL
   webhooks_used: {{interfaces.webhook_ids}} | OPTIONAL
   file_transfers: {{interfaces.file_transfers}} | OPTIONAL
4. Data Objects & Mapping
   entities_or_schemas: {{data.objects}}
   mapping_ref: {{data.mapping_ref}} (expected: {{xref:IXS-05}}) | OPTIONAL
   transform_rules: {{data.transform_rules}} | OPTIONAL
5. Data Flow
   flow:
   {{flow[0]}} (e.g., source → transform → destination)
   {{flow[1]}} | OPTIONAL
6. Auth & Credentials
   auth_method: {{auth.method}}

token_scope_or_claims: {{auth.scopes_or_claims}} | OPTIONAL
credential_policy_ref: {{auth.credential_policy_ref}} (expected: {{xref:IXS-04}}) | OPTIONAL

7. Rate Limits / Quotas
   limits_summary: {{limits.summary}}
   quota_enforcement_ref: {{limits.enforcement_ref}} | OPTIONAL
8. Error Handling
   error_policy_ref: {{errors.policy_ref}} (expected: {{xref:IXS-06}})
   retry_behavior: {{errors.retry_behavior}} | OPTIONAL
9. Observability
   observability_ref: {{obs.ref}} (expected: {{xref:IXS-07}})
   key_metrics: {{obs.key_metrics}} | OPTIONAL
10. Security & Compliance
    security_ref: {{security.ref}} (expected: {{xref:IXS-08}})
    pii_involved: {{security.pii_involved}} | OPTIONAL
11. Notes / Open Questions
    sla_notes: {{notes.sla_notes}} | OPTIONAL
    batch_vs_realtime: {{notes.batch_vs_realtime}} | OPTIONAL
    retention_notes: {{notes.retention_notes}} | OPTIONAL
    open_questions:
    {{open_questions[0]}} | OPTIONAL
12. References
    Inventory: {{xref:IXS-01}}
    Connectivity policy: {{xref:IXS-03}} | OPTIONAL
    Secrets/credentials: {{xref:IXS-04}} | OPTIONAL
    Data mapping: {{xref:IXS-05}} | OPTIONAL
    Error handling: {{xref:IXS-06}} | OPTIONAL
    Observability: {{xref:IXS-07}} | OPTIONAL
    Security/compliance: {{xref:IXS-08}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}}, {{xref:API-01}} | OPTIONAL, {{xref:DATA-06}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IXS-09}}, {{xref:IXS-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Fill identity/purpose/interfaces/flows with UNKNOWN where needed; do not invent IDs.
intermediate: Required. Bind endpoints/schema refs and add auth + error/obs/security refs.
advanced: Required. Add explicit data object lists, rate-limit/SLAs, and traceability pointers.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, category, success definition, endpoint/webhook ids, file transfers, mapping ref, transform rules, auth scopes/claims,

credential policy ref, quota enforcement ref, retry behavior, key metrics, pii involved, notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If integration.id is UNKNOWN → block Completeness Gate.
If errors.policy_ref is UNKNOWN → block Completeness Gate.
If obs.ref is UNKNOWN → block Completeness Gate.
If security.ref is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.IXS
Pass conditions:
required_fields_present == true
integration_id_exists_in_inventory == true
no new endpoint_ids or schema_refs introduced
auth_method_defined == true
refs_to_error_obs_security_present == true
placeholder_resolution == true
no_unapproved_unknowns == true

IXS-03

IXS-03 — Connectivity & Network Policy (timeouts, retries, allowlists)
Header Block
template_id: IXS-03
title: Connectivity & Network Policy (timeouts, retries, allowlists)
type: integration_connectivity_network_policy
template_version: 1.0.0
output_path: 10_app/integrations/IXS-03_Connectivity_Network_Policy.md
compliance_gate_id: TMP-05.PRIMARY.IXS
upstream_dependencies: ["IXS-01", "IXS-02", "API-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "IXS-02", "API-05", "CER-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical connectivity and network policy for integrations: allowed destinations, network timeouts, retry/backoff behavior, DNS/TLS requirements, and handling for degraded connectivity. This template must be consistent with integration specs and must not invent external destinations not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}}
IXS-02 Integration Specs: {{ixs.integration_specs}} | OPTIONAL
API-05 Rate Limit & Abuse Controls: {{api.rate_limits}} | OPTIONAL
CER-02 Retry & Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Destination allowlist policy (domains/IPs)
TLS requirements (min version, cert validation)
Timeout policy (connect/read/total)
Retry policy (retryable failures, backoff)
Circuit breaker policy (stop trying)
Concurrency limits (per integration)
Outbound proxy/VPC egress notes (if applicable)
Inbound connectivity requirements (webhook exposure, IP allowlist)
DNS policy (caching, failover)
Telemetry requirements (latency, failures by destination)

Optional Fields
Regional routing policy | OPTIONAL

Pinned certs / mTLS notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
No outbound connectivity to non-allowlisted destinations.
Retries MUST not amplify incidents; use backoff and circuit breaker rules.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Destination Allowlist
   allowlist_model: {{allowlist.model}} (exact/wildcard/UNKNOWN)
   domains: {{allowlist.domains}}
   ips: {{allowlist.ips}} | OPTIONAL
   per_integration_overrides: {{allowlist.per_integration_overrides}} | OPTIONAL
2. TLS / Transport Security
   tls_min_version: {{tls.min_version}}
   cert_validation_rule: {{tls.cert_validation_rule}}
   mtls_supported: {{tls.mtls_supported}} | OPTIONAL
   pinned_certs: {{tls.pinned_certs}} | OPTIONAL
3. Timeouts
   connect_timeout_ms: {{timeouts.connect_ms}}
   read_timeout_ms: {{timeouts.read_ms}} | OPTIONAL
   total_timeout_ms: {{timeouts.total_ms}}
4. Retries
   retry_supported: {{retry.supported}}
   retryable_failures: {{retry.retryable_failures}}
   backoff_policy: {{retry.backoff_policy}} (exponential/jitter/UNKNOWN)
   max_attempts: {{retry.max_attempts}} | OPTIONAL
5. Circuit Breaker
   breaker_supported: {{breaker.supported}}
   breaker_trigger: {{breaker.trigger}} | OPTIONAL
   cooldown_ms: {{breaker.cooldown_ms}} | OPTIONAL
6. Concurrency Limits
   max_concurrent_requests: {{conc.max_concurrent_requests}}
   per_integration_limits: {{conc.per_integration_limits}} | OPTIONAL
7. Inbound Connectivity (Webhooks)
   public_endpoint_required: {{inbound.public_endpoint_required}} | OPTIONAL
   ip_allowlist_supported: {{inbound.ip_allowlist_supported}} | OPTIONAL
   inbound_allowlist: {{inbound.allowlist}} | OPTIONAL
8. DNS
   dns_cache_policy: {{dns.cache_policy}}
   failover_policy: {{dns.failover_policy}} | OPTIONAL

9. Telemetry
   latency_metric: {{telemetry.latency_metric}}
   failure_metric: {{telemetry.failure_metric}}
   fields: {{telemetry.fields}} | OPTIONAL
10. References
   Integration inventory: {{xref:IXS-01}}
   Integration specs: {{xref:IXS-02}} | OPTIONAL
   Error handling: {{xref:IXS-06}} | OPTIONAL
   Observability: {{xref:IXS-07}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}}, {{xref:IXS-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IXS-06}}, {{xref:IXS-07}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define allowlist + timeouts + basic retry; use UNKNOWN for mTLS/regional routing.
intermediate: Required. Define backoff, breaker, concurrency, and inbound webhook exposure rules.
advanced: Required. Add per-integration overrides, DNS failover specifics, and telemetry fields rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, ips, per-integration overrides, mTLS/pins, read timeout, max attempts, breaker details, per-integration limits, inbound options, dns failover, telemetry fields, regional routing, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If allowlist.domains is UNKNOWN → block Completeness Gate.
If timeouts.total_ms is UNKNOWN → block Completeness Gate.
If retry.backoff_policy is UNKNOWN → block Completeness Gate.
If telemetry.failure_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.IXS
Pass conditions:
required_fields_present == true
allowlist_defined == true
timeouts_defined == true
retry_and_breaker_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

IXS-04

IXS-04 — Secrets & Credential Handling (storage, rotation, access)
Header Block
template_id: IXS-04
title: Secrets & Credential Handling (storage, rotation, access)
type: integration_secrets_credential_handling
template_version: 1.0.0
output_path: 10_app/integrations/IXS-04_Secrets_Credential_Handling.md
compliance_gate_id: TMP-05.PRIMARY.IXS
upstream_dependencies: ["IXS-01", "IXS-02", "SIGN-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "IXS-02", "SIGN-02", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical policy for handling secrets and credentials used by integrations: where they are stored, how they are accessed, rotation schedules, least-privilege access controls, and compromise response. This template must be consistent with security and audit policies and must not expose secret material.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}}
IXS-02 Integration Specs: {{ixs.integration_specs}} | OPTIONAL
SIGN-02 Signing Keys & Rotation Policy: {{sign.keys_policy}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Credential inventory (credential_id list)
credential_id (stable identifier)
integration_id binding (must exist)
credential type (api_key/oauth_client/secret/webhook_secret/cert/UNKNOWN)
storage location policy (secret manager/ci vault/env/UNKNOWN)
access control policy (who/what can read)
rotation policy (schedule + triggers)
revocation/compromise response steps
logging/redaction rules (never log secrets)
environment scoping (dev/stage/prod separation)
secret distribution method (runtime fetch vs build-time inject)
audit logging requirements (access/use events)

Optional Fields
Key ceremony notes | OPTIONAL
mTLS/cert chain notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Never include secret values in this document.
Credentials MUST be scoped per environment unless explicitly justified.
Access MUST follow least privilege and be auditable.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Credential Inventory Summary
   total_credentials: {{inv.total}}
   types_present: {{inv.types_present}} | OPTIONAL
   notes: {{inv.notes}} | OPTIONAL
2. Credential Entries (by credential_id)
   Credential
   credential_id: {{creds[0].credential_id}}
   integration_id: {{creds[0].integration_id}}
   type: {{creds[0].type}}
   env_scope: {{creds[0].env_scope}}
   storage_location: {{creds[0].storage_location}}
   distribution_method: {{creds[0].distribution_method}}
   (runtime_fetch/build_inject/UNKNOWN)
   access_policy: {{creds[0].access_policy}}
   rotation_schedule: {{creds[0].rotation_schedule}}
   rotation_triggers: {{creds[0].rotation_triggers}} | OPTIONAL
   revocation_steps: {{creds[0].revocation_steps}}
   audit_required: {{creds[0].audit_required}}
   redaction_rule: {{creds[0].redaction_rule}}
   notes: {{creds[0].notes}} | OPTIONAL
   open_questions:
   {{creds[0].open_questions[0]}} | OPTIONAL

(Repeat per credential_id.)

3. Global Rules
   no_secret_in_logs_rule: {{global.no_secret_in_logs_rule}}
   no_secret_in_repo_rule: {{global.no_secret_in_repo_rule}}
   separation_of_envs_rule: {{global.separation_of_envs_rule}}

4. Compromise Response
   detection_signals: {{compromise.detection_signals}} | OPTIONAL
   response_steps: {{compromise.response_steps}}
   notification_policy: {{compromise.notification_policy}} | OPTIONAL
5. Audit Logging
   audit_fields: {{audit.fields}}
   audit_retention_policy: {{audit.retention_policy}} | OPTIONAL
6. References
   Integration inventory: {{xref:IXS-01}}
   Integration specs: {{xref:IXS-02}} | OPTIONAL
   Audit trail: {{xref:ADMIN-03}} | OPTIONAL
   Signing/rotation baseline: {{xref:SIGN-02}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}}, {{xref:IXS-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IXS-08}}, {{xref:IXS-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. List credentials and storage locations; use UNKNOWN for rotation triggers if not provided; do not invent.
intermediate: Required. Add access policies, env scoping, distribution method, and audit requirements.
advanced: Required. Add compromise response rigor and rotation procedures aligned to org security.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, inv notes, types present, rotation triggers, detection signals, notification policy, retention policy, key/cert notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If creds[*].credential_id is UNKNOWN → block Completeness Gate.
If creds[].storage_location is UNKNOWN → block Completeness Gate.
If global.no_secret_in_repo_rule is UNKNOWN → block Completeness Gate.
If creds[*].revocation_steps is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.IXS
Pass conditions:
required_fields_present == true
credential_ids_unique == true
integration_ids_exist_in_inventory == true
storage_and_access_defined == true
no_secret_exposure_rules_defined == true

audit_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

IXS-05

IXS-05 — Data Mapping & Transformation Rules (field maps, normalization)
Header Block
template_id: IXS-05
title: Data Mapping & Transformation Rules (field maps, normalization)
type: integration_data_mapping_transforms
template_version: 1.0.0
output_path: 10_app/integrations/IXS-05_Data_Mapping_Transformation_Rules.md
compliance_gate_id: TMP-05.PRIMARY.IXS
upstream_dependencies: ["IXS-02", "DATA-06", "DQV-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-02", "DATA-06", "DQV-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical mapping between external system fields and internal schemas/entities, including transforms/normalization, validation rules, defaulting, and error handling for unmapped or invalid data. This template must be consistent with canonical data schemas and must not invent schema fields beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-02 Integration Specs: {{ixs.integration_specs}}
DATA-06 Canonical Data Schemas: {{data.schemas}} | OPTIONAL
DQV-03 Data Validation Rules: {{data.validation_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
integration_id binding
Mapping registry (map_id list)
map_id (stable identifier)
external object name (vendor object/event)
internal schema/entity ref (schema_ref/entity_id)
primary key mapping (external id → internal id)
field mapping list (external_field → internal_path)
transform rules (trim, case, units, timezone)
defaulting rules (missing values)
validation binding (DQV-03 / schema constraints)
unknown/unmapped field policy
PII handling rules (redaction/storage constraints)

Optional Fields
Bidirectional mapping notes | OPTIONAL
Enum mapping tables | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent schema paths; use only valid refs from {{xref:DATA-06}}.
Transform rules must be deterministic (same input → same output).
Unmapped required fields MUST cause explicit failure or explicit defaulting; no silent drops.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Mapping Registry Summary
   integration_id: {{meta.integration_id}}
   total_maps: {{meta.total_maps}}
   notes: {{meta.notes}} | OPTIONAL
2. Map Entry (repeat per map_id)
   Map
   map_id: {{maps[0].map_id}}
   external_object: {{maps[0].external_object}}
   internal_ref: {{maps[0].internal_ref}} (schema_ref/entity_id)
   primary_key_map: {{maps[0].primary_key_map}}

fields:

- external_field: {{maps[0].fields[0].external_field}}
  internal_path: {{maps[0].fields[0].internal_path}}
  required: {{maps[0].fields[0].required}} | OPTIONAL
  transform: {{maps[0].fields[0].transform}} | OPTIONAL
  default: {{maps[0].fields[0].default}} | OPTIONAL
  validation_ref: {{maps[0].fields[0].validation_ref}} | OPTIONAL

(Repeat field blocks; repeat map blocks.)

3. Transform Rules (Global)
   timezone_rule: {{transform.timezone_rule}} | OPTIONAL
   unit_normalization_rule: {{transform.unit_rule}} | OPTIONAL
   string_normalization_rule: {{transform.string_rule}} | OPTIONAL
4. Validation Binding
   schema_first_or_mapping_first: {{validation.source_of_truth}}
   (schema-first/mapping-first/UNKNOWN)
   dqv_ref: {{validation.dqv_ref}} (expected: {{xref:DQV-03}}) | OPTIONAL

5. Unknown/Unmapped Policy
   unknown_field_behavior: {{unknown.unknown_field_behavior}}
   (ignore/fail/store_raw/UNKNOWN)
   unmapped_required_behavior: {{unknown.unmapped_required_behavior}}
6. PII Handling
   pii_fields: {{pii.fields}} | OPTIONAL
   redaction_rule: {{pii.redaction_rule}}
   storage_constraints: {{pii.storage_constraints}} | OPTIONAL
7. References
   Integration spec: {{xref:IXS-02}}
   Canonical data schemas: {{xref:DATA-06}} | OPTIONAL
   Data validation rules: {{xref:DQV-03}} | OPTIONAL
   Security/compliance: {{xref:IXS-08}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-02}}, {{xref:DATA-06}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IXS-06}}, {{xref:IXS-07}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define map entries and core field mappings; use UNKNOWN for optional transforms.
intermediate: Required. Add deterministic transforms and validation bindings.
advanced: Required. Add enum tables, bidirectional notes, and strict unmapped handling.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, meta notes, field required/transform/default/validation refs, global transform rules, dqv ref, pii fields, storage constraints, enum tables, bidirectional notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If meta.integration_id is UNKNOWN → block Completeness Gate.
If maps[*].internal_ref is UNKNOWN → block Completeness Gate.
If unknown.unmapped_required_behavior is UNKNOWN → block Completeness Gate.
If pii.redaction_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.IXS
Pass conditions:
required_fields_present == true
map_ids_unique == true
integration_id_exists_in_inventory == true
all internal_path references are valid (DATA-06)
unmapped_required_behavior_defined == true

pii_handling_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

IXS-06

IXS-06 — Error Handling & Recovery (retries, fallbacks, operator actions)
Header Block
template_id: IXS-06
title: Error Handling & Recovery (retries, fallbacks, operator actions)
type: integration_error_handling_recovery
template_version: 1.0.0
output_path: 10_app/integrations/IXS-06_Error_Handling_Recovery.md
compliance_gate_id: TMP-05.PRIMARY.IXS
upstream_dependencies: ["IXS-02", "API-03", "CER-02", "JBS-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-02", "API-03", "CER-02", "JBS-04", "EVT-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical error handling and recovery model for integrations: error taxonomy, retry/backoff behavior, idempotency expectations, fallbacks, DLQ/quarantine behavior, and operator actions (replay/backfill/disable). This template must be consistent with API error policy and job/event failure handling rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-02 Integration Specs: {{ixs.integration_specs}}
API-03 Error & Status Code Policy: {{api.error_policy}} | OPTIONAL
CER-02 Retry & Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
JBS-04 Retry/DLQ Policy: {{jobs.retry_dlq}} | OPTIONAL
EVT-07 Event Failure Handling: {{evt.failure_handling}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Error taxonomy (classes)
Retry eligibility rules (by class)
Backoff policy (and max attempts)
Idempotency requirements (per operation)
Timeout handling policy
Fallback behaviors (degraded mode / skip / queue)
Quarantine/DLQ rules (when used, how to drain)
Operator actions matrix (disable integration, replay, backfill, rotate creds)
User impact policy (silent vs surfaced)
Telemetry requirements (error rates, retry counts, DLQ depth)

Optional Fields
Per-integration overrides | OPTIONAL
Vendor-specific error mappings | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not “retry forever”; retries must be bounded and observable.
Idempotency MUST be explicit for any operation that can be replayed.
DLQ/quarantine MUST have a drain/replay policy.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Error Taxonomy
   classes:
   network: {{errors.classes.network}}
   timeout: {{errors.classes.timeout}}
   auth: {{errors.classes.auth}}
   rate_limit: {{errors.classes.rate_limit}}
   vendor_4xx: {{errors.classes.vendor_4xx}}
   vendor_5xx: {{errors.classes.vendor_5xx}}
   schema_validation: {{errors.classes.schema_validation}}
   duplicate: {{errors.classes.duplicate}} | OPTIONAL
2. Retry Eligibility
   retryable_classes: {{retry.retryable_classes}}
   non_retryable_classes: {{retry.non_retryable_classes}} | OPTIONAL
   retry_conditions: {{retry.conditions}} | OPTIONAL
3. Backoff Policy
   policy: {{backoff.policy}} (exponential/jitter/UNKNOWN)
   base_delay_ms: {{backoff.base_delay_ms}} | OPTIONAL
   max_delay_ms: {{backoff.max_delay_ms}} | OPTIONAL
   max_attempts: {{backoff.max_attempts}}
4. Idempotency
   idempotency_required: {{idem.required}}
   idempotency_key_rule: {{idem.key_rule}} | OPTIONAL
   dedupe_window: {{idem.dedupe_window}} | OPTIONAL
5. Timeout Handling
   timeout_budget_ms: {{timeout.budget_ms}}
   on_timeout_behavior: {{timeout.on_timeout_behavior}} | OPTIONAL
6. Fallback Behaviors
   on_rate_limit: {{fallback.on_rate_limit}}
   on_auth_failure: {{fallback.on_auth_failure}}
   on_schema_validation_fail: {{fallback.on_schema_validation_fail}} | OPTIONAL

7. Quarantine / DLQ
   dlq_supported: {{dlq.supported}}
   dlq_trigger_rule: {{dlq.trigger_rule}}
   replay_policy: {{dlq.replay_policy}}
   max_age_ms: {{dlq.max_age_ms}} | OPTIONAL
8. Operator Actions Matrix
   Action
   action_id: {{ops.actions[0].action_id}}
   name: {{ops.actions[0].name}}
   when_allowed: {{ops.actions[0].when_allowed}}
   required_role: {{ops.actions[0].required_role}} | OPTIONAL
   audit_required: {{ops.actions[0].audit_required}}
   notes: {{ops.actions[0].notes}} | OPTIONAL

(Repeat per action.)

9. User Impact Policy
   surface_errors_to_users: {{impact.surface_to_users}}
   user_copy_policy_ref: {{impact.copy_policy_ref}} | OPTIONAL
10. Telemetry
    error_rate_metric: {{telemetry.error_rate_metric}}
    retry_count_metric: {{telemetry.retry_count_metric}} | OPTIONAL
    dlq_depth_metric: {{telemetry.dlq_depth_metric}} | OPTIONAL
    fields: {{telemetry.fields}} | OPTIONAL
11. References
    Integration specs: {{xref:IXS-02}}
    Connectivity policy: {{xref:IXS-03}} | OPTIONAL
    Jobs DLQ policy: {{xref:JBS-04}} | OPTIONAL
    Event failure handling: {{xref:EVT-07}} | OPTIONAL
    Observability: {{xref:IXS-07}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-02}}, {{xref:API-03}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IXS-07}}, {{xref:ADMIN-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define taxonomy + bounded retries + timeout budget; use UNKNOWN for vendor mappings.
intermediate: Required. Define idempotency and DLQ rules and operator actions.
advanced: Required. Add per-integration overrides and rigorous telemetry fields/runbook linkages.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, retry conditions, base/max delays,

idempotency key/dedupe window, timeout on-timeout behavior, schema validation fallback, max age, op required role, notes, user copy policy ref, telemetry fields, overrides, vendor mappings, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If backoff.max_attempts is UNKNOWN → block Completeness Gate.
If timeout.budget_ms is UNKNOWN → block Completeness Gate.
If dlq.trigger_rule is UNKNOWN → block Completeness Gate (when dlq.supported == true).
If telemetry.error_rate_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.IXS
Pass conditions:
required_fields_present == true
bounded_retry_defined == true
idempotency_defined == true
dlq_and_replay_defined == true
operator_actions_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

IXS-07

IXS-07 — Observability (logs/metrics/traces, dashboards, alerts)
Header Block
template_id: IXS-07
title: Observability (logs/metrics/traces, dashboards, alerts)
type: integration_observability
template_version: 1.0.0
output_path: 10_app/integrations/IXS-07_Observability.md
compliance_gate_id: TMP-05.PRIMARY.IXS
upstream_dependencies: ["IXS-01", "IXS-02", "IXS-06", "CER-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "IXS-02", "IXS-06", "CER-05", "SMD-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical observability requirements for integrations: what must be logged, which metrics must exist, trace/correlation rules, dashboards and alerting, and redaction/privacy constraints. This template must be consistent with client/server logging policies and must not introduce telemetry that violates privacy constraints.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}}
IXS-02 Integration Specs: {{ixs.integration_specs}}
IXS-06 Error Handling & Recovery: {{ixs.error_recovery}} | OPTIONAL
CER-05 Logging & Crash Reporting: {{cer.logging}} | OPTIONAL
SMD-06 Client Data Layer Observability: {{smd.observability}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Metric taxonomy (success/failure/latency/throughput)
Per-integration metric requirements (integration_id → metrics)
Log field requirements (integration_id, operation, request_id, error_class)
Tracing/correlation rules (trace_id/span_id propagation)
Redaction rules (no secrets/PII)
Dashboards minimum set (integration health overview)
Alerts rules (failure spikes, latency, DLQ depth)
SLO/SLA monitoring fields (if applicable)
Runbook linkage (where to look when alert fires)

Optional Fields
Sampling policy | OPTIONAL

High-cardinality controls | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
No PII/secrets in logs/metrics; identifiers must be hashed or replaced as required.
Metrics must use stable identifiers (integration_id) and avoid high-cardinality labels.
Alerts must be actionable (include runbook references).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Core Metrics (Global)
   success_count_metric: {{metrics.success_count}}
   failure_count_metric: {{metrics.failure_count}}
   latency_ms_metric: {{metrics.latency_ms}}
   throughput_metric: {{metrics.throughput}} | OPTIONAL
2. Per-Integration Metrics
   Integration Metrics
   integration_id: {{per[0].integration_id}}
   operations: {{per[0].operations}} | OPTIONAL
   success_metric: {{per[0].success_metric}}
   failure_metric: {{per[0].failure_metric}}
   latency_metric: {{per[0].latency_metric}}
   dlq_depth_metric: {{per[0].dlq_depth_metric}} | OPTIONAL
   quota_metric: {{per[0].quota_metric}} | OPTIONAL
   notes: {{per[0].notes}} | OPTIONAL

(Repeat per integration_id.)

3. Logs — Required Fields
   timestamp: {{logs.timestamp}}
   integration_id: {{logs.integration_id}}
   operation: {{logs.operation}}
   direction: {{logs.direction}} | OPTIONAL
   request_id: {{logs.request_id}} | OPTIONAL
   trace_id: {{logs.trace_id}} | OPTIONAL
   error_class: {{logs.error_class}} | OPTIONAL
   status_code: {{logs.status_code}} | OPTIONAL
   duration_ms: {{logs.duration_ms}} | OPTIONAL
   redaction_applied: {{logs.redaction_applied}}
4. Tracing / Correlation
   propagation_rule: {{trace.propagation_rule}}
   span_naming_rule: {{trace.span_naming_rule}} | OPTIONAL

5. Redaction / Privacy
   no_pii_rule: {{privacy.no_pii_rule}}
   no_secrets_rule: {{privacy.no_secrets_rule}}
   field_allowlist: {{privacy.field_allowlist}} | OPTIONAL
6. Dashboards
   integration_health_dashboard: {{dash.health_dashboard}}
   minimum_panels: {{dash.minimum_panels}}
   panel_list: {{dash.panel_list}} | OPTIONAL
7. Alerts
   failure_spike_alert: {{alerts.failure_spike}}
   latency_spike_alert: {{alerts.latency_spike}} | OPTIONAL
   dlq_stuck_alert: {{alerts.dlq_stuck}} | OPTIONAL
   threshold_model: {{alerts.threshold_model}} (static/baseline/UNKNOWN)
   routing_policy: {{alerts.routing_policy}} | OPTIONAL
8. Runbooks
   runbook_location: {{runbooks.location}}
   per_integration_runbooks: {{runbooks.per_integration}} | OPTIONAL
9. References
   Integration inventory: {{xref:IXS-01}}
   Integration specs: {{xref:IXS-02}}
   Error handling/DLQ: {{xref:IXS-06}} | OPTIONAL
   Logging/redaction: {{xref:CER-05}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}}, {{xref:IXS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IXS-10}}, {{xref:ADMIN-06}} | OPTIONAL
Standards: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define core metrics + required log fields + no-PII rules; use UNKNOWN for dashboards tooling details.
intermediate: Required. Define per-integration metrics, dashboards, and failure alerts.
advanced: Required. Add tracing propagation rules, runbook mapping, and threshold model rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, throughput metric, per ops list, optional per metrics, notes, optional log fields, span naming, field allowlist, panel list, routing policy, sampling/high-cardinality notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If logs.integration_id is UNKNOWN → block Completeness Gate.
If privacy.no_pii_rule is UNKNOWN → block Completeness Gate.

If dash.minimum_panels is UNKNOWN → block Completeness Gate.
If alerts.failure_spike is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.IXS
Pass conditions:
required_fields_present == true
core_metrics_defined == true
per_integration_metrics_defined == true
log_fields_defined == true
privacy_rules_defined == true
dashboards_and_alerts_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

IXS-08

IXS-08 — Security & Compliance (PII, retention, vendor risk)
Header Block
template_id: IXS-08
title: Security & Compliance (PII, retention, vendor risk)
type: integration_security_compliance
template_version: 1.0.0
output_path: 10_app/integrations/IXS-08_Security_Compliance.md
compliance_gate_id: TMP-05.PRIMARY.IXS
upstream_dependencies: ["IXS-01", "IXS-02", "IXS-04", "CSec-02", "FPMP-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "IXS-02", "IXS-04", "CSec-02", "FPMP-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical security and compliance requirements for integrations: data classification, PII handling, retention/deletion expectations, vendor risk controls, encryption requirements, and auditability. This template must be consistent with client/server data protection policies and must not invent compliance guarantees beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}}
IXS-02 Integration Specs: {{ixs.integration_specs}}
IXS-04 Secrets & Credential Handling: {{ixs.secrets_policy}} | OPTIONAL
CSec-02 Client Data Protection: {{csec.data_protection}} | OPTIONAL
FPMP-06 Security & Compliance for Files: {{fpmp.file_security}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Data classification model (what classes exist)
Per-integration data classification (integration_id → classes)
PII handling rules (minimize, redact, encrypt)
Retention rules (data stored externally, for how long)
Deletion/right-to-delete handling (if applicable)
Encryption requirements (in transit/at rest)
Access control requirements (least privilege)
Vendor risk checklist (SLA, breach notification, sub-processors)
Audit logging requirements (access, sends/receives)
Compliance constraints (PCI/PHI/etc. if applicable)

Optional Fields
Data residency constraints | OPTIONAL
DPA/contract references | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
No integration may transmit/store sensitive data classes unless explicitly approved and documented.
PII must be minimized; only necessary fields may be sent.
Retention must be explicit (do not leave “forever”).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Data Classification Model
   classes: {{class.model}}
   definitions: {{class.definitions}} | OPTIONAL
2. Per-Integration Classification
   Classification
   integration_id: {{per[0].integration_id}}
   data_classes: {{per[0].data_classes}}
   pii_fields: {{per[0].pii_fields}} | OPTIONAL
   notes: {{per[0].notes}} | OPTIONAL

(Repeat per integration_id.)

3. PII Handling Rules
   minimize_rule: {{pii.minimize_rule}}
   redaction_rule: {{pii.redaction_rule}}
   encryption_rule: {{pii.encryption_rule}} | OPTIONAL
4. Retention & Deletion
   retention_policy: {{retention.policy}}
   delete_propagation_rule: {{retention.delete_propagation_rule}} | OPTIONAL
   right_to_delete_supported: {{retention.rtd_supported}} | OPTIONAL
5. Encryption Requirements
   in_transit: {{crypto.in_transit}}
   at_rest: {{crypto.at_rest}} | OPTIONAL
   key_management_notes: {{crypto.key_mgmt_notes}} | OPTIONAL
6. Access Control
   least_privilege_rule: {{access.least_privilege_rule}}
   credential_policy_ref: {{access.credential_policy_ref}} (expected: {{xref:IXS-04}}) | OPTIONAL

7. Vendor Risk Checklist
   sla_required: {{vendor.sla_required}}
   breach_notification_rule: {{vendor.breach_notification_rule}}
   subprocessor_policy: {{vendor.subprocessor_policy}} | OPTIONAL
   security_review_required: {{vendor.security_review_required}} | OPTIONAL
8. Audit Logging
   audit_required: {{audit.required}}
   audit_fields: {{audit.fields}} | OPTIONAL
   retention_policy: {{audit.retention_policy}} | OPTIONAL
9. Compliance Constraints
   pci_in_scope: {{compliance.pci_in_scope}} | OPTIONAL
   phi_in_scope: {{compliance.phi_in_scope}} | OPTIONAL
   other_constraints: {{compliance.other_constraints}} | OPTIONAL
10. References
    Integration inventory: {{xref:IXS-01}}
    Integration specs: {{xref:IXS-02}}
    Secrets policy: {{xref:IXS-04}} | OPTIONAL
    Client data protection: {{xref:CSec-02}} | OPTIONAL
    File security: {{xref:FPMP-06}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}}, {{xref:IXS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IXS-10}} | OPTIONAL
Standards: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define classification + per-integration mapping; use UNKNOWN for contract refs.
intermediate: Required. Define PII handling + retention/deletion + vendor checklist.
advanced: Required. Add data residency, encryption/key management detail, and audit rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, definitions, pii fields, notes, encryption at rest, key mgmt notes, credential policy ref, subprocessor/security review, audit fields/retention, compliance flags, residency/contract refs, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If class.model is UNKNOWN → block Completeness Gate.
If pii.redaction_rule is UNKNOWN → block Completeness Gate.
If retention.policy is UNKNOWN → block Completeness Gate.
If vendor.breach_notification_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.IXS

Pass conditions:
required_fields_present == true
classification_defined == true
per_integration_mapping_defined == true
pii_and_retention_defined == true
vendor_risk_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

IXS-09

IXS-09 — Sandbox/Test Environment Strategy (stubs, mocks, fixtures)
Header Block
template_id: IXS-09
title: Sandbox/Test Environment Strategy (stubs, mocks, fixtures)
type: integration_sandbox_test_strategy
template_version: 1.0.0
output_path: 10_app/integrations/IXS-09_Sandbox_Test_Strategy.md
compliance_gate_id: TMP-05.PRIMARY.IXS
upstream_dependencies: ["IXS-01", "IXS-02", "IXS-10"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "IXS-02", "IXS-05", "TESTPLAN-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical strategy for testing integrations safely: which vendors have sandboxes, how stubs/mocks are used, fixture datasets, replayable test events, and CI-safe verification. This template must ensure tests do not hit production systems and must not invent test environments that don't exist in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}}
IXS-02 Integration Specs: {{ixs.integration_specs}}
IXS-05 Data Mapping Rules: {{ixs.data_mapping}} | OPTIONAL
Existing test plans/notes: {{inputs.test_notes}} | OPTIONAL

Required Fields
Sandbox availability map (integration_id → sandbox available yes/no)
Sandbox endpoints/credentials policy (no secrets, references only)
Stub/mock strategy (when to stub vs use sandbox)
Fixture data strategy (schemas, sample payloads)
Replay strategy (recorded webhooks/events)
CI execution rules (what runs in CI)
Safety rules (never call prod, allowlist only)
Test coverage expectations (happy path + failure classes)
Verification commands/policies (how to run tests)

Optional Fields
Contract testing approach | OPTIONAL
Rate-limited test windows | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
No production endpoints or production credentials may be used in tests.
Fixtures must conform to mapping rules and data schemas.
If sandbox does not exist, tests must use stubs/mocks with deterministic fixtures.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Sandbox Availability
   Integration Sandbox
   integration_id: {{sandboxes[0].integration_id}}
   sandbox_available: {{sandboxes[0].available}}
   sandbox_endpoint: {{sandboxes[0].endpoint}} | OPTIONAL
   sandbox_notes: {{sandboxes[0].notes}} | OPTIONAL

(Repeat per integration_id.)

2. Stub / Mock Strategy
   use_sandbox_when: {{stubs.use_sandbox_when}}
   use_stub_when: {{stubs.use_stub_when}}
   mock_fidelity_rule: {{stubs.fidelity_rule}} | OPTIONAL
3. Fixtures
   fixture_location: {{fixtures.location}}
   fixture_format: {{fixtures.format}} (json/yaml/UNKNOWN)
   schema_binding_ref: {{fixtures.schema_binding_ref}} (expected: {{xref:DATA-06}}) | OPTIONAL
   mapping_binding_ref: {{fixtures.mapping_binding_ref}} (expected: {{xref:IXS-05}}) | OPTIONAL
4. Replay Strategy
   replay_supported: {{replay.supported}}
   recording_rules: {{replay.recording_rules}} | OPTIONAL
   replay_runner: {{replay.runner}} | OPTIONAL
5. CI Rules
   ci_tests_enabled: {{ci.enabled}}
   ci_scope: {{ci.scope}}
   ci_secrets_policy: {{ci.secrets_policy}} | OPTIONAL
6. Safety Rules
   prod_call_block_rule: {{safety.prod_call_block_rule}}
   destination_allowlist_ref: {{safety.allowlist_ref}} (expected: {{xref:IXS-03}}) | OPTIONAL
7. Coverage Expectations
   must_cover:
   {{coverage.must_cover[0]}}
   {{coverage.must_cover[1]}} | OPTIONAL

8. Verification Commands
   how_to_run: {{verify.how_to_run}}
   expected_outputs: {{verify.expected_outputs}} | OPTIONAL
9. References
   Integration inventory: {{xref:IXS-01}}
   Integration specs: {{xref:IXS-02}}
   Connectivity allowlist: {{xref:IXS-03}} | OPTIONAL
   Data mapping: {{xref:IXS-05}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}}, {{xref:IXS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IXS-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define sandbox availability + CI safety rules; use UNKNOWN for replay tooling if not present.
intermediate: Required. Define stubs/fixtures and how-to-run verification.
advanced: Required. Add contract testing and replay runner rigor with deterministic fixtures.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, sandbox endpoint/notes, fidelity rule, schema/mapping refs, replay details, ci secrets policy, allowlist ref, extra coverage items, expected outputs, contract testing, rate-limited windows, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If ci.enabled is UNKNOWN → block Completeness Gate.
If safety.prod_call_block_rule is UNKNOWN → block Completeness Gate.
If fixtures.location is UNKNOWN → block Completeness Gate.
If verify.how_to_run is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.IXS
Pass conditions:
required_fields_present == true
sandbox_map_defined == true
ci_and_safety_rules_defined == true
fixtures_defined == true
verification_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

IXS-10

IXS-10 — Versioning & Change Management (breaking changes, deprecation)
Header Block
template_id: IXS-10
title: Versioning & Change Management (breaking changes, deprecation)
type: integration_versioning_change_management
template_version: 1.0.0
output_path: 10_app/integrations/IXS-10_Versioning_Change_Management.md
compliance_gate_id: TMP-05.PRIMARY.IXS
upstream_dependencies: ["IXS-01", "IXS-02", "IXS-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "IXS-02", "IXS-05", "EVT-02", "WHCP-09"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical change management policy for integrations: how external API/schema versions are tracked, how breaking changes are detected, how deprecations are handled, and how compatibility is maintained across environments. This template must be consistent with schema/versioning rules for events/webhooks and must not invent versioning strategies not supported by upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}}
IXS-02 Integration Specs: {{ixs.integration_specs}}
IXS-05 Data Mapping Rules: {{ixs.data_mapping}} | OPTIONAL
EVT-02 Event Schema Spec (versioning): {{evt.schema_spec}} | OPTIONAL
WHCP-09 Payload Versioning (webhooks): {{whcp.webhook_versioning}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Version surfaces list (API version, webhook payload version, file schema version)
Per-integration current version tracking (integration_id → versions)
Compatibility policy (backwards compatible window)
Breaking change definition (what counts)
Deprecation policy (notice period, removal steps)
Migration strategy (dual-write/dual-read, adapters)
Change detection strategy (tests, monitoring, vendor notices)
Rollout strategy for changes (staged, flags)
Rollback strategy (version pin, revert)
Audit/communication requirements (who is notified)

Optional Fields
Contract testing requirement | OPTIONAL
Vendor roadmap tracking | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Version surfaces MUST be explicit and referenced by integration_id.
Breaking changes MUST have a migration and rollback plan before rollout.
Deprecations MUST be tracked; “surprise removals” are not allowed.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Version Surfaces
   surfaces: {{surfaces.list}}
   examples: {{surfaces.examples}} | OPTIONAL
2. Per-Integration Version Tracking
   Version Entry
   integration_id: {{versions[0].integration_id}}
   api_version: {{versions[0].api_version}} | OPTIONAL
   webhook_payload_version: {{versions[0].webhook_payload_version}} | OPTIONAL
   file_schema_version: {{versions[0].file_schema_version}} | OPTIONAL
   notes: {{versions[0].notes}} | OPTIONAL

(Repeat per integration_id.)

3. Compatibility Policy
   compat_window: {{compat.window}}
   compat_rules: {{compat.rules}} | OPTIONAL
4. Breaking Change Definition
   breaking_change_rules: {{breaking.rules}}
   examples: {{breaking.examples}} | OPTIONAL
5. Deprecation Policy
   notice_period_days: {{deprec.notice_period_days}}
   deprec_process: {{deprec.process}}
   sunset_steps: {{deprec.sunset_steps}} | OPTIONAL
6. Migration Strategy
   migration_models: {{migrate.models}} (dual_read/dual_write/adapter/UNKNOWN)
   preferred_model: {{migrate.preferred_model}} | OPTIONAL
   data_mapping_ref: {{migrate.data_mapping_ref}} (expected: {{xref:IXS-05}}) | OPTIONAL
7. Change Detection
   vendor_notice_rule: {{detect.vendor_notice_rule}}

contract_tests_required: {{detect.contract_tests_required}} | OPTIONAL
monitoring_signals: {{detect.monitoring_signals}} | OPTIONAL

8. Rollout & Rollback
   rollout_model: {{rollout.model}} (staged/flags/UNKNOWN)
   rollout_steps: {{rollout.steps}} | OPTIONAL
   rollback_model: {{rollback.model}} (version_pin/revert/UNKNOWN)
   rollback_steps: {{rollback.steps}} | OPTIONAL
9. Communication / Audit
   notify_who: {{comms.notify_who}}
   audit_required: {{comms.audit_required}}
   where_logged: {{comms.where_logged}} | OPTIONAL
10. References
    Integration inventory: {{xref:IXS-01}}
    Integration specs: {{xref:IXS-02}}
    Data mapping: {{xref:IXS-05}} | OPTIONAL
    Event schema versioning: {{xref:EVT-02}} | OPTIONAL
    Webhook payload versioning: {{xref:WHCP-09}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}}, {{xref:IXS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:RELEASE-GATE}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Track current versions and define deprecation notice period; use UNKNOWN for contract tests if missing.
intermediate: Required. Define compatibility window, breaking change definition, and migration/rollback models.
advanced: Required. Add staged rollout steps, monitoring signals, and audit/communications rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, examples, optional version fields, notes, compat rules, breaking examples, sunset steps, preferred model, mapping ref, contract tests, monitoring signals, rollout/rollback steps, where logged, vendor roadmap, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If surfaces.list is UNKNOWN → block Completeness Gate.
If deprec.notice_period_days is UNKNOWN → block Completeness Gate.
If breaking.rules is UNKNOWN → block Completeness Gate.
If rollout.model is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.IXS
Pass conditions:

required_fields_present == true
version_surfaces_defined == true
per_integration_versions_defined == true
breaking_and_deprecation_defined == true
migration_and_rollback_defined == true
communication_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Third-Party Auth & SSO (SSO)

Third-Party Auth & SSO (SSO)
SSO-01 Provider Inventory (by provider_id)
SSO-02 SSO Flow Spec (OIDC/SAML, redirects, state/nonce)
SSO-03 Claim Mapping & Role Provisioning (claims → roles/permissions)
SSO-04 Session & Token Exchange Rules (refresh, expiry, rotation)
SSO-05 SCIM Provisioning Spec (users/groups lifecycle)
SSO-06 Account Linking & Identity Merge Rules
SSO-07 MFA/Step-Up Auth Rules (when required)
SSO-08 Failure Handling (lockouts, partial provision, retries)
SSO-09 Security Controls (PKCE, signature validation, replay defense)
SSO-10 Audit & Compliance (login events, admin actions, retention)

SSO-01

SSO-01 — Provider Inventory (by provider_id)
Header Block
template_id: SSO-01
title: Provider Inventory (by provider_id)
type: sso_provider_inventory
template_version: 1.0.0
output_path: 10_app/sso/SSO-01_Provider_Inventory.md
compliance_gate_id: TMP-05.PRIMARY.SSO
upstream_dependencies: ["IXS-01", "API-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "API-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical inventory of all third-party auth/SSO providers, indexed by provider_id, including protocol type, environments enabled, and which flows/features depend on each provider. This document must not invent provider_ids not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}} | OPTIONAL
API-04 AuthZ Rules (roles/claims): {{api.authz_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Provider registry (provider_id list)
provider_id (stable identifier)
provider name/vendor
protocol (OIDC/SAML/LDAP/UNKNOWN)
tenant model (single-tenant/multi-tenant/UNKNOWN)
login entrypoints supported (web/mobile)
environments enabled (dev/stage/prod)
scopes/claims expected (summary)
user provisioning mode (JIT/SCIM/manual/UNKNOWN)
criticality (low/med/high)
owner (team/role)
security notes pointer (SSO-09)

Optional Fields
Branding/custom UI notes | OPTIONAL

SLA/uptime notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new provider_ids; use only {{spec.sso_providers_by_id}} if present, else mark UNKNOWN and flag.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Inventory Summary
   total_providers: {{inv.total}}
   protocols_present: {{inv.protocols_present}} | OPTIONAL
   notes: {{inv.notes}} | OPTIONAL
2. Provider Entries (by provider_id)
   Provider
   provider_id: {{providers[0].provider_id}}
   name: {{providers[0].name}}
   protocol: {{providers[0].protocol}}
   tenant_model: {{providers[0].tenant_model}}
   entrypoints: {{providers[0].entrypoints}}
   env_enabled: {{providers[0].env_enabled}}
   claims_summary: {{providers[0].claims_summary}}
   provisioning_mode: {{providers[0].provisioning_mode}}
   criticality: {{providers[0].criticality}}
   owner: {{providers[0].owner}}
   security_ref: {{providers[0].security_ref}} (expected: {{xref:SSO-09}}) | OPTIONAL
   branding_notes: {{providers[0].branding_notes}} | OPTIONAL
   sla_notes: {{providers[0].sla_notes}} | OPTIONAL
   open_questions:
   {{providers[0].open_questions[0]}} | OPTIONAL

(Repeat per provider_id.)

3. References
   SSO flow spec: {{xref:SSO-02}} | OPTIONAL
   Claim mapping/provisioning: {{xref:SSO-03}} | OPTIONAL
   Session/token exchange: {{xref:SSO-04}} | OPTIONAL
   SCIM provisioning: {{xref:SSO-05}} | OPTIONAL
   Security controls: {{xref:SSO-09}} | OPTIONAL
   Audit/compliance: {{xref:SSO-10}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}} | OPTIONAL, {{xref:API-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SSO-02}}, {{xref:SSO-03}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Populate provider registry and core fields; do not invent provider_ids.
intermediate: Required. Add entrypoints, provisioning modes, and claims summary.
advanced: Required. Add tenant model clarity, criticality/ownership, and security/audit references.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, inv notes, tenant model, entrypoints, claims summary, provisioning mode, security ref, branding/sla notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If providers[*].provider_id is UNKNOWN → block Completeness Gate.
If providers[*].protocol is UNKNOWN → block Completeness Gate.
If providers[*].env_enabled is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SSO
Pass conditions:
required_fields_present == true
provider_ids_unique == true
no new provider_ids introduced
placeholder_resolution == true
no_unapproved_unknowns == true

SSO-02

SSO-02 — SSO Flow Spec (OIDC/SAML, redirects, state/nonce)
Header Block
template_id: SSO-02
title: SSO Flow Spec (OIDC/SAML, redirects, state/nonce)
type: sso_flow_spec
template_version: 1.0.0
output_path: 10_app/sso/SSO-02_SSO_Flow_Spec.md
compliance_gate_id: TMP-05.PRIMARY.SSO
upstream_dependencies: ["SSO-01", "ROUTE-03", "CSec-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SSO-01", "ROUTE-03", "ROUTE-04", "CSec-05", "CER-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical SSO login flow(s) per provider and protocol (OIDC/SAML): endpoints, redirects/callbacks, state/nonce handling, PKCE rules (if applicable), session creation, and failure cases. This template must align with secure deep link handling and route guard rules and must not invent providers or callback routes not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SSO-01 Provider Inventory: {{sso.providers}}
ROUTE-03 Deep Link Map (callback routing): {{route.deep_link_map}} | OPTIONAL
ROUTE-04 Guard Rules: {{route.guard_rules}} | OPTIONAL
CSec-05 Secure Deep Link Handling: {{csec.deep_link_security}} | OPTIONAL
CER-04 Session Expiry Handling: {{cer.session_expiry}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
provider_id binding (must exist)
protocol (OIDC/SAML)
entrypoint routes (login start)
callback routes (redirect URI / ACS endpoint)
state handling rule (CSRF protection)
nonce handling rule (OIDC)
PKCE rules (if OIDC public clients)
token/assertion validation rules (signature, issuer, audience)
session creation rules (what claims stored)
error cases and UX handling
logout flow behavior (SP-initiated/logout redirect)
telemetry requirements (success/fail, error classes)

Optional Fields
Multi-tenant discovery rules | OPTIONAL
IdP-initiated SAML behavior | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Callback routes MUST be allowlisted and validated; no open redirects.
State MUST be required for all redirect-based flows.
Token/assertion signatures MUST be validated; do not skip verification.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Flow Registry (per provider_id)
   Flow
   provider_id: {{flows[0].provider_id}}
   protocol: {{flows[0].protocol}}
   login_entry_route: {{flows[0].login_entry_route}}
   callback_route: {{flows[0].callback_route}}
   logout_route: {{flows[0].logout_route}} | OPTIONAL
2. Redirect / Callback Details
   redirect_uri: {{flows[0].redirect_uri}} | OPTIONAL
   acs_url: {{flows[0].acs_url}} | OPTIONAL
   allowed_redirects: {{flows[0].allowed_redirects}} | OPTIONAL
3. State / Nonce / PKCE
   state_rule: {{flows[0].state_rule}}
   nonce_rule: {{flows[0].nonce_rule}} | OPTIONAL
   pkce_supported: {{flows[0].pkce_supported}} | OPTIONAL
   pkce_rule: {{flows[0].pkce_rule}} | OPTIONAL
4. Validation
   issuer_rule: {{flows[0].validate.issuer_rule}}
   audience_rule: {{flows[0].validate.audience_rule}}
   signature_rule: {{flows[0].validate.signature_rule}}
   clock_skew_seconds: {{flows[0].validate.clock_skew_seconds}} | OPTIONAL
5. Session Creation
   claims_used: {{flows[0].session.claims_used}}
   session_duration_policy: {{flows[0].session.duration_policy}} | OPTIONAL
   refresh_policy_ref: {{flows[0].session.refresh_policy_ref}} (expected: {{xref:SSO-04}}) | OPTIONAL
6. Error Handling
   error_taxonomy: {{flows[0].errors.taxonomy}}
   user_ux_rule: {{flows[0].errors.user_ux_rule}}
   retry_rule: {{flows[0].errors.retry_rule}} | OPTIONAL

7. Logout
   logout_supported: {{flows[0].logout.supported}}
   logout_behavior: {{flows[0].logout.behavior}} | OPTIONAL
8. Telemetry
   login_success_metric: {{flows[0].telemetry.success_metric}}
   login_failure_metric: {{flows[0].telemetry.failure_metric}} | OPTIONAL
   fields: {{flows[0].telemetry.fields}} | OPTIONAL

(Repeat flow blocks per provider_id.)

9. References
   Provider inventory: {{xref:SSO-01}}
   Claim mapping/provisioning: {{xref:SSO-03}} | OPTIONAL
   Session/token exchange: {{xref:SSO-04}} | OPTIONAL
   Security controls: {{xref:SSO-09}} | OPTIONAL
   Audit/compliance: {{xref:SSO-10}} | OPTIONAL
   Secure link handling: {{xref:CSec-05}} | OPTIONAL
   Guard rules: {{xref:ROUTE-04}} | OPTIONAL

Cross-References
Upstream: {{xref:SSO-01}}, {{xref:ROUTE-03}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SSO-03}}, {{xref:SSO-04}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define entry/callback + state + basic validation; use UNKNOWN for multi-tenant discovery if not applicable.
intermediate: Required. Define nonce/PKCE rules (as applicable) and error taxonomy + telemetry.
advanced: Required. Add logout rigor, clock skew, and protocol-specific edge cases (IdP-initiated, tenant discovery).

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, redirect_uri/acs_url/allowed redirects, nonce/pkce fields, clock skew, session duration/refresh ref, failure metric/fields, logout behavior, multi-tenant/idp initiated, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If flows[*].state_rule is UNKNOWN → block Completeness Gate.*
*If flows[*].validate.signature_rule is UNKNOWN → block Completeness Gate.
If flows[*].errors.user_ux_rule is UNKNOWN → block Completeness Gate.*
*If flows[*].telemetry.success_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SSO

Pass conditions:
required_fields_present == true
provider_ids_exist_in_SSO_01 == true
callback_routes_allowlisted == true
state_and_signature_validation_defined == true
error_and_telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SSO-03

SSO-03 — Claim Mapping & Role Provisioning (claims → roles/permissions)
Header Block
template_id: SSO-03
title: Claim Mapping & Role Provisioning (claims → roles/permissions)
type: sso_claim_mapping_role_provisioning
template_version: 1.0.0
output_path: 10_app/sso/SSO-03_Claim_Mapping_Role_Provisioning.md
compliance_gate_id: TMP-05.PRIMARY.SSO
upstream_dependencies: ["SSO-01", "API-04", "ROUTE-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SSO-01", "SSO-02", "API-04", "ROUTE-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical mapping from identity provider claims/attributes to internal roles/permissions, including provisioning rules, default roles, group mappings, and handling of missing/invalid claims. This template must be consistent with AuthZ rules and must not invent roles/permissions not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SSO-01 Provider Inventory: {{sso.providers}}
SSO-02 Flow Spec: {{sso.flows}} | OPTIONAL
API-04 AuthZ Rules (roles/claims): {{api.authz_rules}}
ROUTE-04 Guard Rules (client gating): {{route.guard_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
provider_id binding
claim sources (id_token / saml attributes / userinfo)
required claim list (must exist)
claim normalization rules (case, arrays, delims)
role mapping rules (claim → role_id)
permission mapping rules (claim → permission/claim)
default role rule (if mapping missing)
group mapping rules (IdP groups → internal groups)
provisioning behavior (create user? update roles?)
deprovisioning behavior (disable user?)
error handling (missing/invalid claims)
audit/telemetry requirements (role assignment events)

Optional Fields
Multi-tenant org mapping | OPTIONAL
Just-in-time provisioning notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Do not invent role_ids/permissions; bind only to {{xref:API-04}}.
Never grant elevated roles by default unless explicitly required.
Missing required claims MUST cause a safe failure or safe default (least privilege).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Claim Sources
   provider_id: {{meta.provider_id}}
   sources: {{claims.sources}}
   required_claims: {{claims.required}}
   optional_claims: {{claims.optional}} | OPTIONAL
2. Normalization Rules
   normalize_rule: {{normalize.rule}}
   group_parse_rule: {{normalize.group_parse_rule}} | OPTIONAL
3. Role Mapping
   Mapping
   claim_key: {{roles.map[0].claim_key}}
   match: {{roles.map[0].match}}
   assign_role_id: {{roles.map[0].assign_role_id}}
   notes: {{roles.map[0].notes}} | OPTIONAL

(Repeat per mapping.)

4. Permission / Entitlement Mapping
   Mapping
   claim_key: {{perms.map[0].claim_key}}
   match: {{perms.map[0].match}}
   assign_permissions: {{perms.map[0].assign_permissions}}
   notes: {{perms.map[0].notes}} | OPTIONAL

(Repeat per mapping.)

5. Defaulting / Fallback
   default_role_id: {{defaults.default_role_id}}
   when_default_applies: {{defaults.when_default_applies}}
   deny_on_missing_required_claims: {{defaults.deny_on_missing_required_claims}}

6. Group Mapping
   group_mapping_supported: {{groups.supported}}
   idp_group_key: {{groups.idp_group_key}} | OPTIONAL
   map: {{groups.map}} | OPTIONAL
7. Provisioning / Deprovisioning
   jit_create_user: {{prov.jit_create_user}}
   update_roles_on_login: {{prov.update_roles_on_login}} | OPTIONAL
   deprovision_rule: {{prov.deprovision_rule}}
8. Error Handling
   missing_claim_behavior: {{errors.missing_claim_behavior}}
   invalid_claim_behavior: {{errors.invalid_claim_behavior}} | OPTIONAL
9. Audit / Telemetry
   role_assignment_metric: {{telemetry.role_assignment_metric}}
   deprovision_metric: {{telemetry.deprovision_metric}} | OPTIONAL
   audit_fields: {{telemetry.audit_fields}} | OPTIONAL
10. References
    AuthZ rules: {{xref:API-04}}
    Flow spec: {{xref:SSO-02}} | OPTIONAL
    Security controls: {{xref:SSO-09}} | OPTIONAL
    Audit/compliance: {{xref:SSO-10}} | OPTIONAL
    Client guards: {{xref:ROUTE-04}} | OPTIONAL

Cross-References
Upstream: {{xref:SSO-01}}, {{xref:API-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SSO-04}}, {{xref:SSO-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define required claims and default role rule; do not invent roles.
intermediate: Required. Define role/permission maps and provisioning behavior.
advanced: Required. Add multi-tenant org mapping and strict audit/telemetry fields.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, optional claims, parse rules, mapping notes, group keys/maps, update roles on login, invalid claim behavior, optional metrics/audit fields, org mapping, jit notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If claims.required is UNKNOWN → block Completeness Gate.
If defaults.default_role_id is UNKNOWN → block Completeness Gate.
If defaults.deny_on_missing_required_claims is UNKNOWN → block Completeness Gate.
If telemetry.role_assignment_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SSO

Pass conditions:
required_fields_present == true
provider_id_exists_in_SSO_01 == true
all role_ids/permissions exist in API-04
least_privilege_defaults_defined == true
audit_and_telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SSO-04

SSO-04 — Session & Token Exchange Rules (refresh, expiry, rotation)
Header Block
template_id: SSO-04
title: Session & Token Exchange Rules (refresh, expiry, rotation)
type: sso_session_token_exchange_rules
template_version: 1.0.0
output_path: 10_app/sso/SSO-04_Session_Token_Exchange_Rules.md
compliance_gate_id: TMP-05.PRIMARY.SSO
upstream_dependencies: ["SSO-02", "CSec-01", "CER-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SSO-01", "SSO-02", "CSec-01", "CER-04", "API-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for exchanging SSO assertions/tokens into application sessions, including session duration, refresh behavior, token storage/rotation, revocation handling, and logout semantics. This template must be consistent with client token storage and session expiry handling and must not invent auth mechanisms not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SSO-01 Provider Inventory: {{sso.providers}}
SSO-02 Flow Spec: {{sso.flows}}
CSec-01 Token Storage Policy: {{csec.token_storage}} | OPTIONAL
CER-04 Session Expiry Handling: {{cer.session_expiry}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
provider_id binding
session model (cookie/session token/jwt/UNKNOWN)
session creation rule (what is issued, where)
session duration policy (absolute/idle)
refresh supported (yes/no/UNKNOWN)
refresh trigger rules (when to refresh)
refresh failure behavior (re-auth/logout)
token rotation rules (if applicable)
revocation handling (IdP revoked, user disabled)
logout behavior (local + IdP logout)
secure storage binding (CSec-01)
telemetry requirements (refresh success/fail, expiry)

Optional Fields
Backchannel logout (OIDC) | OPTIONAL
Session binding to device/browser | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Tokens MUST be stored/handled per {{xref:CSec-01}}.
Session expiry behavior MUST align to {{xref:CER-04}}.
Refresh loops must be prevented; failures must degrade safely.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Session Model
   provider_id: {{meta.provider_id}}
   session_model: {{session.model}} (cookie/jwt/session_token/UNKNOWN)
   issued_artifacts: {{session.issued_artifacts}}
   storage_ref: {{session.storage_ref}} (expected: {{xref:CSec-01}}) | OPTIONAL
2. Session Duration
   absolute_ttl_seconds: {{ttl.absolute_ttl_seconds}}
   idle_ttl_seconds: {{ttl.idle_ttl_seconds}} | OPTIONAL
   renew_on_activity: {{ttl.renew_on_activity}} | OPTIONAL
3. Refresh
   refresh_supported: {{refresh.supported}}
   trigger_rule: {{refresh.trigger_rule}} | OPTIONAL
   max_refresh_attempts: {{refresh.max_attempts}} | OPTIONAL
   on_refresh_fail: {{refresh.on_fail}}
4. Rotation
   rotation_supported: {{rotation.supported}}
   rotation_rule: {{rotation.rule}} | OPTIONAL
5. Revocation Handling
   revocation_signals: {{revoke.signals}}
   on_revoked_behavior: {{revoke.on_revoked_behavior}}
6. Logout
   local_logout_clears: {{logout.local_clears}}
   idp_logout_supported: {{logout.idp_supported}} | OPTIONAL
   idp_logout_rule: {{logout.idp_rule}} | OPTIONAL
7. Telemetry
   session_created_metric: {{telemetry.session_created_metric}}
   refresh_success_metric: {{telemetry.refresh_success_metric}} | OPTIONAL
   refresh_failure_metric: {{telemetry.refresh_failure_metric}} | OPTIONAL
   session_expired_metric: {{telemetry.session_expired_metric}} | OPTIONAL

8. References
   Flow spec: {{xref:SSO-02}}
   Claim mapping/provisioning: {{xref:SSO-03}} | OPTIONAL
   Token storage policy: {{xref:CSec-01}} | OPTIONAL
   Session expiry handling: {{xref:CER-04}} | OPTIONAL
   Security controls: {{xref:SSO-09}} | OPTIONAL

Cross-References
Upstream: {{xref:SSO-01}}, {{xref:SSO-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SSO-05}}, {{xref:SSO-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define session model + TTL + refresh fail behavior; use UNKNOWN for rotation if not used.
intermediate: Required. Define refresh trigger rules and revocation handling and logout behavior.
advanced: Required. Add backchannel/device binding and strict telemetry fields and loop prevention.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, storage ref, idle ttl/renewal, refresh trigger/max attempts, rotation rule, idp logout details, optional telemetry metrics, backchannel/device binding, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If session.model is UNKNOWN → block Completeness Gate.
If ttl.absolute_ttl_seconds is UNKNOWN → block Completeness Gate.
If refresh.on_fail is UNKNOWN → block Completeness Gate (when refresh.supported == true).
If revoke.on_revoked_behavior is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SSO
Pass conditions:
required_fields_present == true
provider_id_exists_in_SSO_01 == true
session_model_and_ttl_defined == true
refresh_or_no_refresh_defined == true
revocation_and_logout_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SSO-05

SSO-05 — SCIM Provisioning Spec (users/groups lifecycle)
Header Block
template_id: SSO-05
title: SCIM Provisioning Spec (users/groups lifecycle)
type: sso_scim_provisioning_spec
template_version: 1.0.0
output_path: 10_app/sso/SSO-05_SCIM_Provisioning_Spec.md
compliance_gate_id: TMP-05.PRIMARY.SSO
upstream_dependencies: ["SSO-01", "SSO-03", "API-02", "API-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SSO-01", "SSO-03", "API-01", "API-02", "API-04", "IXS-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical SCIM provisioning implementation: supported resources (Users/Groups), lifecycle operations (create/update/deactivate), attribute mapping, authentication, rate limits, idempotency, and failure handling. This template must be consistent with role/claim mapping and AuthZ rules and must not invent endpoints/resources beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SSO-01 Provider Inventory: {{sso.providers}}
SSO-03 Claim/Role Mapping: {{sso.claim_mapping}} | OPTIONAL
API-01 Endpoint Catalog: {{api.endpoint_catalog}} | OPTIONAL
API-02 Endpoint Specs: {{api.endpoint_specs}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
IXS-04 Secrets/Credentials Policy: {{ixs.secrets_policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
provider_id binding
SCIM enabled (true/false/UNKNOWN)
Supported resources (Users/Groups)
Supported operations (POST/PUT/PATCH/DELETE)
Auth method for SCIM (bearer token/basic/mTLS/UNKNOWN)
Attribute mapping (SCIM → internal fields)
Group membership handling rules
Idempotency rules (externalId, resourceId)
Deprovisioning rules (disable/delete)
Rate limit policy (for SCIM endpoints)

Failure handling (retries, error responses)
Audit logging requirements (who/what changed)

Optional Fields
Multi-tenant partitioning rules | OPTIONAL
Schema extensions supported | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
SCIM changes MUST be auditable and least-privilege.
Idempotency MUST be enforced for create/update operations.
Deprovisioning MUST not orphan privileged access.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. SCIM Enablement
   provider_id: {{meta.provider_id}}
   scim_enabled: {{scim.enabled}}
   base_path: {{scim.base_path}} | OPTIONAL
2. Resources Supported
   users_supported: {{resources.users_supported}}
   groups_supported: {{resources.groups_supported}} | OPTIONAL
3. Operations Supported
   users_ops: {{ops.users_ops}}
   groups_ops: {{ops.groups_ops}} | OPTIONAL
4. Auth
   auth_method: {{auth.method}}
   credential_ref: {{auth.credential_ref}} (expected: {{xref:IXS-04}}) | OPTIONAL
5. Attribute Mapping
   user_attribute_map: {{map.user_attributes}}
   group_attribute_map: {{map.group_attributes}} | OPTIONAL
6. Group Membership Handling
   membership_update_rule: {{groups.membership_update_rule}}
   conflict_rule: {{groups.conflict_rule}} | OPTIONAL
7. Idempotency
   idempotency_key_rule: {{idem.key_rule}}
   external_id_usage: {{idem.external_id_usage}} | OPTIONAL
8. Deprovisioning
   deactivate_rule: {{deprov.deactivate_rule}}
   delete_rule: {{deprov.delete_rule}} | OPTIONAL

9. Rate Limits
   rate_limit_ref: {{limits.ref}} (expected: {{xref:RLIM-02}}) | OPTIONAL
   limits_summary: {{limits.summary}} | OPTIONAL
10. Failure Handling
    error_response_rules: {{fail.error_response_rules}}
    retry_rules: {{fail.retry_rules}} | OPTIONAL
11. Audit Logging
    audit_required: {{audit.required}}
    audit_fields: {{audit.fields}} | OPTIONAL
12. References
    Provider inventory: {{xref:SSO-01}}
    Claim/role mapping: {{xref:SSO-03}} | OPTIONAL
    AuthZ rules: {{xref:API-04}} | OPTIONAL
    Secrets policy: {{xref:IXS-04}} | OPTIONAL
    Error handling: {{xref:IXS-06}} | OPTIONAL

Cross-References
Upstream: {{xref:SSO-01}}, {{xref:API-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SSO-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define whether SCIM enabled and which resources/ops; use UNKNOWN for extensions.
intermediate: Required. Define attribute mapping, idempotency, and deprovisioning rules.
advanced: Required. Add multi-tenant partitioning and strict audit/rate-limit bindings.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, base path, groups supported/ops, credential ref, group attribute map, conflict rule, external id usage, delete rule, rate limit ref/summary, retry rules, audit fields, extensions/tenant rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If scim.enabled is UNKNOWN → block Completeness Gate.
If auth.method is UNKNOWN → block Completeness Gate (when scim.enabled == true).
If idem.key_rule is UNKNOWN → block Completeness Gate (when scim.enabled == true).
If audit.required is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SSO
Pass conditions:
required_fields_present == true
provider_id_exists_in_SSO_01 == true
scim_enablement_defined == true
if scim_enabled then resources_ops_mapping_defined == true

idempotency_defined == true
audit_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SSO-06

SSO-06 — Account Linking & Identity Merge Rules
Header Block
template_id: SSO-06
title: Account Linking & Identity Merge Rules
type: sso_account_linking_identity_merge
template_version: 1.0.0
output_path: 10_app/sso/SSO-06_Account_Linking_Identity_Merge_Rules.md
compliance_gate_id: TMP-05.PRIMARY.SSO
upstream_dependencies: ["SSO-02", "SSO-03", "API-04", "DATA-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SSO-01", "SSO-02", "SSO-03", "API-04", "DATA-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for linking external identities (SSO providers) to internal user accounts, including when accounts auto-link, when user confirmation is required, how identity merges are performed, and how to prevent account takeover. This template must be consistent with AuthZ and data schema rules and must not invent identity fields beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SSO-01 Provider Inventory: {{sso.providers}}
SSO-02 Flow Spec: {{sso.flows}} | OPTIONAL
SSO-03 Claim Mapping: {{sso.claim_mapping}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
DATA-06 Data Schemas (identity/user): {{data.schemas}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Identity keys used for linking (email/sub/externalId/UNKNOWN)
Auto-link policy (allowed/blocked)
User confirmation rules (when required)
Linking constraints (one-to-one, many-to-one)
Merge eligibility rules (what can merge)
Merge strategy (fields precedence)
Audit logging requirements (link/merge events)
Security controls (prevent takeover, verified email rules)
Unlink policy (is unlink allowed?)
Telemetry requirements (link success/fail, merges)

Optional Fields
Cross-tenant linking rules | OPTIONAL
Admin-assisted merge rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Never link accounts based on unverified identifiers.
Merges MUST be auditable and reversible if possible.
Linking must not grant elevated roles beyond claim mapping rules.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Linking Keys
   keys: {{link.keys}}
   verified_identifier_rule: {{link.verified_identifier_rule}}
2. Auto-Link Policy
   auto_link_enabled: {{auto.enabled}}
   auto_link_conditions: {{auto.conditions}} | OPTIONAL
3. Confirmation Rules
   confirmation_required_when: {{confirm.required_when}}
   confirmation_ui_pattern: {{confirm.ui_pattern}} | OPTIONAL
4. Linking Constraints
   constraint_model: {{constraints.model}} (one_to_one/many_to_one/UNKNOWN)
   duplicate_identity_rule: {{constraints.duplicate_identity_rule}} | OPTIONAL
5. Merge Eligibility
   eligible_when: {{merge.eligible_when}}
   ineligible_when: {{merge.ineligible_when}} | OPTIONAL
6. Merge Strategy
   field_precedence: {{merge.field_precedence}}
   conflict_resolution: {{merge.conflict_resolution}} | OPTIONAL
7. Unlink Policy
   unlink_supported: {{unlink.supported}}
   unlink_rules: {{unlink.rules}} | OPTIONAL
8. Security Controls
   prevent_takeover_rules: {{security.prevent_takeover_rules}}
   role_escalation_prevention: {{security.role_escalation_prevention}} | OPTIONAL
9. Audit / Telemetry
   audit_required: {{audit.required}}
   audit_fields: {{audit.fields}} | OPTIONAL
   link_success_metric: {{telemetry.link_success_metric}}
   merge_metric: {{telemetry.merge_metric}} | OPTIONAL

10. References
    Flow spec: {{xref:SSO-02}} | OPTIONAL
    Claim mapping: {{xref:SSO-03}} | OPTIONAL
    Security controls: {{xref:SSO-09}} | OPTIONAL
    Audit/compliance: {{xref:SSO-10}} | OPTIONAL

Cross-References
Upstream: {{xref:SSO-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SSO-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define linking keys + auto-link policy + verified identifier rule.
intermediate: Required. Define merge strategy and unlink policy and security controls.
advanced: Required. Add cross-tenant/admin merge rules and reversibility guarantees.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, auto conditions, ui pattern, duplicate rule, ineligible rules, conflict resolution, unlink rules, role escalation prevention, audit fields, merge metric, cross-tenant/admin rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If link.keys is UNKNOWN → block Completeness Gate.
If link.verified_identifier_rule is UNKNOWN → block Completeness Gate.
If telemetry.link_success_metric is UNKNOWN → block Completeness Gate.
If audit.required is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SSO
Pass conditions:
required_fields_present == true
linking_keys_and_verification_defined == true
auto_link_or_confirmation_defined == true
security_controls_defined == true
audit_and_telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SSO-07

SSO-07 — MFA/Step-Up Auth Rules (when required)
Header Block
template_id: SSO-07
title: MFA/Step-Up Auth Rules (when required)
type: sso_mfa_step_up_rules
template_version: 1.0.0
output_path: 10_app/sso/SSO-07_MFA_Step_Up_Auth_Rules.md
compliance_gate_id: TMP-05.PRIMARY.SSO
upstream_dependencies: ["SSO-02", "API-04", "ADMIN-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SSO-01", "SSO-02", "API-04", "ADMIN-01", "ROUTE-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical MFA and step-up authentication rules: what triggers step-up, what assurance levels are required for sensitive actions, how the client/server enforce it, and what the user experience is when step-up is required. This template must be consistent with AuthZ rules and admin capability risk levels and must not invent roles/actions not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SSO-01 Provider Inventory: {{sso.providers}}
SSO-02 Flow Spec: {{sso.flows}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
ADMIN-01 Admin Capabilities Matrix: {{admin.capabilities}} | OPTIONAL
ROUTE-04 Guard Rules: {{route.guard_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
MFA supported (yes/no/UNKNOWN)
Assurance level model (AAL/LOA/UNKNOWN)
Step-up triggers (actions, risk level, location)
Per-action requirement map (capability/action → required level)
How step-up is enforced (server-side gate)
How step-up is requested (redirect/challenge)
Session binding for step-up (duration, re-use window)
Failure handling (deny, retry, lockout)
Telemetry requirements (step-up challenged, success/fail)

Optional Fields
Remembered device policy | OPTIONAL
Risk engine inputs | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Step-up MUST be enforced server-side for privileged actions.
Do not rely on client-only checks for MFA/assurance enforcement.
If MFA is not supported, privileged actions requiring step-up MUST be blocked or redesigned.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. MFA Support
   mfa_supported: {{mfa.supported}}
   providers_supporting_mfa: {{mfa.providers_supporting_mfa}} | OPTIONAL
2. Assurance Model
   model: {{assurance.model}} (AAL/LOA/UNKNOWN)
   levels: {{assurance.levels}} | OPTIONAL
3. Step-Up Triggers
   triggers: {{stepup.triggers}}
   risk_level_binding: {{stepup.risk_level_binding}} | OPTIONAL
4. Per-Action Requirements
   Requirement
   action_id: {{reqs[0].action_id}}
   action_ref: {{reqs[0].action_ref}} (e.g., {{xref:ADMIN-01}} capability_id) | OPTIONAL
   required_assurance_level: {{reqs[0].required_level}}
   notes: {{reqs[0].notes}} | OPTIONAL

(Repeat per action.)

5. Enforcement
   server_enforced: {{enforce.server_enforced}}
   enforcement_point: {{enforce.point}} (middleware/endpoint/UNKNOWN)
   deny_on_insufficient_assurance: {{enforce.deny_rule}}
6. Step-Up Request Flow
   request_method: {{flow.request_method}} (redirect/challenge/UNKNOWN)
   return_to_rule: {{flow.return_to_rule}} | OPTIONAL
   max_challenge_attempts: {{flow.max_attempts}} | OPTIONAL
7. Step-Up Session Binding
   reuse_window_seconds: {{session.reuse_window_seconds}}
   reauth_required_after: {{session.reauth_required_after}} | OPTIONAL

8. Failure Handling
   on_fail_behavior: {{fail.on_fail_behavior}}
   lockout_policy: {{fail.lockout_policy}} | OPTIONAL
9. Telemetry
   challenge_shown_metric: {{telemetry.challenge_shown_metric}}
   challenge_success_metric: {{telemetry.challenge_success_metric}} | OPTIONAL
   challenge_fail_metric: {{telemetry.challenge_fail_metric}} | OPTIONAL
10. References
    Flow spec: {{xref:SSO-02}} | OPTIONAL
    AuthZ rules: {{xref:API-04}} | OPTIONAL
    Admin capabilities/risk: {{xref:ADMIN-01}} | OPTIONAL
    Guard rules: {{xref:ROUTE-04}} | OPTIONAL
    Security controls: {{xref:SSO-09}} | OPTIONAL

Cross-References
Upstream: {{xref:SSO-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SSO-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define whether MFA supported and a trigger list; do not invent action ids.
intermediate: Required. Define per-action map and server enforcement points.
advanced: Required. Add risk engine inputs, remembered device policy, and strict telemetry.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, providers supporting mfa, assurance levels, risk bindings, action refs/notes, enforcement point, return-to rule, max attempts, reauth after, lockout policy, optional metrics, remembered device/risk engine, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If mfa.supported is UNKNOWN → block Completeness Gate.
If enforce.server_enforced is UNKNOWN → block Completeness Gate.
If enforce.deny_rule is UNKNOWN → block Completeness Gate.
If telemetry.challenge_shown_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SSO
Pass conditions:
required_fields_present == true
mfa_support_defined == true
per_action_requirements_defined == true
server_enforcement_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SSO-08

SSO-08 — Failure Handling (lockouts, partial provision, retries)
Header Block
template_id: SSO-08
title: Failure Handling (lockouts, partial provision, retries)
type: sso_failure_handling
template_version: 1.0.0
output_path: 10_app/sso/SSO-08_Failure_Handling.md
compliance_gate_id: TMP-05.PRIMARY.SSO
upstream_dependencies: ["SSO-02", "SSO-03", "SSO-05", "CER-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SSO-01", "SSO-02", "SSO-03", "SSO-05", "CER-02", "CER-04", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical failure handling for SSO and provisioning: login failures, token/assertion validation failures, account linking/provisioning errors, lockouts, retries, and safe user/operator remediation. This template must be consistent with retry/session-expiry patterns and must not introduce unsafe fallbacks.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SSO-01 Provider Inventory: {{sso.providers}}
SSO-02 Flow Spec: {{sso.flows}}
SSO-03 Claim Mapping: {{sso.claim_mapping}} | OPTIONAL
SSO-05 SCIM Provisioning: {{sso.scim}} | OPTIONAL
CER-02 Retry/Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
CER-04 Session Expiry Handling: {{cer.session_expiry}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Failure taxonomy (authn/authz/claims/provisioning/network)
User-facing handling rules (what user sees)
Operator-facing handling rules (what support/admin does)
Retry rules (when retry allowed vs forbidden)
Lockout rules (thresholds, cool-down)
Partial provisioning handling (rollback vs quarantine)
Account linking conflicts handling
Session loop prevention rules
Telemetry requirements (failure rates, lockouts)
Audit logging requirements (security events)

Optional Fields
Provider-specific error mapping | OPTIONAL
Self-serve recovery flow (contact IT) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Never “fail open” (no granting access on uncertain identity).
Lockouts must not be bypassable via retries.
Partial provisioning must not result in elevated access.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Failure Taxonomy
   classes:
   authn_failed: {{fail.classes.authn_failed}}
   assertion_invalid: {{fail.classes.assertion_invalid}}
   claims_missing: {{fail.classes.claims_missing}}
   provisioning_failed: {{fail.classes.provisioning_failed}}
   account_link_conflict: {{fail.classes.account_link_conflict}}
   network_timeout: {{fail.classes.network_timeout}} | OPTIONAL
2. User Handling
   user_error_ux_rule: {{user.ux_rule}}
   when_to_prompt_retry: {{user.prompt_retry_rule}} | OPTIONAL
   contact_support_rule: {{user.contact_support_rule}} | OPTIONAL
3. Operator Handling
   operator_actions: {{ops.actions}}
   runbook_location: {{ops.runbook_location}} | OPTIONAL
4. Retry Rules
   retry_allowed_classes: {{retry.allowed_classes}}
   max_attempts: {{retry.max_attempts}} | OPTIONAL
   backoff_policy: {{retry.backoff_policy}} | OPTIONAL
5. Lockout Rules
   lockout_supported: {{lockout.supported}}
   threshold: {{lockout.threshold}}
   cooldown_minutes: {{lockout.cooldown_minutes}} | OPTIONAL
   reset_policy: {{lockout.reset_policy}} | OPTIONAL
6. Partial Provisioning
   policy: {{prov.partial_policy}} (rollback/quarantine/UNKNOWN)
   quarantine_behavior: {{prov.quarantine_behavior}} | OPTIONAL
   rollback_steps: {{prov.rollback_steps}} | OPTIONAL

7. Account Linking Conflicts
   conflict_behavior: {{link.conflict_behavior}}
   manual_resolution_allowed: {{link.manual_resolution_allowed}} | OPTIONAL
8. Session Loop Prevention
   max_login_attempts_per_window: {{loops.max_login_attempts}}
   cooldown_ms: {{loops.cooldown_ms}} | OPTIONAL
9. Telemetry
   login_failure_metric: {{telemetry.login_failure_metric}}
   lockout_metric: {{telemetry.lockout_metric}} | OPTIONAL
   provisioning_failure_metric: {{telemetry.provisioning_failure_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
10. Audit Logging
    audit_required: {{audit.required}}
    audit_fields: {{audit.fields}} | OPTIONAL
    retention_policy: {{audit.retention_policy}} | OPTIONAL
11. References
    Flow spec: {{xref:SSO-02}}
    Claim mapping: {{xref:SSO-03}} | OPTIONAL
    SCIM provisioning: {{xref:SSO-05}} | OPTIONAL
    Session expiry: {{xref:CER-04}} | OPTIONAL
    Security controls: {{xref:SSO-09}} | OPTIONAL
    Audit trail: {{xref:ADMIN-03}} | OPTIONAL

Cross-References
Upstream: {{xref:SSO-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SSO-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define taxonomy + user UX rule + lockout basics; use UNKNOWN for provider mappings.
intermediate: Required. Define partial provisioning policy and retry/backoff.
advanced: Required. Add operator runbooks, precise thresholds, and audit/telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, prompt retry, contact support, runbook location, retry max/backoff, cooldown/reset, quarantine/rollback, manual resolution, telemetry fields, audit fields/retention, provider mapping, self-serve recovery, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If user.ux_rule is UNKNOWN → block Completeness Gate.
If lockout.supported is UNKNOWN → block Completeness Gate.
If lockout.threshold is UNKNOWN → block Completeness Gate (when lockout.supported == true).

If telemetry.login_failure_metric is UNKNOWN → block Completeness Gate.
If audit.required is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SSO
Pass conditions:
required_fields_present == true
failure_taxonomy_defined == true
lockout_policy_defined == true
partial_provisioning_policy_defined == true
telemetry_defined == true
audit_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SSO-09

SSO-09 — Security Controls (PKCE, signature validation, replay defense)
Header Block
template_id: SSO-09
title: Security Controls (PKCE, signature validation, replay defense)
type: sso_security_controls
template_version: 1.0.0
output_path: 10_app/sso/SSO-09_Security_Controls.md
compliance_gate_id: TMP-05.PRIMARY.SSO
upstream_dependencies: ["SSO-02", "CSec-01", "CSec-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SSO-01", "SSO-02", "SSO-04", "CSec-01", "CSec-05", "CER-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical security control requirements for SSO: PKCE usage, redirect/callback hardening, token/assertion signature validation, replay defenses (state/nonce), secure cookie/session settings, and logging/redaction. This template must be consistent with client token storage and deep link security and must not invent controls not applicable to the chosen protocol/provider.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SSO-01 Provider Inventory: {{sso.providers}}
SSO-02 Flow Spec: {{sso.flows}}
SSO-04 Session/Token Exchange: {{sso.session_rules}} | OPTIONAL
CSec-01 Token Storage: {{csec.token_storage}} | OPTIONAL
CSec-05 Secure Deep Link Handling: {{csec.deep_link_security}} | OPTIONAL
CER-05 Logging/Redaction: {{cer.logging}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Provider/protocol applicability map (provider_id → controls)
Redirect URI allowlist rule
State required rule (CSRF)
Nonce required rule (OIDC)
PKCE required rule (where applicable)
Signature validation rule (JWT/SAML)
Issuer/audience validation rules
Replay prevention window rules (nonce/state TTL)
Secure session settings (cookie flags / token storage constraints)
Key rotation/JWKS caching rules

Logging/redaction rules (no tokens/PII)
Security telemetry requirements (validation failures)

Optional Fields
mTLS between app and IdP | OPTIONAL
Advanced token binding (DPoP/MTLS OIDC) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
No skipping signature validation.
No dynamic redirects; redirect URIs must be fixed/allowlisted.
Controls must be deterministic and enforceable.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Applicability Map
   Provider Controls
   provider_id: {{controls[0].provider_id}}
   protocol: {{controls[0].protocol}}
   pkce_required: {{controls[0].pkce_required}} | OPTIONAL
   state_required: {{controls[0].state_required}}
   nonce_required: {{controls[0].nonce_required}} | OPTIONAL
   signature_validation: {{controls[0].signature_validation}}
   issuer_validation: {{controls[0].issuer_validation}}
   audience_validation: {{controls[0].audience_validation}}
   jwks_cache_rule: {{controls[0].jwks_cache_rule}} | OPTIONAL
   notes: {{controls[0].notes}} | OPTIONAL

(Repeat per provider_id.)

2. Redirect & Callback Hardening
   redirect_allowlist_rule: {{redirect.allowlist_rule}}
   no_open_redirects_rule: {{redirect.no_open_redirects_rule}}
   callback_path_validation: {{redirect.callback_path_validation}} | OPTIONAL
3. State / Nonce / Replay
   state_ttl_seconds: {{replay.state_ttl_seconds}}
   nonce_ttl_seconds: {{replay.nonce_ttl_seconds}} | OPTIONAL
   single_use_nonce: {{replay.single_use_nonce}} | OPTIONAL
4. PKCE
   pkce_method: {{pkce.method}} (S256/plain/UNKNOWN) | OPTIONAL
   required_for_clients: {{pkce.required_for_clients}} | OPTIONAL

5. Signature / Claims Validation
   signature_rule: {{validate.signature_rule}}
   issuer_rule: {{validate.issuer_rule}}
   audience_rule: {{validate.audience_rule}}
   alg_allowlist: {{validate.alg_allowlist}} | OPTIONAL
   clock_skew_seconds: {{validate.clock_skew_seconds}} | OPTIONAL
6. Key Rotation / JWKS
   jwks_fetch_rule: {{keys.jwks_fetch_rule}} | OPTIONAL
   jwks_rotation_handling: {{keys.rotation_handling}} | OPTIONAL
7. Session Security
   session_storage_ref: {{session.storage_ref}} (expected: {{xref:CSec-01}}) | OPTIONAL
   cookie_flags_rule: {{session.cookie_flags_rule}} | OPTIONAL
   token_redaction_rule: {{session.token_redaction_rule}}
8. Logging / Redaction
   no_token_in_logs: {{logs.no_token_in_logs}}
   no_pii_in_logs: {{logs.no_pii_in_logs}} | OPTIONAL
   log_fields_allowlist: {{logs.field_allowlist}} | OPTIONAL
9. Security Telemetry
   signature_fail_metric: {{telemetry.signature_fail_metric}}
   state_nonce_fail_metric: {{telemetry.state_nonce_fail_metric}} | OPTIONAL
   issuer_audience_fail_metric: {{telemetry.issuer_audience_fail_metric}} | OPTIONAL
10. References
    Flow spec: {{xref:SSO-02}}
    Session/token rules: {{xref:SSO-04}} | OPTIONAL
    Token storage: {{xref:CSec-01}} | OPTIONAL
    Secure deep links: {{xref:CSec-05}} | OPTIONAL
    Logging/redaction: {{xref:CER-05}} | OPTIONAL

Cross-References
Upstream: {{xref:SSO-01}}, {{xref:SSO-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SSO-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define state + signature validation + redirect allowlist; use UNKNOWN for advanced token binding.
intermediate: Required. Define nonce/PKCE applicability and replay TTLs and JWKS caching.
advanced: Required. Add algorithm allowlists, rotation handling, and security telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, pkce required, nonce required, jwks cache, notes, callback path validation, nonce ttl, single-use nonce, pkce method/clients, alg allowlist, clock skew, jwks fetch/rotation, session storage/cookie flags, log allowlist, optional telemetry metrics, mtls/dpop, open_questions
If any Required Field is UNKNOWN, allow only if:

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If redirect.allowlist_rule is UNKNOWN → block Completeness Gate.
If validate.signature_rule is UNKNOWN → block Completeness Gate.
If validate.issuer_rule is UNKNOWN → block Completeness Gate.
If session.token_redaction_rule is UNKNOWN → block Completeness Gate.
If telemetry.signature_fail_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SSO
Pass conditions:
required_fields_present == true
redirect_and_replay_controls_defined == true
signature_and_claims_validation_defined == true
session_security_defined == true
security_telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SSO-10

SSO-10 — Audit & Compliance (login events, admin actions, retention)
Header Block
template_id: SSO-10
title: Audit & Compliance (login events, admin actions, retention)
type: sso_audit_compliance
template_version: 1.0.0
output_path: 10_app/sso/SSO-10_Audit_Compliance.md
compliance_gate_id: TMP-05.PRIMARY.SSO
upstream_dependencies: ["SSO-02", "SSO-03", "ADMIN-03", "CER-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SSO-01", "SSO-02", "SSO-03", "SSO-05", "ADMIN-03", "CER-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical audit and compliance requirements for SSO: what events must be recorded (logins, failures, lockouts, provisioning, role assignments), required fields, retention, redaction, and how audit data is accessed for investigations. This template must be consistent with global audit trail and logging/redaction rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SSO-01 Provider Inventory: {{sso.providers}}
SSO-02 Flow Spec: {{sso.flows}} | OPTIONAL
SSO-03 Claim/Role Mapping: {{sso.claim_mapping}} | OPTIONAL
SSO-05 SCIM Provisioning: {{sso.scim}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}}
CER-05 Logging/Crash Reporting: {{cer.logging}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Audited event types list (login, failure, lockout, role assigned, provisioned)
Event schema (required fields)
Correlation fields (request_id, trace_id)
Identity fields policy (hashing, redaction)
Retention policy (days)
Access policy (who can view)
Export policy (for compliance)
Alerting triggers (suspicious patterns)
Tamper-evidence expectations (append-only)
Deletion policy for audit logs (if any)

Optional Fields
SIEM forwarding rules | OPTIONAL
Geo/IP capture policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Audit logs must be append-only and tamper-evident where possible.
Do not store raw secrets/tokens; sensitive identifiers should be hashed as required.
Retention must be explicit and enforced.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Audited Event Types
   events: {{audit.events}}
2. Event Schema (Required Fields)
   timestamp: {{schema.timestamp}}
   event_type: {{schema.event_type}}
   provider_id: {{schema.provider_id}} | OPTIONAL
   user_id_hash: {{schema.user_id_hash}} | OPTIONAL
   subject_identifier_hash: {{schema.subject_id_hash}} | OPTIONAL
   result: {{schema.result}}
   reason_code: {{schema.reason_code}} | OPTIONAL
   request_id: {{schema.request_id}} | OPTIONAL
   trace_id: {{schema.trace_id}} | OPTIONAL
   ip_hash_or_policy: {{schema.ip_policy}} | OPTIONAL
3. Correlation Rules
   correlate_with_server_logs: {{corr.with_server}}
   correlation_id_sources: {{corr.id_sources}} | OPTIONAL
4. Identity Fields Policy
   hashing_rule: {{identity.hashing_rule}}
   redaction_rule: {{identity.redaction_rule}} | OPTIONAL
5. Retention
   retention_days: {{retention.days}}
   archive_policy: {{retention.archive_policy}} | OPTIONAL
6. Access Policy
   who_can_view: {{access.who_can_view}}
   approval_required: {{access.approval_required}} | OPTIONAL
7. Export Policy
   export_supported: {{export.supported}}
   export_format: {{export.format}} | OPTIONAL
   export_audit_required: {{export.export_audit_required}} | OPTIONAL

8. Alerting
   suspicious_login_alert: {{alerts.suspicious_login_alert}}
   lockout_spike_alert: {{alerts.lockout_spike_alert}} | OPTIONAL
9. Tamper Evidence
   append_only_rule: {{tamper.append_only_rule}}
   integrity_checks: {{tamper.integrity_checks}} | OPTIONAL
10. Deletion Policy
    delete_audit_logs: {{delete.delete_audit_logs}}
    delete_exceptions: {{delete.exceptions}} | OPTIONAL
11. References
    Global audit trail: {{xref:ADMIN-03}}
    Logging/redaction: {{xref:CER-05}} | OPTIONAL
    Failure handling: {{xref:SSO-08}} | OPTIONAL

Cross-References
Upstream: {{xref:ADMIN-03}}, {{xref:SSO-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-02}} | OPTIONAL
Standards: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define event types + retention + append-only rule; use UNKNOWN for SIEM forwarding.
intermediate: Required. Define schema fields, access policy, and alert triggers.
advanced: Required. Add export policy, integrity checks, and correlation rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, provider/user hashes, reason code, ids, ip policy, id sources, redaction rule, archive policy, approval required, export fields, optional alerts, integrity checks, delete exceptions, siem/geo policy, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If audit.events is UNKNOWN → block Completeness Gate.
If retention.days is UNKNOWN → block Completeness Gate.
If access.who_can_view is UNKNOWN → block Completeness Gate.
If tamper.append_only_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SSO
Pass conditions:
required_fields_present == true
audit_events_defined == true
schema_defined == true
retention_defined == true
access_policy_defined == true

tamper_evidence_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CRM/ERP Integrations (CRMERP)

CRM/ERP Integrations (CRMERP)
CRMERP-01 System Inventory (by system_id)
CRMERP-02 Object/Entity Mapping Catalog (objects ↔ entities, keys)
CRMERP-03 Sync Direction Rules (push/pull/bidirectional)
CRMERP-04 Sync Scheduling & Triggers (cron/events/manual)
CRMERP-05 Conflict Resolution Rules (source of truth, LWW, prompts)
CRMERP-06 Rate Limits & Quotas (per vendor, backoff)
CRMERP-07 Data Quality & Validation (required fields, dedupe)
CRMERP-08 Error Handling & Reconciliation (replay, backfill)
CRMERP-09 Security & Compliance (PII, least privilege, audit)
CRMERP-10 Observability & Runbooks (dashboards, alerts, operator steps)

CRMERP-01

CRMERP-01 — System Inventory (by system_id)
Header Block
template_id: CRMERP-01
title: System Inventory (by system_id)
type: crmerp_system_inventory
template_version: 1.0.0
output_path: 10_app/crmerp/CRMERP-01_System_Inventory.md
compliance_gate_id: TMP-05.PRIMARY.CRMERP
upstream_dependencies: ["IXS-01", "IXS-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "IXS-02", "API-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical inventory of all CRM/ERP systems integrated with the application, indexed by system_id, including which objects/entities are synced, directionality, environments, and ownership. This document must not invent system_ids not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}} | OPTIONAL
IXS-02 Integration Specs: {{ixs.integration_specs}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
System registry (system_id list)
system_id (stable identifier)
system name/vendor (e.g., Salesforce, NetSuite)
system type (CRM/ERP/UNKNOWN)
integration_id binding (from IXS-01)
sync direction (push/pull/bidirectional)
objects/entities covered (external object names + internal entity_id/schema_ref)
primary keys (external id fields)
environments enabled (dev/stage/prod)
criticality (low/med/high)
owner (team/role)
data classes involved (PII/financial/etc.)
references to detailed specs (CRMERP-02..10)

Optional Fields
Status (planned/active/deprecated) | OPTIONAL
SLA/uptime notes | OPTIONAL
Cost/billing owner | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new system_ids; use only {{spec.crmerp_systems_by_id}} if present, else mark UNKNOWN and flag.
system_id MUST bind to an integration_id in {{xref:IXS-01}} (or be flagged UNKNOWN with reason).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Inventory Summary
   total_systems: {{inv.total}}
   types_present: {{inv.types_present}} | OPTIONAL
   notes: {{inv.notes}} | OPTIONAL
2. System Entries (by system_id)
   System
   system_id: {{systems[0].system_id}}
   name: {{systems[0].name}}
   vendor: {{systems[0].vendor}} | OPTIONAL
   type: {{systems[0].type}}
   integration_id: {{systems[0].integration_id}}
   sync_direction: {{systems[0].sync_direction}}
   objects_covered: {{systems[0].objects_covered}}
   primary_key_fields: {{systems[0].primary_key_fields}}
   env_enabled: {{systems[0].env_enabled}}
   criticality: {{systems[0].criticality}}
   owner: {{systems[0].owner}}
   data_classes: {{systems[0].data_classes}}
   spec_refs:
- {{systems[0].spec_refs[0]}} (e.g., {{xref:CRMERP-02}} / {{xref:CRMERP-03}})
- {{systems[0].spec_refs[1]}} | OPTIONAL
  status: {{systems[0].status}} | OPTIONAL
  sla_notes: {{systems[0].sla_notes}} | OPTIONAL
  cost_owner: {{systems[0].cost_owner}} | OPTIONAL
  open_questions:
  {{systems[0].open_questions[0]}} | OPTIONAL

(Repeat per system_id.)

3. References
   Entity/object mapping: {{xref:CRMERP-02}} | OPTIONAL
   Sync direction rules: {{xref:CRMERP-03}} | OPTIONAL
   Scheduling/triggers: {{xref:CRMERP-04}} | OPTIONAL
   Conflict resolution: {{xref:CRMERP-05}} | OPTIONAL
   Quotas/rate limits: {{xref:CRMERP-06}} | OPTIONAL
   Data quality: {{xref:CRMERP-07}} | OPTIONAL
   Reconciliation/backfills: {{xref:CRMERP-08}} | OPTIONAL
   Security/compliance: {{xref:CRMERP-09}} | OPTIONAL
   Observability/runbooks: {{xref:CRMERP-10}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}} | OPTIONAL, {{xref:IXS-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CRMERP-02}}, {{xref:CRMERP-03}}, {{xref:CRMERP-04}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Populate system registry and core fields; do not invent IDs.
intermediate: Required. Add object/entity coverage and primary key fields and spec_refs.
advanced: Required. Add criticality/ownership and data class flags with traceability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, vendor, inv notes, status, sla_notes, cost_owner, secondary spec refs, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If systems[*].system_id is UNKNOWN → block Completeness Gate.*
*If systems[*].integration_id is UNKNOWN → block Completeness Gate.
If systems[*].sync_direction is UNKNOWN → block Completeness Gate.*
*If systems[*].spec_refs is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CRMERP
Pass conditions:
required_fields_present == true
system_ids_unique == true
integration_ids_exist_in_IXS_01 == true
objects_covered_defined == true
spec_refs_present == true
placeholder_resolution == true
no_unapproved_unknowns == true

CRMERP-02

CRMERP-02 — Object/Entity Mapping Catalog (objects ↔ entities, keys)
Header Block
template_id: CRMERP-02
title: Object/Entity Mapping Catalog (objects ↔ entities, keys)
type: crmerp_object_entity_mapping_catalog
template_version: 1.0.0
output_path: 10_app/crmerp/CRMERP-02_Object_Entity_Mapping_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.CRMERP
upstream_dependencies: ["CRMERP-01", "IXS-05", "DATA-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CRMERP-01", "IXS-05", "DATA-06", "DQV-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical catalog mapping CRM/ERP objects to internal entities/schemas, including key mappings, field-level mapping references, and normalization/validation bindings. This template must be consistent with canonical schemas and integration mapping rules and must not invent object names or schema paths beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CRMERP-01 System Inventory: {{crmerp.systems}}
IXS-05 Data Mapping & Transformation Rules: {{ixs.data_mapping}} | OPTIONAL
DATA-06 Canonical Data Schemas: {{data.schemas}} | OPTIONAL
DQV-03 Data Validation Rules: {{data.validation_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
system_id binding
Mapping registry (mapping_id list)
external_object name
internal entity/schema ref (entity_id/schema_ref)
primary key mapping (external_id → internal_id)
foreign key/reference mapping (relationships)
field mapping reference (link to IXS-05 map_id or inline summary)
sync direction per object (push/pull/bidirectional)
create/update semantics (upsert keys)
delete semantics (soft delete, archive)
validation binding (DQV-03)
PII/data class notes

Optional Fields
Lookup resolution rules | OPTIONAL
Object-specific constraints | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent schema refs; only use valid refs from {{xref:DATA-06}}.
Each external object MUST map to exactly one primary internal entity/schema (or explicitly document composite mapping).
Key mapping MUST be explicit; no “match by name” without deterministic rules.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Mapping Summary
   system_id: {{meta.system_id}}
   total_objects_mapped: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Object ↔ Entity Entries
   Entry
   mapping_id: {{maps[0].mapping_id}}
   external_object: {{maps[0].external_object}}
   internal_ref: {{maps[0].internal_ref}}
   sync_direction: {{maps[0].sync_direction}}
   primary_key_map: {{maps[0].primary_key_map}}
   foreign_keys: {{maps[0].foreign_keys}} | OPTIONAL
   upsert_rule: {{maps[0].upsert_rule}}
   delete_rule: {{maps[0].delete_rule}}
   field_map_ref: {{maps[0].field_map_ref}} (expected: {{xref:IXS-05}}) | OPTIONAL
   validation_ref: {{maps[0].validation_ref}} (expected: {{xref:DQV-03}}) | OPTIONAL
   pii_notes: {{maps[0].pii_notes}} | OPTIONAL
   constraints: {{maps[0].constraints}} | OPTIONAL
   open_questions:
   {{maps[0].open_questions[0]}} | OPTIONAL

(Repeat per mapping_id.)

3. Lookup Resolution Rules (Optional)
   lookup_strategy: {{lookup.strategy}} | OPTIONAL
   missing_reference_behavior: {{lookup.missing_reference_behavior}} | OPTIONAL
4. References
   System inventory: {{xref:CRMERP-01}}
   Integration mapping rules: {{xref:IXS-05}} | OPTIONAL

Canonical schemas: {{xref:DATA-06}} | OPTIONAL
Validation rules: {{xref:DQV-03}} | OPTIONAL
Conflict resolution: {{xref:CRMERP-05}} | OPTIONAL

Cross-References
Upstream: {{xref:CRMERP-01}}, {{xref:DATA-06}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CRMERP-03}}, {{xref:CRMERP-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define object/entity mappings and primary keys; use UNKNOWN for optional foreign keys.
intermediate: Required. Define upsert/delete semantics and validation refs.
advanced: Required. Add lookup resolution rules, constraints, and strict mapping traceability to IXS-05.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, meta notes, foreign keys, field map/validation refs, pii notes, constraints, lookup section, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If meta.system_id is UNKNOWN → block Completeness Gate.
If maps[*].internal_ref is UNKNOWN → block Completeness Gate.
If maps[*].primary_key_map is UNKNOWN → block Completeness Gate.
If maps[*].upsert_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CRMERP
Pass conditions:
required_fields_present == true
system_id_exists_in_CRMERP_01 == true
mapping_ids_unique == true
all internal_ref values valid (DATA-06)
primary_keys_and_upsert_rules_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CRMERP-03

CRMERP-03 — Sync Direction Rules (push/pull/bidirectional)
Header Block
template_id: CRMERP-03
title: Sync Direction Rules (push/pull/bidirectional)
type: crmerp_sync_direction_rules
template_version: 1.0.0
output_path: 10_app/crmerp/CRMERP-03_Sync_Direction_Rules.md
compliance_gate_id: TMP-05.PRIMARY.CRMERP
upstream_dependencies: ["CRMERP-01", "CRMERP-02", "OFS-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CRMERP-01", "CRMERP-02", "IXS-06", "OFS-02", "EVT-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules that govern sync direction (push, pull, bidirectional) for CRM/ERP integrations, including source-of-truth designation per object/field, write permissions, conflict triggers, and safe handling for bidirectional sync. This template must be consistent with object/entity mapping and conflict/reconciliation policies.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CRMERP-01 System Inventory: {{crmerp.systems}}
CRMERP-02 Object/Entity Mapping: {{crmerp.mapping}}
IXS-06 Error Handling & Recovery: {{ixs.error_recovery}} | OPTIONAL
OFS-02 Sync Model (queues/conflicts): {{ofs.sync_model}} | OPTIONAL
EVT-03 Producer/Consumer Map: {{evt.producer_consumer}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
system_id binding
Supported sync modes (push/pull/bidirectional)
Per-object sync mode assignment (external_object → mode)
Source of truth rules (system vs app)
Per-field write authority rules (field-level exceptions)
Outbound write constraints (what is allowed to be pushed)
Inbound pull constraints (filters/criteria)
Conflict trigger definitions (what counts as conflict)
Idempotency constraints for writes
Audit/telemetry requirements (sync decisions)

Optional Fields
User-initiated sync rules | OPTIONAL
Rate-limit aware behavior | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Bidirectional sync MUST have explicit source-of-truth rules; no “both own it” ambiguity.
Field-level exceptions must be documented and minimal.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Supported Modes
   system_id: {{meta.system_id}}
   modes_supported: {{modes.supported}} (push/pull/bidirectional/UNKNOWN)
2. Per-Object Mode Assignment
   Assignment
   external_object: {{objects[0].external_object}}
   mode: {{objects[0].mode}}
   source_of_truth: {{objects[0].source_of_truth}} (external/internal/UNKNOWN)
   notes: {{objects[0].notes}} | OPTIONAL

(Repeat per object.)

3. Field-Level Write Authority (Optional but recommended)
   Rule
   external_object: {{fields[0].external_object}}
   field: {{fields[0].field}}
   write_authority: {{fields[0].write_authority}} (external/internal/UNKNOWN)
   notes: {{fields[0].notes}} | OPTIONAL

(Repeat per rule.)

4. Outbound Write Constraints
   allowed_writes: {{outbound.allowed_writes}}
   blocked_writes: {{outbound.blocked_writes}} | OPTIONAL
5. Inbound Pull Constraints
   pull_filters: {{inbound.pull_filters}}
   incremental_sync_rule: {{inbound.incremental_sync_rule}} | OPTIONAL
6. Conflict Triggers
   conflict_triggers: {{conflicts.triggers}}
   version_field_rule: {{conflicts.version_field_rule}} | OPTIONAL

7. Idempotency
   idempotency_required: {{idem.required}}
   idempotency_key_rule: {{idem.key_rule}} | OPTIONAL
8. Audit / Telemetry
   sync_decision_metric: {{telemetry.sync_decision_metric}}
   conflict_metric_ref: {{telemetry.conflict_metric_ref}} (expected: {{xref:CRMERP-05}}/{{xref:OFS-02}}) | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
9. References
   System inventory: {{xref:CRMERP-01}}
   Object/entity mapping: {{xref:CRMERP-02}}
   Conflict resolution: {{xref:CRMERP-05}} | OPTIONAL
   Error handling: {{xref:IXS-06}} | OPTIONAL
   Offline sync model: {{xref:OFS-02}} | OPTIONAL

Cross-References
Upstream: {{xref:CRMERP-01}}, {{xref:CRMERP-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CRMERP-04}}, {{xref:CRMERP-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define per-object mode and source-of-truth; use UNKNOWN for field-level overrides.
intermediate: Required. Define outbound/inbound constraints and conflict triggers.
advanced: Required. Add field-level authority map, idempotency key rules, and telemetry field rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, object notes, field rules, blocked writes, incremental sync, version field rule, idempotency key, conflict metric ref, telemetry fields, user-initiated/rate-limit behavior, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If meta.system_id is UNKNOWN → block Completeness Gate.
If modes.supported is UNKNOWN → block Completeness Gate.
If objects[*].source_of_truth is UNKNOWN → block Completeness Gate (for bidirectional).
If telemetry.sync_decision_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CRMERP
Pass conditions:
required_fields_present == true
system_id_exists_in_CRMERP_01 == true
per_object_modes_defined == true
source_of_truth_defined == true

conflict_triggers_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CRMERP-04

CRMERP-04 — Sync Scheduling & Triggers (cron/event-driven, env enablement)
Header Block
template_id: CRMERP-04
title: Sync Scheduling & Triggers (cron/event-driven, env enablement)
type: crmerp_sync_scheduling_triggers
template_version: 1.0.0
output_path: 10_app/crmerp/CRMERP-04_Sync_Scheduling_Triggers.md
compliance_gate_id: TMP-05.PRIMARY.CRMERP
upstream_dependencies: ["CRMERP-03", "JBS-03", "FFCFG-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CRMERP-01", "CRMERP-03", "JBS-01", "JBS-03", "FFCFG-04", "RLIM-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical scheduling and trigger model for CRM/ERP sync: when sync runs (cron/event/manual), what triggers it, per-environment enablement, backoff under vendor limits, and operator controls. This template must be consistent with job scheduling rules, rate limits, and feature/config matrices.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CRMERP-01 System Inventory: {{crmerp.systems}}
CRMERP-03 Sync Direction Rules: {{crmerp.sync_direction}} | OPTIONAL
JBS-01 Jobs Inventory: {{jobs.inventory}} | OPTIONAL
JBS-03 Scheduling Rules: {{jobs.scheduling_rules}} | OPTIONAL
FFCFG-04 Config Matrix: {{cfg.matrix}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
system_id binding
Trigger types supported (cron/event/manual)
Per-object trigger assignment (object → trigger)
Cron schedule(s) (if used)
Event triggers list (event_id/source) (if used)
Manual trigger rules (who can run, rate limits)
Env enablement rules (dev/stage/prod)
Backoff policy under vendor rate limits
Concurrency rules (avoid overlap)
Telemetry requirements (runs, duration, failures)

Optional Fields
Time window constraints (business hours) | OPTIONAL
Catch-up/backfill trigger rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Scheduling MUST respect vendor rate limits and internal concurrency limits.
Env enablement must be explicit (no accidental prod runs).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Trigger Types
   system_id: {{meta.system_id}}
   supported: {{triggers.supported}} (cron/event/manual/UNKNOWN)
2. Per-Object Trigger Assignment
   Assignment
   external_object: {{assign[0].external_object}}
   trigger_type: {{assign[0].trigger_type}}
   schedule: {{assign[0].schedule}} | OPTIONAL
   event_ref: {{assign[0].event_ref}} | OPTIONAL
   notes: {{assign[0].notes}} | OPTIONAL

(Repeat per object.)

3. Cron Schedules (if applicable)
   default_cron: {{cron.default}} | OPTIONAL
   per_object_cron: {{cron.per_object}} | OPTIONAL
4. Event Triggers (if applicable)
   events: {{events.list}} | OPTIONAL
   producer_ref: {{events.producer_ref}} | OPTIONAL
5. Manual Triggers
   who_can_trigger: {{manual.who_can_trigger}}
   manual_rate_limit_rule: {{manual.rate_limit_rule}} | OPTIONAL
   audit_required: {{manual.audit_required}} | OPTIONAL
6. Environment Enablement
   dev_enabled: {{env.dev_enabled}}
   stage_enabled: {{env.stage_enabled}} | OPTIONAL
   prod_enabled: {{env.prod_enabled}}
   config_ref: {{env.config_ref}} (expected: {{xref:FFCFG-04}}) | OPTIONAL
7. Backoff & Rate Limits
   rate_limit_ref: {{limits.ref}} (expected: {{xref:RLIM-01}}) | OPTIONAL

backoff_policy: {{limits.backoff_policy}}
max_attempts: {{limits.max_attempts}} | OPTIONAL

8. Concurrency
   prevent_overlap: {{conc.prevent_overlap}}
   lock_key_rule: {{conc.lock_key_rule}} | OPTIONAL
9. Telemetry
   sync_run_metric: {{telemetry.run_metric}}
   sync_duration_metric: {{telemetry.duration_metric}} | OPTIONAL
   sync_failure_metric: {{telemetry.failure_metric}} | OPTIONAL
10. References
    System inventory: {{xref:CRMERP-01}}
    Sync direction rules: {{xref:CRMERP-03}} | OPTIONAL
    Jobs scheduling rules: {{xref:JBS-03}} | OPTIONAL
    Config matrix: {{xref:FFCFG-04}} | OPTIONAL
    Rate limit policy: {{xref:RLIM-01}} | OPTIONAL
    Observability/runbooks: {{xref:CRMERP-10}} | OPTIONAL

Cross-References
Upstream: {{xref:CRMERP-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CRMERP-08}}, {{xref:CRMERP-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define supported triggers and env enablement; use UNKNOWN for event triggers if not used.
intermediate: Required. Define backoff, overlap prevention, and telemetry metrics.
advanced: Required. Add catch-up/backfill rules and time-window constraints with explicit schedules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, object notes, schedules/event refs, cron details, events list/producer ref, manual rate limit/audit, config ref, max attempts, lock key, optional telemetry metrics, time windows/backfill, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If triggers.supported is UNKNOWN → block Completeness Gate.
If env.prod_enabled is UNKNOWN → block Completeness Gate.
If limits.backoff_policy is UNKNOWN → block Completeness Gate.
If telemetry.run_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CRMERP
Pass conditions:
required_fields_present == true
trigger_assignments_defined == true

env_enablement_defined == true
rate_limit_backoff_defined == true
concurrency_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CRMERP-05

CRMERP-05 — Conflict Resolution Rules (source of truth, LWW, prompts)
Header Block
template_id: CRMERP-05
title: Conflict Resolution Rules (source of truth, LWW, prompts)
type: crmerp_conflict_resolution_rules
template_version: 1.0.0
output_path: 10_app/crmerp/CRMERP-05_Conflict_Resolution_Rules.md
compliance_gate_id: TMP-05.PRIMARY.CRMERP
upstream_dependencies: ["CRMERP-03", "OFS-04", "IXS-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CRMERP-01", "CRMERP-03", "CRMERP-02", "OFS-04", "IXS-05", "FE-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical conflict detection and resolution rules for CRM/ERP sync, including source-of-truth per object/field, LWW/merge/prompt policies, and what happens when conflicts cannot be resolved automatically. This template must be consistent with sync direction rules and offline reconciliation patterns.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CRMERP-01 System Inventory: {{crmerp.systems}}
CRMERP-03 Sync Direction Rules: {{crmerp.sync_direction}}
CRMERP-02 Object/Entity Mapping: {{crmerp.mapping}} | OPTIONAL
OFS-04 Reconciliation Rules: {{ofs.recon_rules}} | OPTIONAL
IXS-05 Data Mapping Rules: {{ixs.data_mapping}} | OPTIONAL
FE-07 Error/Recovery UX: {{fe.error_ux}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
system_id binding
Conflict detection signals (version/etag/updated_at)
Conflict taxonomy (types)
Default resolution policy (LWW/merge/prompt)
Per-object resolution rules (object → policy)
Per-field overrides (field → authority)
User prompt rules (if prompting allowed)
Automatic merge rules (if merge used)
Unresolvable conflict handling (quarantine/manual)
Telemetry requirements (conflict rate, outcomes)

Optional Fields
Admin resolution workflow ref | OPTIONAL
Batch conflict handling | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Bidirectional objects MUST define resolution policy; no UNKNOWN allowed for those.
Resolution must be deterministic unless explicitly prompt-based.
Prompts must not grant write authority beyond source-of-truth rules.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Conflict Detection
   system_id: {{meta.system_id}}
   signals: {{detect.signals}}
   version_field_rule: {{detect.version_field_rule}} | OPTIONAL
2. Conflict Taxonomy
   types: {{conflicts.types}}
   examples: {{conflicts.examples}} | OPTIONAL
3. Default Policy
   default_policy: {{policy.default}} (LWW/merge/prompt/UNKNOWN)
   rationale: {{policy.rationale}} | OPTIONAL
4. Per-Object Rules
   Rule
   external_object: {{rules[0].external_object}}
   policy: {{rules[0].policy}}
   source_of_truth: {{rules[0].source_of_truth}} (external/internal/UNKNOWN)
   notes: {{rules[0].notes}} | OPTIONAL

(Repeat per object.)

5. Per-Field Overrides (Optional)
   Override
   external_object: {{fields[0].external_object}}
   field: {{fields[0].field}}
   write_authority: {{fields[0].write_authority}} (external/internal/UNKNOWN)
   notes: {{fields[0].notes}} | OPTIONAL
6. Prompt Rules (if applicable)
   prompt_supported: {{prompt.supported}}
   prompt_ui_pattern: {{prompt.ui_pattern}} | OPTIONAL
   prompt_copy_ref: {{prompt.copy_ref}} | OPTIONAL

7. Merge Rules (if applicable)
   merge_strategy: {{merge.strategy}} (field_level/object_level/UNKNOWN)
   merge_precedence: {{merge.precedence}} | OPTIONAL
8. Unresolvable Conflicts
   unresolvable_behavior: {{unresolvable.behavior}} (quarantine/manual/UNKNOWN)
   admin_workflow_ref: {{unresolvable.admin_workflow_ref}} | OPTIONAL
9. Telemetry
   conflict_metric: {{telemetry.conflict_metric}}
   resolution_outcome_metric: {{telemetry.outcome_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
10. References
    Sync direction: {{xref:CRMERP-03}}
    Object mapping: {{xref:CRMERP-02}} | OPTIONAL
    Offline reconciliation: {{xref:OFS-04}} | OPTIONAL
    Error UX: {{xref:FE-07}} | OPTIONAL
    Reconciliation/backfills: {{xref:CRMERP-08}} | OPTIONAL

Cross-References
Upstream: {{xref:CRMERP-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CRMERP-08}}, {{xref:CRMERP-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define detection signals + default policy; use UNKNOWN for per-field overrides.
intermediate: Required. Define per-object policies and unresolvable behavior and telemetry.
advanced: Required. Add merge/prompt rigor and admin workflows and batch handling rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, version field rule, examples, rationale, notes, field overrides, prompt ui/copy refs, merge precedence, admin workflow ref, telemetry fields, batch handling, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If policy.default is UNKNOWN → block Completeness Gate.
If unresolvable.behavior is UNKNOWN → block Completeness Gate.
If telemetry.conflict_metric is UNKNOWN → block Completeness Gate.
If detect.signals is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CRMERP
Pass conditions:
required_fields_present == true
conflict_detection_defined == true
resolution_policies_defined == true

unresolvable_handling_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CRMERP-06

CRMERP-06 — Rate Limits & Quotas (per vendor, backoff)
Header Block
template_id: CRMERP-06
title: Rate Limits & Quotas (per vendor, backoff)
type: crmerp_rate_limits_quotas
template_version: 1.0.0
output_path: 10_app/crmerp/CRMERP-06_Rate_Limits_Quotas.md
compliance_gate_id: TMP-05.PRIMARY.CRMERP
upstream_dependencies: ["CRMERP-01", "IXS-03", "RLIM-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CRMERP-01", "IXS-03", "RLIM-01", "RLIM-02", "CER-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical vendor rate limit and quota rules for CRM/ERP integrations, including documented vendor limits, internal concurrency caps, backoff behavior, and how sync scheduling adapts under throttling. This template must be consistent with integration connectivity policy and internal rate limit governance.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CRMERP-01 System Inventory: {{crmerp.systems}}
IXS-03 Connectivity & Network Policy: {{ixs.network_policy}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}} | OPTIONAL
RLIM-02 Rate Limit Catalog: {{rlim.catalog}} | OPTIONAL
CER-02 Retry/Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
system_id binding
Vendor limit registry (limit_id list)
Vendor documented limits (requests/minute, daily caps)
Quota reset behavior (window type)
Retry-after handling (429 behavior)
Backoff policy (exponential/jitter)
Internal concurrency caps (per system/object)
Scheduling adaptation rules (slowdown, pause)
Exemptions/allowlist policy (if any)
Telemetry requirements (throttles, quota usage)

Optional Fields
Cost/usage budgeting notes | OPTIONAL
Burst allowance rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not exceed vendor limits; must degrade safely under throttling.
Backoff must be bounded; do not spin.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Vendor Limits
   system_id: {{meta.system_id}}
   limits:
   Limit
   limit_id: {{limits[0].limit_id}}
   scope: {{limits[0].scope}} (global/per_object/UNKNOWN)
   value: {{limits[0].value}}
   unit: {{limits[0].unit}}
   window: {{limits[0].window}} (per_minute/per_day/UNKNOWN)
   reset_rule: {{limits[0].reset_rule}} | OPTIONAL
   notes: {{limits[0].notes}} | OPTIONAL

(Repeat per limit.)

2. 429 / Retry-After Handling
   retry_after_respected: {{r429.respected}}
   header_name: {{r429.header_name}} | OPTIONAL
   fallback_delay_ms: {{r429.fallback_delay_ms}} | OPTIONAL
3. Backoff Policy
   policy: {{backoff.policy}} (exponential/jitter/UNKNOWN)
   base_delay_ms: {{backoff.base_delay_ms}} | OPTIONAL
   max_delay_ms: {{backoff.max_delay_ms}}
   max_attempts: {{backoff.max_attempts}} | OPTIONAL
4. Internal Caps
   max_concurrent_requests: {{caps.max_concurrent_requests}}
   per_object_caps: {{caps.per_object_caps}} | OPTIONAL
5. Scheduling Adaptation
   on_throttle_behavior: {{adapt.on_throttle_behavior}} (slowdown/pause/UNKNOWN)
   cooldown_rule: {{adapt.cooldown_rule}} | OPTIONAL
   resume_rule: {{adapt.resume_rule}} | OPTIONAL

6. Exemptions
   exemptions_supported: {{exempt.supported}} | OPTIONAL
   allowlist_policy: {{exempt.allowlist_policy}} | OPTIONAL
7. Telemetry
   throttle_metric: {{telemetry.throttle_metric}}
   quota_usage_metric: {{telemetry.quota_usage_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
8. References
   System inventory: {{xref:CRMERP-01}}
   Connectivity policy: {{xref:IXS-03}} | OPTIONAL
   Rate limit policy: {{xref:RLIM-01}} | OPTIONAL
   Rate limit catalog: {{xref:RLIM-02}} | OPTIONAL
   Retry patterns: {{xref:CER-02}} | OPTIONAL

Cross-References
Upstream: {{xref:CRMERP-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CRMERP-04}}, {{xref:CRMERP-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. List vendor limits and basic 429 handling; use UNKNOWN for caps if missing.
intermediate: Required. Define backoff and internal caps and adaptation behavior.
advanced: Required. Add exemptions policy and budgeting/burst rules with telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, reset rules/notes, 429 header/fallback, base delay/max attempts, per-object caps, cooldown/resume, exemptions, quota usage metric/fields, budgeting/burst, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If limits list is UNKNOWN → block Completeness Gate.
If backoff.max_delay_ms is UNKNOWN → block Completeness Gate.
If adapt.on_throttle_behavior is UNKNOWN → block Completeness Gate.
If telemetry.throttle_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CRMERP
Pass conditions:
required_fields_present == true
vendor_limits_defined == true
429_and_backoff_defined == true
internal_caps_defined == true
adaptation_defined == true
telemetry_defined == true

placeholder_resolution == true
no_unapproved_unknowns == true

CRMERP-07

CRMERP-07 — Data Quality & Validation (required fields, dedupe)
Header Block
template_id: CRMERP-07
title: Data Quality & Validation (required fields, dedupe)
type: crmerp_data_quality_validation
template_version: 1.0.0
output_path: 10_app/crmerp/CRMERP-07_Data_Quality_Validation.md
compliance_gate_id: TMP-05.PRIMARY.CRMERP
upstream_dependencies: ["CRMERP-02", "IXS-05", "DQV-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CRMERP-01", "CRMERP-02", "IXS-05", "DQV-03", "API-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical data quality and validation rules for CRM/ERP sync: required fields, schema validation, normalization, deduplication, and how invalid records are handled (reject/quarantine/repair). This template must be consistent with mapping rules and canonical validation standards and must not invent data fields beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CRMERP-01 System Inventory: {{crmerp.systems}}
CRMERP-02 Object/Entity Mapping: {{crmerp.mapping}}
IXS-05 Data Mapping & Transforms: {{ixs.data_mapping}} | OPTIONAL
DQV-03 Data Validation Rules: {{dqv.rules}} | OPTIONAL
API-03 Error/Status Code Policy: {{api.error_policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
system_id binding
Validation stages (ingest, transform, pre-write, post-write)
Per-object required field rules (external_object → required fields)
Schema validation rules (canonical schema binding)
Normalization rules (strings, dates, units)
Deduplication keys and window
Invalid record handling (reject/quarantine/repair)
Data repair workflow (operator actions)
Telemetry requirements (invalid rate, dedupe hits)
Audit requirements for repairs

Optional Fields
Sampling/quality reporting cadence | OPTIONAL
PII redaction during validation | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Required fields must be explicit and consistent with canonical schemas.
Deduplication rules must be deterministic; avoid fuzzy matching unless defined.
Invalid records must not silently pass into system-of-record.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Validation Stages
   system_id: {{meta.system_id}}
   stages: {{stages.list}}
2. Required Fields (Per Object)
   Rule
   external_object: {{required[0].external_object}}
   required_fields: {{required[0].required_fields}}
   missing_field_behavior: {{required[0].missing_field_behavior}}
   notes: {{required[0].notes}} | OPTIONAL

(Repeat per object.)

3. Schema Validation
   schema_binding_ref: {{schema.ref}} (expected: {{xref:DATA-06}}/{{xref:DQV-03}}) | OPTIONAL
   validation_method: {{schema.method}} (strict/lenient/UNKNOWN)
   on_validation_fail: {{schema.on_fail}}
4. Normalization
   string_rules: {{norm.string_rules}} | OPTIONAL
   date_time_rules: {{norm.date_time_rules}} | OPTIONAL
   unit_rules: {{norm.unit_rules}} | OPTIONAL
5. Deduplication
   dedupe_keys: {{dedupe.keys}}
   dedupe_window: {{dedupe.window}} | OPTIONAL
   dedupe_behavior: {{dedupe.behavior}} (drop/merge/quarantine/UNKNOWN)
6. Invalid Record Handling
   invalid_behavior: {{invalid.behavior}} (reject/quarantine/repair/UNKNOWN)
   quarantine_location: {{invalid.quarantine_location}} | OPTIONAL
   repair_allowed: {{invalid.repair_allowed}} | OPTIONAL

7. Repair Workflow
   operator_actions: {{repair.operator_actions}}
   audit_required: {{repair.audit_required}}
8. Telemetry
   invalid_record_metric: {{telemetry.invalid_record_metric}}
   dedupe_hit_metric: {{telemetry.dedupe_hit_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
9. Audit
   audit_fields: {{audit.fields}} | OPTIONAL
   retention_policy: {{audit.retention_policy}} | OPTIONAL
10. References
    System inventory: {{xref:CRMERP-01}}
    Object/entity mapping: {{xref:CRMERP-02}}
    Data mapping: {{xref:IXS-05}} | OPTIONAL
    Validation rules: {{xref:DQV-03}} | OPTIONAL
    Reconciliation/backfills: {{xref:CRMERP-08}} | OPTIONAL

Cross-References
Upstream: {{xref:CRMERP-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CRMERP-08}}, {{xref:CRMERP-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define required fields and invalid behavior and a dedupe key list.
intermediate: Required. Define validation stages, schema method, and repair workflow/audit.
advanced: Required. Add normalization rules, reporting cadence, and PII-safe validation handling.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, schema ref, normalization rules, dedupe window, quarantine location, repair allowed, telemetry fields, audit fields/retention, cadence/pii handling, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If stages.list is UNKNOWN → block Completeness Gate.
If required list is UNKNOWN → block Completeness Gate.
If schema.on_fail is UNKNOWN → block Completeness Gate.
If invalid.behavior is UNKNOWN → block Completeness Gate.
If telemetry.invalid_record_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CRMERP
Pass conditions:
required_fields_present == true
validation_stages_defined == true

required_fields_per_object_defined == true
dedupe_defined == true
invalid_handling_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CRMERP-08

CRMERP-08 — Error Handling & Reconciliation (replay, backfill)
Header Block
template_id: CRMERP-08
title: Error Handling & Reconciliation (replay, backfill)
type: crmerp_error_handling_reconciliation
template_version: 1.0.0
output_path: 10_app/crmerp/CRMERP-08_Error_Handling_Reconciliation.md
compliance_gate_id: TMP-05.PRIMARY.CRMERP
upstream_dependencies: ["CRMERP-06", "CRMERP-07", "IXS-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CRMERP-01", "CRMERP-04", "CRMERP-06", "CRMERP-07", "IXS-06", "JBS-04", "EVT-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical reconciliation and recovery procedures for CRM/ERP sync: how failed sync attempts are retried, how DLQ/quarantine is handled, how replay works, and how backfills are executed safely with auditability. This template must be consistent with integration error handling and job/event failure policies.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CRMERP-01 System Inventory: {{crmerp.systems}}
CRMERP-04 Scheduling/Triggers: {{crmerp.scheduling}} | OPTIONAL
CRMERP-06 Rate Limits/Quotas: {{crmerp.limits}} | OPTIONAL
CRMERP-07 Data Quality/Validation: {{crmerp.data_quality}} | OPTIONAL
IXS-06 Integration Error Handling: {{ixs.error_recovery}}
JBS-04 Retry/DLQ Policy: {{jobs.retry_dlq}} | OPTIONAL
EVT-07 Event Failure Handling: {{evt.failure_handling}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
system_id binding
Failure sources list (vendor errors, validation, throttling)
Retry policy binding (max attempts/backoff)
Quarantine/DLQ handling rules (what goes there)
Replay policy (who can replay, how scoped)
Backfill policy (who can backfill, safety limits)
Reconciliation strategy (diff checks, sampling)
Operator runbook steps (triage → fix → replay)
Audit logging requirements (replay/backfill actions)
Telemetry requirements (replay success, backlog size)

Optional Fields
Automated backfill triggers | OPTIONAL
Data repair workflow refs | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Replays/backfills MUST be idempotent or guarded with idempotency keys.
Backfills MUST have safety limits (time range, max records) and must be auditable.
DLQ/quarantine MUST have a drain policy; no permanent limbo.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Failure Sources
   system_id: {{meta.system_id}}
   sources: {{fail.sources}}
2. Retry Binding
   retry_policy_ref: {{retry.ref}} (expected: {{xref:IXS-06}}/{{xref:JBS-04}}) | OPTIONAL
   max_attempts: {{retry.max_attempts}} | OPTIONAL
   backoff_policy: {{retry.backoff_policy}} | OPTIONAL
3. Quarantine / DLQ
   dlq_supported: {{dlq.supported}}
   dlq_trigger_rule: {{dlq.trigger_rule}}
   dlq_location: {{dlq.location}} | OPTIONAL
   drain_policy: {{dlq.drain_policy}}
4. Replay Policy
   replay_supported: {{replay.supported}}
   who_can_replay: {{replay.who_can_replay}}
   scope_rules: {{replay.scope_rules}}
   safety_checks: {{replay.safety_checks}} | OPTIONAL
5. Backfill Policy
   backfill_supported: {{backfill.supported}}
   who_can_backfill: {{backfill.who_can_backfill}}
   max_range_rule: {{backfill.max_range_rule}}
   max_records_rule: {{backfill.max_records_rule}} | OPTIONAL
   approval_required: {{backfill.approval_required}} | OPTIONAL
6. Reconciliation Strategy
   diff_method: {{recon.diff_method}} (spot_check/full_diff/UNKNOWN)
   sampling_rules: {{recon.sampling_rules}} | OPTIONAL
   success_criteria: {{recon.success_criteria}}
7. Operator Runbook
   runbook_location: {{ops.runbook_location}}
   steps:

{{ops.steps[0]}}
{{ops.steps[1]}} | OPTIONAL

8. Audit Logging
   audit_required: {{audit.required}}
   audit_fields: {{audit.fields}} | OPTIONAL
9. Telemetry
   backlog_depth_metric: {{telemetry.backlog_depth_metric}}
   replay_success_metric: {{telemetry.replay_success_metric}} | OPTIONAL
   backfill_success_metric: {{telemetry.backfill_success_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
10. References
    System inventory: {{xref:CRMERP-01}}
    Scheduling/triggers: {{xref:CRMERP-04}} | OPTIONAL
    Rate limits: {{xref:CRMERP-06}} | OPTIONAL
    Data quality: {{xref:CRMERP-07}} | OPTIONAL
    Integration error handling: {{xref:IXS-06}}
    Observability/runbooks: {{xref:CRMERP-10}} | OPTIONAL

Cross-References
Upstream: {{xref:CRMERP-06}}, {{xref:IXS-06}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CRMERP-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define DLQ + replay/backfill support flags and safety rules.
intermediate: Required. Define scope rules, reconciliation success criteria, and audit requirements.
advanced: Required. Add approvals/limits rigor and automated backfill triggers with telemetry fields.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, retry refs/details, dlq location, replay safety checks, max records/approvals, sampling rules, extra runbook steps, audit fields, telemetry fields, automated triggers/repair refs, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If dlq.trigger_rule is UNKNOWN → block Completeness Gate (when dlq.supported == true).
If dlq.drain_policy is UNKNOWN → block Completeness Gate (when dlq.supported == true).
If backfill.max_range_rule is UNKNOWN → block Completeness Gate (when backfill.supported == true).
If recon.success_criteria is UNKNOWN → block Completeness Gate.
If telemetry.backlog_depth_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CRMERP

Pass conditions:
required_fields_present == true
dlq_and_drain_defined == true
replay_and_backfill_policies_defined == true
reconciliation_defined == true
audit_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CRMERP-09

CRMERP-09 — Security & Compliance (PII, least privilege, audit)
Header Block
template_id: CRMERP-09
title: Security & Compliance (PII, least privilege, audit)
type: crmerp_security_compliance
template_version: 1.0.0
output_path: 10_app/crmerp/CRMERP-09_Security_Compliance.md
compliance_gate_id: TMP-05.PRIMARY.CRMERP
upstream_dependencies: ["CRMERP-01", "IXS-08", "IXS-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CRMERP-01", "IXS-08", "IXS-04", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical security and compliance requirements specific to CRM/ERP integrations: least privilege access, PII/financial data handling, audit requirements, retention/deletion expectations, and vendor risk controls. This template must be consistent with the general integration security baseline and must not invent compliance controls not supported by upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CRMERP-01 System Inventory: {{crmerp.systems}}
IXS-08 Integration Security & Compliance: {{ixs.security_compliance}} | OPTIONAL
IXS-04 Secrets/Credential Handling: {{ixs.secrets_policy}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
system_id binding
Data classes handled (PII/financial/etc.)
Least privilege policy (scopes/roles)
Credential handling reference (IXS-04)
Encryption requirements (in transit/at rest)
Audit requirements (sync writes/reads, admin changes)
Retention rules (external system storage)
Deletion propagation (if applicable)
Vendor risk controls (SLA, breach notification)
Compliance flags (PCI/PHI/etc. if applicable)

Optional Fields
Data residency constraints | OPTIONAL
DPA/contract references | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
CRM/ERP credentials must be scoped to minimum permissions needed for defined sync direction.
Audit logs must include system_id/object/key for all writes.
Retention and deletion must be explicit for regulated data.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Data Classes
   system_id: {{meta.system_id}}
   data_classes: {{data.classes}}
   pii_fields_summary: {{data.pii_fields_summary}} | OPTIONAL
2. Least Privilege
   access_scopes: {{access.scopes}}
   role_model: {{access.role_model}} | OPTIONAL
   blocked_permissions: {{access.blocked_permissions}} | OPTIONAL
3. Credential Handling
   credential_policy_ref: {{creds.ref}} (expected: {{xref:IXS-04}}) | OPTIONAL
   rotation_required: {{creds.rotation_required}} | OPTIONAL
4. Encryption
   in_transit: {{crypto.in_transit}}
   at_rest: {{crypto.at_rest}} | OPTIONAL
5. Audit
   audit_required: {{audit.required}}
   audit_fields: {{audit.fields}}
   retention_policy: {{audit.retention_policy}} | OPTIONAL
6. Retention / Deletion
   retention_policy: {{retention.policy}}
   delete_propagation_rule: {{retention.delete_propagation_rule}} | OPTIONAL
7. Vendor Risk
   sla_required: {{vendor.sla_required}}
   breach_notification_rule: {{vendor.breach_notification_rule}}
   security_review_required: {{vendor.security_review_required}} | OPTIONAL
8. Compliance Flags
   pci_in_scope: {{compliance.pci_in_scope}} | OPTIONAL
   phi_in_scope: {{compliance.phi_in_scope}} | OPTIONAL
   other_constraints: {{compliance.other_constraints}} | OPTIONAL

9. References
   General integration security: {{xref:IXS-08}} | OPTIONAL
   Audit trail baseline: {{xref:ADMIN-03}} | OPTIONAL
   Error handling/reconciliation: {{xref:CRMERP-08}} | OPTIONAL

Cross-References
Upstream: {{xref:CRMERP-01}}, {{xref:IXS-08}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CRMERP-10}} | OPTIONAL
Standards: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define data classes + encryption in transit + audit required fields.
intermediate: Required. Define least privilege scopes and retention/deletion rules.
advanced: Required. Add data residency/contract references and vendor risk governance rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, pii summary, role model, blocked permissions, credential ref/rotation, at rest encryption, audit retention, delete propagation, security review, compliance flags, residency/contract refs, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If data.classes is UNKNOWN → block Completeness Gate.
If access.scopes is UNKNOWN → block Completeness Gate.
If audit.fields is UNKNOWN → block Completeness Gate.
If vendor.breach_notification_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CRMERP
Pass conditions:
required_fields_present == true
data_classes_defined == true
least_privilege_defined == true
audit_defined == true
retention_defined == true
vendor_risk_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CRMERP-10

CRMERP-10 — Observability & Runbooks (dashboards, alerts, operator steps)
Header Block
template_id: CRMERP-10
title: Observability & Runbooks (dashboards, alerts, operator steps)
type: crmerp_observability_runbooks
template_version: 1.0.0
output_path: 10_app/crmerp/CRMERP-10_Observability_Runbooks.md
compliance_gate_id: TMP-05.PRIMARY.CRMERP
upstream_dependencies: ["CRMERP-04", "CRMERP-06", "CRMERP-08", "IXS-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CRMERP-01", "CRMERP-04", "CRMERP-06", "CRMERP-08", "IXS-07", "ADMIN-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical observability and operator runbooks for CRM/ERP sync: required dashboards, alerts, triage steps, and operational actions (pause sync, replay, backfill, credential rotation). This template must be consistent with integration observability and error/reconciliation policies.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CRMERP-01 System Inventory: {{crmerp.systems}}
CRMERP-04 Scheduling/Triggers: {{crmerp.scheduling}} | OPTIONAL
CRMERP-06 Rate Limits/Quotas: {{crmerp.limits}} | OPTIONAL
CRMERP-08 Reconciliation/Backfill: {{crmerp.recon}} | OPTIONAL
IXS-07 Integration Observability: {{ixs.observability}} | OPTIONAL
ADMIN-02 Support Tools Spec: {{admin.support_tools}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
system_id binding
Dashboard requirements (minimum panels)
Key metrics list (success/fail, latency, throttles, backlog)
Alert definitions (failure spike, throttle spike, backlog stuck)
Alert routing policy (oncall/team)
Runbook location and structure
Triage flow (step-by-step)
Operator actions (pause/resume, replay, backfill, rotate creds)
User/customer impact policy (if incidents)
Post-incident review requirements (RCA notes)

Optional Fields
Per-object dashboards | OPTIONAL
SLO definitions | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Alerts must be actionable and map to runbook steps.
Operator actions must require proper permissions and be auditable.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Dashboards
   system_id: {{meta.system_id}}
   dashboard_name: {{dash.name}}
   minimum_panels: {{dash.minimum_panels}}
   panel_list: {{dash.panel_list}} | OPTIONAL
2. Key Metrics
   metrics: {{metrics.list}}
3. Alerts
   Alert
   alert_id: {{alerts[0].alert_id}}
   condition: {{alerts[0].condition}}
   severity: {{alerts[0].severity}} (low/med/high/UNKNOWN)
   routing: {{alerts[0].routing}}
   runbook_ref: {{alerts[0].runbook_ref}} | OPTIONAL

(Repeat per alert.)

4. Runbook Location
   runbook_location: {{runbook.location}}
   format: {{runbook.format}} (md/wiki/UNKNOWN) | OPTIONAL
5. Triage Flow
   steps:
   {{triage.steps[0]}}
   {{triage.steps[1]}}
   {{triage.steps[2]}} | OPTIONAL
6. Operator Actions
   actions:
   {{ops.actions[0]}}
   {{ops.actions[1]}} | OPTIONAL
   permissions_ref: {{ops.permissions_ref}} (expected: {{xref:API-04}}/{{xref:ADMIN-01}}) | OPTIONAL

7. Impact & Comms
   user_impact_policy: {{impact.user_impact_policy}}
   comms_owner: {{impact.comms_owner}} | OPTIONAL
8. Post-Incident
   rca_required: {{post.rca_required}}
   rca_location: {{post.rca_location}} | OPTIONAL
9. References
   Integration observability baseline: {{xref:IXS-07}} | OPTIONAL
   Scheduling: {{xref:CRMERP-04}} | OPTIONAL
   Rate limits: {{xref:CRMERP-06}} | OPTIONAL
   Reconciliation/backfill: {{xref:CRMERP-08}} | OPTIONAL
   Support tools: {{xref:ADMIN-02}} | OPTIONAL

Cross-References
Upstream: {{xref:CRMERP-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-02}}, {{xref:ADMIN-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define dashboards + key metrics + triage steps; use UNKNOWN for SLOs.
intermediate: Required. Define alert routing and operator actions.
advanced: Required. Add per-object dashboards, SLOs, and RCA rigor and comms ownership.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, panel list, optional triage steps, runbook format, permissions ref, comms owner, rca location, per-object dashboards, SLOs, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If dash.minimum_panels is UNKNOWN → block Completeness Gate.
If metrics.list is UNKNOWN → block Completeness Gate.
If triage.steps[0] is UNKNOWN → block Completeness Gate.
If alerts list is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CRMERP
Pass conditions:
required_fields_present == true
dashboards_defined == true
alerts_defined == true
triage_and_operator_actions_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Webhooks Consumers & Providers (WHCP)

Webhooks Consumers & Providers (WHCP)
WHCP-01 Webhook Event Catalog (by webhook_id/event_id)
WHCP-02 Outbound Webhook Producer Spec (signing, retries, dedupe)
WHCP-03 Inbound Webhook Consumer Spec (verification, idempotency)
WHCP-04 Delivery Semantics (ordering, replay, backoff)
WHCP-05 Security Rules (signatures, secrets rotation, allowlists)
WHCP-06 Endpoint Registration & Management (subscriptions, secrets)
WHCP-07 Error Handling (DLQ, quarantine, manual replay)
WHCP-08 Observability (delivery rate, latency, failures)
WHCP-09 Payload Versioning & Compatibility (schema evolution)
WHCP-10 Testing & Sandbox Strategy (test hooks, simulators)

WHCP-01

WHCP-01 — Webhook Event Catalog (by webhook_id/event_id)

Header Block

template_id: WHCP-01
title: Webhook Event Catalog (by webhook_id/event_id)
type: webhook_event_catalog
template_version: 1.0.0
output_path: 10_app/webhooks/WHCP-01_Webhook_Event_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.WHCP
upstream_dependencies: ["EVT-01", "IXS-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "EVT-01", "EVT-02", "IXS-01", "ROUTE-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Create the canonical catalog of webhook events (inbound and outbound), indexed by webhook_id and/or event_id, including producers/consumers, payload schema references, security requirements, and delivery semantics pointers. This template must not invent webhook IDs or event IDs beyond upstream inputs.

Inputs Required

SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
EVT-01 Event Catalog: {{evt.catalog}} | OPTIONAL
EVT-02 Event Schema Spec: {{evt.schemas}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}} | OPTIONAL
ROUTE-03 Deep Link Map (if webhooks trigger links): {{route.deep_link_map}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields

Webhook registry (webhook_id list)
webhook_id (stable identifier)
direction (inbound/outbound)
event_id binding (if applicable)
name/purpose
producer (system/service)
consumer (system/service)
payload schema ref (EVT-02 / WHCP-09)
delivery semantics ref (WHCP-04)
security policy ref (WHCP-05)
retry/dedupe policy ref (WHCP-04/WHCP-02/03)
environments enabled (dev/stage/prod)

Optional Fields
Subscription/endpoint registration ref (WHCP-06) | OPTIONAL
PII flags | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent webhook_id/event_id; use only upstream catalogs when present.
Every webhook entry MUST reference schema + security + delivery semantics (or approved UNKNOWN).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Catalog Summary
   total_webhooks: {{catalog.total}}
   inbound_count: {{catalog.inbound_count}} | OPTIONAL
   outbound_count: {{catalog.outbound_count}} | OPTIONAL

2. Webhook Entries (by webhook_id)
   Webhook
   webhook_id: {{items[0].webhook_id}}
   direction: {{items[0].direction}}
   event_id: {{items[0].event_id}} | OPTIONAL
   name: {{items[0].name}}
   purpose: {{items[0].purpose}}
   producer: {{items[0].producer}}
   consumer: {{items[0].consumer}}
   schema_ref: {{items[0].schema_ref}}
   delivery_semantics_ref: {{items[0].delivery_semantics_ref}} (expected: {{xref:WHCP-04}})
   security_ref: {{items[0].security_ref}} (expected: {{xref:WHCP-05}})
   retry_dedupe_ref: {{items[0].retry_dedupe_ref}} (expected: {{xref:WHCP-02}}/{{xref:WHCP-03}}/{{xref:WHCP-04}})
   env_enabled: {{items[0].env_enabled}}
   registration_ref: {{items[0].registration_ref}} | OPTIONAL
   pii_flags: {{items[0].pii_flags}} | OPTIONAL
   open_questions:
   {{items[0].open_questions[0]}} | OPTIONAL

(Repeat per webhook_id.)

3. References
   Outbound producer spec: {{xref:WHCP-02}} | OPTIONAL
   Inbound consumer spec: {{xref:WHCP-03}} | OPTIONAL
   Delivery semantics: {{xref:WHCP-04}} | OPTIONAL
   Security rules: {{xref:WHCP-05}} | OPTIONAL
   Endpoint registration: {{xref:WHCP-06}} | OPTIONAL
   Failure handling: {{xref:WHCP-07}} | OPTIONAL
   Observability: {{xref:WHCP-08}} | OPTIONAL
   Versioning: {{xref:WHCP-09}} | OPTIONAL
   Testing/sandbox: {{xref:WHCP-10}} | OPTIONAL

Cross-References
Upstream: {{xref:EVT-01}} | OPTIONAL, {{xref:IXS-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:WHCP-02}}, {{xref:WHCP-03}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Populate webhook registry and core fields; do not invent IDs.
intermediate: Required. Add schema/security/delivery refs and retry/dedupe linkage.
advanced: Required. Add env enablement rigor, registration refs, and PII flags/traceability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, event_id, registration ref, pii flags, open_questions
If any Required Field is UNKNOWN, allow only if: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If items[*].webhook_id is UNKNOWN → block Completeness Gate.*
*If items[*].schema_ref is UNKNOWN → block Completeness Gate.
If items[*].security_ref is UNKNOWN → block Completeness Gate.*
*If items[*].delivery_semantics_ref is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.WHCP
Pass conditions:
required_fields_present == true
webhook_ids_unique == true
schema_security_delivery_refs_present == true
placeholder_resolution == true
no_unapproved_unknowns == true

WHCP-02

WHCP-02 — Outbound Webhook Producer Spec (signing, retries, dedupe)
Header Block
template_id: WHCP-02
title: Outbound Webhook Producer Spec (signing, retries, dedupe)
type: webhook_outbound_producer_spec
template_version: 1.0.0
output_path: 10_app/webhooks/WHCP-02_Outbound_Webhook_Producer_Spec.md
compliance_gate_id: TMP-05.PRIMARY.WHCP
upstream_dependencies: ["WHCP-01", "EVT-02", "IXS-03", "IXS-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "WHCP-01", "EVT-02", "WHCP-04", "WHCP-05", "WHCP-06", "IXS-03", "IXS-04", "RLIM-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical outbound webhook producer behavior: how events are selected and serialized, signature/signing rules, endpoint targeting/subscriptions, retries/backoff, deduplication/idempotency, and failure handling. This template must be consistent with webhook catalog, delivery semantics, and security rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
WHCP-01 Webhook Event Catalog: {{whcp.catalog}}
EVT-02 Event Schema Spec: {{evt.schema_spec}} | OPTIONAL
WHCP-04 Delivery Semantics: {{whcp.delivery_semantics}} | OPTIONAL
WHCP-05 Security Rules: {{whcp.security_rules}} | OPTIONAL
WHCP-06 Registration/Management: {{whcp.registration}} | OPTIONAL
IXS-03 Connectivity Policy: {{ixs.network_policy}} | OPTIONAL
IXS-04 Secrets/Credentials Policy: {{ixs.secrets_policy}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Producer identity (service/component)
Outbound webhook selection rules (which webhook_ids emitted)
Payload schema versioning rule
Target resolution (subscription lookup)
HTTP method/headers policy
Signing supported (yes/no/UNKNOWN)
Signing scheme (HMAC/JWS/UNKNOWN)
Signature header fields
Retries supported + backoff rules

Max attempts and dead-letter/quarantine behavior
Deduplication/idempotency strategy (event_id + delivery_id)
Rate caps (per subscriber)
Telemetry requirements (delivered/failed, latency)

Optional Fields
Batching support | OPTIONAL
Compression support | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Do not send PII unless explicitly allowed by schema/security policies.
Retries must be bounded; dedupe must prevent duplicate side effects.
Signing secrets must be handled per {{xref:IXS-04}}; never log them.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Producer Identity
   producer_name: {{producer.name}}
   service_owner: {{producer.owner}} | OPTIONAL
2. Emitted Webhooks
   webhook_ids_emitted: {{emit.webhook_ids}}
   selection_rules: {{emit.selection_rules}} | OPTIONAL
3. Payload Versioning
   schema_ref: {{payload.schema_ref}} (expected: {{xref:EVT-02}}/{{xref:WHCP-09}}) | OPTIONAL
   version_field: {{payload.version_field}} | OPTIONAL
4. Target Resolution
   subscription_lookup_rule: {{targets.lookup_rule}}
   endpoint_fields: {{targets.endpoint_fields}} | OPTIONAL
5. HTTP Policy
   method: {{http.method}}
   required_headers: {{http.required_headers}}
   timeout_ref: {{http.timeout_ref}} (expected: {{xref:IXS-03}}) | OPTIONAL
6. Signing
   signing_supported: {{sign.supported}}
   scheme: {{sign.scheme}} | OPTIONAL
   signature_headers: {{sign.signature_headers}} | OPTIONAL
   canonical_string_rule: {{sign.canonical_string_rule}} | OPTIONAL
   secret_ref: {{sign.secret_ref}} (expected: {{xref:IXS-04}}) | OPTIONAL
7. Retries / Backoff
   retry_supported: {{retry.supported}}

backoff_policy: {{retry.backoff_policy}}
max_attempts: {{retry.max_attempts}}
retryable_status_codes: {{retry.retryable_status_codes}} | OPTIONAL

8. Dedupe / Idempotency
   delivery_id_rule: {{dedupe.delivery_id_rule}}
   dedupe_key_rule: {{dedupe.key_rule}}
   dedupe_window: {{dedupe.window}} | OPTIONAL
9. DLQ / Quarantine
   dlq_supported: {{dlq.supported}}
   dlq_trigger_rule: {{dlq.trigger_rule}}
   replay_policy_ref: {{dlq.replay_policy_ref}} (expected: {{xref:WHCP-07}}) | OPTIONAL
10. Rate Caps
    per_subscriber_caps: {{caps.per_subscriber_caps}}
    enforcement_ref: {{caps.enforcement_ref}} (expected: {{xref:RLIM-01}}) | OPTIONAL
11. Telemetry
    delivery_success_metric: {{telemetry.success_metric}}
    delivery_failure_metric: {{telemetry.failure_metric}} | OPTIONAL
    latency_metric: {{telemetry.latency_metric}} | OPTIONAL
    fields: {{telemetry.fields}} | OPTIONAL
12. References
    Webhook catalog: {{xref:WHCP-01}}
    Delivery semantics: {{xref:WHCP-04}} | OPTIONAL
    Security rules: {{xref:WHCP-05}} | OPTIONAL
    Registration/management: {{xref:WHCP-06}} | OPTIONAL
    Failure handling: {{xref:WHCP-07}} | OPTIONAL
    Observability: {{xref:WHCP-08}} | OPTIONAL

Cross-References
Upstream: {{xref:WHCP-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:WHCP-07}}, {{xref:WHCP-08}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define emitted webhook_ids + retries + max attempts; use UNKNOWN for signing scheme if not used.
intermediate: Required. Define signing fields (if supported), dedupe keys, and DLQ rules.
advanced: Required. Add batching/compression, subscriber caps, and telemetry fields rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, owner, selection rules, schema ref/version field, endpoint fields, timeout ref, signing scheme/headers/canonical rule/secret ref, retryable status codes, dedupe window, replay policy ref, enforcement ref, telemetry fields, batching/compression, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

If emit.webhook_ids is UNKNOWN → block Completeness Gate.
If retry.max_attempts is UNKNOWN → block Completeness Gate.
If dedupe.key_rule is UNKNOWN → block Completeness Gate.
If sign.supported is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.WHCP
Pass conditions:
required_fields_present == true
emitted_webhooks_defined == true
bounded_retry_defined == true
dedupe_defined == true
dlq_policy_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

WHCP-03

WHCP-03 — Inbound Webhook Consumer Spec (verification, idempotency)
Header Block
template_id: WHCP-03
title: Inbound Webhook Consumer Spec (verification, idempotency)
type: webhook_inbound_consumer_spec
template_version: 1.0.0
output_path: 10_app/webhooks/WHCP-03_Inbound_Webhook_Consumer_Spec.md
compliance_gate_id: TMP-05.PRIMARY.WHCP
upstream_dependencies: ["WHCP-01", "WHCP-05", "API-02", "IXS-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "WHCP-01", "WHCP-04", "WHCP-05", "WHCP-06", "API-01", "API-02", "IXS-04", "IXS-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical inbound webhook consumer behavior: endpoint contract, signature verification, allowlists, idempotency/deduplication, parsing/validation, acknowledgement rules, and failure handling (DLQ/quarantine/replay). This template must be consistent with webhook catalog and security rules and must not invent endpoints or secrets beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
WHCP-01 Webhook Event Catalog: {{whcp.catalog}}
WHCP-04 Delivery Semantics: {{whcp.delivery_semantics}} | OPTIONAL
WHCP-05 Security Rules: {{whcp.security_rules}}
WHCP-06 Registration/Management: {{whcp.registration}} | OPTIONAL
API-01 Endpoint Catalog: {{api.endpoint_catalog}} | OPTIONAL
API-02 Endpoint Specs: {{api.endpoint_specs}} | OPTIONAL
IXS-04 Secrets/Credentials Policy: {{ixs.secrets_policy}} | OPTIONAL
IXS-06 Error Handling & Recovery: {{ixs.error_recovery}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Inbound endpoint registry (endpoint_id or path list)
Which webhook_ids accepted per endpoint
Verification required (yes/no/UNKNOWN)
Verification scheme (HMAC/JWS/mTLS/UNKNOWN)
Secret/cert reference (IXS-04)
Allowlist rules (IP allowlist if supported)
Payload parsing rules (schema validation)
Idempotency key rules (event_id/delivery_id)
Ack rules (2xx on accept vs after processing)

Backpressure rules (queueing)
Failure handling (DLQ/quarantine)
Telemetry requirements (accepted/rejected/invalid sig)

Optional Fields
Async processing model (job queue) | OPTIONAL
Replay endpoint support | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Never process unverified webhooks when verification is required.
Idempotency MUST be enforced; inbound duplicates must not cause repeated side effects.
Ack behavior must be explicit and consistent with delivery semantics.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Inbound Endpoints
   Endpoint
   endpoint_id: {{endpoints[0].endpoint_id}} | OPTIONAL
   path: {{endpoints[0].path}} | OPTIONAL
   accepted_webhook_ids: {{endpoints[0].accepted_webhook_ids}}
   notes: {{endpoints[0].notes}} | OPTIONAL

(Repeat per endpoint.)

2. Verification
   verification_required: {{verify.required}}
   scheme: {{verify.scheme}}
   secret_or_cert_ref: {{verify.secret_or_cert_ref}} (expected: {{xref:IXS-04}}) | OPTIONAL
   signature_header_rule: {{verify.signature_header_rule}} | OPTIONAL
3. Allowlists
   ip_allowlist_supported: {{allow.ip_allowlist_supported}} | OPTIONAL
   ip_allowlist: {{allow.ip_allowlist}} | OPTIONAL
4. Parsing / Validation
   schema_ref_rule: {{parse.schema_ref_rule}} (expected: {{xref:WHCP-09}}/{{xref:EVT-02}}) | OPTIONAL
   on_parse_fail: {{parse.on_parse_fail}}
   validation_error_response: {{parse.validation_error_response}} | OPTIONAL
5. Idempotency
   idempotency_required: {{idem.required}}
   idempotency_key_rule: {{idem.key_rule}}
   dedupe_window: {{idem.dedupe_window}} | OPTIONAL

6. Ack Rules
   ack_model: {{ack.model}} (ack_on_accept/ack_on_processed/UNKNOWN)
   response_codes: {{ack.response_codes}} | OPTIONAL
7. Backpressure / Queueing
   async_processing: {{queue.async_processing}} | OPTIONAL
   queue_ref: {{queue.queue_ref}} | OPTIONAL
   max_inflight: {{queue.max_inflight}} | OPTIONAL
8. Failure Handling
   dlq_supported: {{fail.dlq_supported}}
   dlq_trigger_rule: {{fail.dlq_trigger_rule}}
   replay_policy_ref: {{fail.replay_policy_ref}} (expected: {{xref:WHCP-07}}) | OPTIONAL
9. Telemetry
   accepted_metric: {{telemetry.accepted_metric}}
   rejected_metric: {{telemetry.rejected_metric}} | OPTIONAL
   invalid_signature_metric: {{telemetry.invalid_signature_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
10. References
    Webhook catalog: {{xref:WHCP-01}}
    Delivery semantics: {{xref:WHCP-04}} | OPTIONAL
    Security rules: {{xref:WHCP-05}}
    Endpoint registration: {{xref:WHCP-06}} | OPTIONAL
    Failure handling: {{xref:WHCP-07}} | OPTIONAL
    Observability: {{xref:WHCP-08}} | OPTIONAL

Cross-References
Upstream: {{xref:WHCP-01}}, {{xref:WHCP-05}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:WHCP-07}}, {{xref:WHCP-08}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define accepted webhook_ids + verification required + idempotency key rule.
intermediate: Required. Define ack model, parsing failure response, and DLQ trigger rule.
advanced: Required. Add async processing details, IP allowlists, and telemetry field rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, endpoint id/path, notes, secret ref, signature header, ip allowlist, schema ref rule, validation response, dedupe window, response codes, queue details, replay policy ref, telemetry fields, replay endpoint, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If verify.required is UNKNOWN → block Completeness Gate.
If verify.scheme is UNKNOWN → block Completeness Gate (when verify.required == true).
If idem.key_rule is UNKNOWN → block Completeness Gate.

If ack.model is UNKNOWN → block Completeness Gate.
If telemetry.accepted_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.WHCP
Pass conditions:
required_fields_present == true
endpoints_defined == true
verification_and_idempotency_defined == true
ack_and_failure_handling_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

WHCP-04

WHCP-04 — Delivery Semantics (ordering, replay, backoff)
Header Block
template_id: WHCP-04
title: Delivery Semantics (ordering, replay, backoff)
type: webhook_delivery_semantics
template_version: 1.0.0
output_path: 10_app/webhooks/WHCP-04_Delivery_Semantics.md
compliance_gate_id: TMP-05.PRIMARY.WHCP
upstream_dependencies: ["WHCP-02", "WHCP-03", "EVT-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "WHCP-01", "WHCP-02", "WHCP-03", "EVT-04", "CER-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical delivery semantics for webhooks: ordering guarantees, retry/backoff strategy, deduplication expectations, replay behavior, and acknowledgement semantics across producers and consumers. This template must be consistent with event delivery semantics and must not invent guarantees beyond what the system can enforce.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
WHCP-01 Webhook Catalog: {{whcp.catalog}}
WHCP-02 Outbound Producer Spec: {{whcp.outbound}} | OPTIONAL
WHCP-03 Inbound Consumer Spec: {{whcp.inbound}} | OPTIONAL
EVT-04 Delivery Semantics (events): {{evt.delivery_semantics}} | OPTIONAL
CER-02 Retry/Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Delivery model (at-least-once / at-most-once / UNKNOWN)
Ordering guarantees (none/per-endpoint/per-key/UNKNOWN)
Deduplication expectation (consumer responsibility vs producer)
Retry schedule (attempts, backoff)
Retry stop conditions (max age/max attempts)
Handling of 429/503 (retry-after)
Replay support (yes/no)
Replay scope rules (by time/event_id)
Poison message rules (quarantine)
Telemetry requirements (attempts, latency, success rate)

Optional Fields
Per-webhook overrides | OPTIONAL
Batch delivery semantics | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
If delivery is at-least-once, consumer idempotency MUST be required.
Ordering guarantees must be explicitly scoped; do not imply global ordering.
Retry must be bounded and must not cause storms.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Delivery Model
   delivery_model: {{delivery.model}} (at_least_once/at_most_once/UNKNOWN)
2. Ordering
   ordering: {{ordering.model}} (none/per_endpoint/per_key/UNKNOWN)
   ordering_key_rule: {{ordering.key_rule}} | OPTIONAL
3. Deduplication Expectation
   dedupe_owner: {{dedupe.owner}} (producer/consumer/UNKNOWN)
   idempotency_required: {{dedupe.idempotency_required}}
   dedupe_key_rule: {{dedupe.key_rule}} | OPTIONAL
4. Retry Schedule
   retry_supported: {{retry.supported}}
   backoff_policy: {{retry.backoff_policy}}
   max_attempts: {{retry.max_attempts}}
   max_age_ms: {{retry.max_age_ms}} | OPTIONAL
5. Stop Conditions
   stop_on: {{stop.conditions}}
   quarantine_on_poison: {{stop.quarantine_on_poison}} | OPTIONAL
6. 429/503 Handling
   retry_after_respected: {{r429.respected}}
   fallback_delay_ms: {{r429.fallback_delay_ms}} | OPTIONAL
7. Replay
   replay_supported: {{replay.supported}}
   replay_scope_rules: {{replay.scope_rules}}
   replay_authorization_rule: {{replay.authorization_rule}} | OPTIONAL
8. Per-Webhook Overrides (Optional)
   overrides: {{overrides.list}} | OPTIONAL
9. Telemetry
   delivery_attempt_metric: {{telemetry.attempt_metric}}
   delivery_success_metric: {{telemetry.success_metric}} | OPTIONAL

delivery_latency_metric: {{telemetry.latency_metric}} | OPTIONAL
fields: {{telemetry.fields}} | OPTIONAL

10. References
    Webhook catalog: {{xref:WHCP-01}}
    Outbound producer: {{xref:WHCP-02}} | OPTIONAL
    Inbound consumer: {{xref:WHCP-03}} | OPTIONAL
    Event delivery semantics: {{xref:EVT-04}} | OPTIONAL
    Failure handling: {{xref:WHCP-07}} | OPTIONAL

Cross-References
Upstream: {{xref:WHCP-02}}, {{xref:WHCP-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:WHCP-07}}, {{xref:WHCP-08}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define delivery model, ordering model, and max attempts.
intermediate: Required. Define replay scope rules and poison/quarantine behavior and telemetry.
advanced: Required. Add per-webhook overrides and storm-safe stop conditions and retry-after nuances.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, ordering key rule, dedupe key rule, max age, quarantine flag, fallback delay, replay authorization, overrides, optional telemetry metrics/fields, batch semantics, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If delivery.model is UNKNOWN → block Completeness Gate.
If retry.max_attempts is UNKNOWN → block Completeness Gate.
If replay.supported is UNKNOWN → block Completeness Gate.
If telemetry.attempt_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.WHCP
Pass conditions:
required_fields_present == true
delivery_and_ordering_defined == true
bounded_retry_defined == true
replay_scope_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

WHCP-05

WHCP-05 — Security Rules (signatures, secrets rotation, allowlists)
Header Block
template_id: WHCP-05
title: Security Rules (signatures, secrets rotation, allowlists)
type: webhook_security_rules
template_version: 1.0.0
output_path: 10_app/webhooks/WHCP-05_Security_Rules.md
compliance_gate_id: TMP-05.PRIMARY.WHCP
upstream_dependencies: ["WHCP-02", "WHCP-03", "IXS-04", "IXS-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "WHCP-01", "WHCP-02", "WHCP-03", "IXS-04", "IXS-03", "CSec-05", "CER-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical security controls for webhooks: signature verification/signing, secret storage/rotation, endpoint allowlists, replay defense, request validation, and safe logging. This template must be consistent with integration secrets/network policies and must not introduce security gaps between inbound and outbound flows.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
WHCP-01 Webhook Catalog: {{whcp.catalog}}
WHCP-02 Outbound Producer Spec: {{whcp.outbound}} | OPTIONAL
WHCP-03 Inbound Consumer Spec: {{whcp.inbound}} | OPTIONAL
IXS-04 Secrets/Credential Handling: {{ixs.secrets_policy}}
IXS-03 Connectivity/Allowlists: {{ixs.network_policy}} | OPTIONAL
CSec-05 Secure Link Handling (patterns): {{csec.deep_link_security}} | OPTIONAL
CER-05 Logging/Redaction: {{cer.logging}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Security baseline statement
Signature required rule (inbound/outbound)
Signature scheme(s) supported (HMAC/JWS/mTLS/UNKNOWN)
Canonical string/signature construction rule
Timestamp/replay protection rule (if used)
Secret storage/rotation binding (IXS-04)
IP allowlist policy (if supported)
Request size limits and content-type policy
Schema validation rule (payload)
Failure response policy (what status codes)

Logging/redaction rules (never log secrets/payload PII)
Telemetry requirements (invalid signature, blocked IP)

Optional Fields
Key rotation overlap window | OPTIONAL
mTLS config notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Inbound verification MUST happen before processing.
Secrets must never be exposed in logs or responses.
Reject requests that fail signature/allowlist/validation with safe responses.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Baseline
   baseline_statement: {{baseline.statement}}
2. Signature Requirements
   inbound_requires_signature: {{sig.inbound_required}}
   outbound_signs_payloads: {{sig.outbound_signs}}
   schemes_supported: {{sig.schemes_supported}}
3. Canonical String / Construction
   canonical_string_rule: {{sig.canonical_string_rule}}
   signature_header_rule: {{sig.signature_header_rule}} | OPTIONAL
   timestamp_header_rule: {{sig.timestamp_header_rule}} | OPTIONAL
4. Replay Protection
   replay_protection_supported: {{replay.supported}}
   max_skew_seconds: {{replay.max_skew_seconds}} | OPTIONAL
   nonce_rule: {{replay.nonce_rule}} | OPTIONAL
5. Secrets & Rotation
   secret_policy_ref: {{secrets.ref}} (expected: {{xref:IXS-04}})
   rotation_required: {{secrets.rotation_required}}
   overlap_window_seconds: {{secrets.overlap_window_seconds}} | OPTIONAL
6. Allowlists
   ip_allowlist_supported: {{allow.ip_allowlist_supported}}
   ip_allowlist_rule: {{allow.ip_allowlist_rule}} | OPTIONAL
   egress_allowlist_ref: {{allow.egress_allowlist_ref}} (expected: {{xref:IXS-03}}) | OPTIONAL
7. Request Constraints
   max_body_bytes: {{req.max_body_bytes}}
   allowed_content_types: {{req.allowed_content_types}}
   reject_unknown_content_type: {{req.reject_unknown_content_type}} | OPTIONAL

8. Schema Validation
   schema_validation_required: {{schema.required}}
   schema_ref_rule: {{schema.ref_rule}} | OPTIONAL
9. Failure Responses
   on_invalid_signature: {{fail.on_invalid_signature}}
   on_blocked_ip: {{fail.on_blocked_ip}} | OPTIONAL
   on_invalid_schema: {{fail.on_invalid_schema}} | OPTIONAL
10. Logging / Redaction
    no_secrets_in_logs: {{logs.no_secrets_in_logs}}
    pii_redaction_rule: {{logs.pii_redaction_rule}} | OPTIONAL
    log_field_allowlist: {{logs.field_allowlist}} | OPTIONAL
11. Telemetry
    invalid_signature_metric: {{telemetry.invalid_signature_metric}}
    blocked_ip_metric: {{telemetry.blocked_ip_metric}} | OPTIONAL
    schema_fail_metric: {{telemetry.schema_fail_metric}} | OPTIONAL
12. References
    Webhook catalog: {{xref:WHCP-01}}
    Outbound spec: {{xref:WHCP-02}} | OPTIONAL
    Inbound spec: {{xref:WHCP-03}} | OPTIONAL
    Secrets policy: {{xref:IXS-04}}
    Connectivity allowlists: {{xref:IXS-03}} | OPTIONAL
    Observability: {{xref:WHCP-08}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:WHCP-06}}, {{xref:WHCP-07}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define baseline + signature required + secret policy ref + max body size.
intermediate: Required. Define canonical string rules, replay/allowlist policies, and failure responses.
advanced: Required. Add rotation overlap window, mTLS notes, and telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, header rules, max skew/nonce, overlap window, ip allowlist rule, egress allowlist ref, reject unknown content type, schema ref rule, optional fail policies, pii redaction/log allowlist, optional telemetry, mtls notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If baseline.statement is UNKNOWN → block Completeness Gate.
If sig.inbound_required is UNKNOWN → block Completeness Gate.
If secrets.ref is UNKNOWN → block Completeness Gate.

If req.max_body_bytes is UNKNOWN → block Completeness Gate.
If telemetry.invalid_signature_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.WHCP
Pass conditions:
required_fields_present == true
signature_and_secret_policies_defined == true
request_constraints_defined == true
failure_responses_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

WHCP-06

WHCP-06 — Endpoint Registration & Management (subscriptions, secrets)
Header Block
template_id: WHCP-06
title: Endpoint Registration & Management (subscriptions, secrets)
type: webhook_endpoint_registration_management
template_version: 1.0.0
output_path: 10_app/webhooks/WHCP-06_Endpoint_Registration_Management.md
compliance_gate_id: TMP-05.PRIMARY.WHCP
upstream_dependencies: ["WHCP-01", "WHCP-05", "ADMIN-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "WHCP-01", "WHCP-05", "IXS-04", "API-04", "ADMIN-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical subscription/endpoint registration and lifecycle management for webhooks: how consumers register endpoints, choose events, manage secrets, rotate credentials, and how admins operate and audit these subscriptions. This template must be consistent with webhook security rules and AuthZ/admin capability controls.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
WHCP-01 Webhook Catalog: {{whcp.catalog}}
WHCP-05 Security Rules: {{whcp.security_rules}}
IXS-04 Secrets/Credential Handling: {{ixs.secrets_policy}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
ADMIN-01 Admin Capabilities Matrix: {{admin.capabilities}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Subscription entity model (subscription_id, owner)
Registration flow (API/UI/admin)
Endpoint URL validation rules (https only, allowlist)
Event selection rules (which webhook_ids can subscribe)
Secret management rules (create/rotate/revoke)
Verification handshake rules (challenge/response) if used
Pause/resume rules
Deletion/unsubscribe rules
Access control rules (who can manage subscriptions)
Audit logging requirements (create/update/rotate/delete)
Telemetry requirements (active subs, failures by sub)

Optional Fields
Test delivery (“send test webhook”) | OPTIONAL
Per-tenant subscription limits | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Registration MUST validate endpoint URLs and must not permit open redirects or non-HTTPS endpoints (unless explicitly allowed).
Secrets MUST be handled per {{xref:IXS-04}} and rotated safely.
Access control MUST be enforced server-side per {{xref:API-04}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Subscription Model
   subscription_id_field: {{model.subscription_id_field}}
   owner_field: {{model.owner_field}}
   tenant_field: {{model.tenant_field}} | OPTIONAL
   status_field: {{model.status_field}} | OPTIONAL
2. Registration Flow
   supported_surfaces: {{reg.surfaces}} (api/ui/admin/UNKNOWN)
   registration_steps: {{reg.steps}}
   verification_handshake: {{reg.verification_handshake}} | OPTIONAL
3. Endpoint URL Validation
   https_required: {{url.https_required}}
   allowed_domains_rule: {{url.allowed_domains_rule}} | OPTIONAL
   url_normalization_rule: {{url.normalization_rule}} | OPTIONAL
4. Event Selection
   allowed_webhook_ids: {{events.allowed_webhook_ids}}
   default_enabled_webhook_ids: {{events.default_enabled}} | OPTIONAL
5. Secrets Management
   secret_created_on_register: {{secrets.created_on_register}}
   rotation_supported: {{secrets.rotation_supported}}
   rotation_process: {{secrets.rotation_process}} | OPTIONAL
   secret_ref_policy: {{secrets.secret_ref_policy}} (expected: {{xref:IXS-04}}) | OPTIONAL
6. Pause / Resume / Delete
   pause_supported: {{lifecycle.pause_supported}}
   resume_supported: {{lifecycle.resume_supported}} | OPTIONAL
   delete_supported: {{lifecycle.delete_supported}}
   delete_behavior: {{lifecycle.delete_behavior}} | OPTIONAL
7. Access Control
   who_can_register: {{access.who_can_register}}

who_can_manage: {{access.who_can_manage}}
admin_override_supported: {{access.admin_override_supported}} | OPTIONAL

8. Audit Logging
   audit_required: {{audit.required}}
   audit_events: {{audit.events}}
   audit_fields: {{audit.fields}} | OPTIONAL
9. Telemetry
   active_subscriptions_metric: {{telemetry.active_subscriptions_metric}}
   delivery_failures_by_sub_metric: {{telemetry.failures_by_sub_metric}} | OPTIONAL
10. References
    Webhook catalog: {{xref:WHCP-01}}
    Security rules: {{xref:WHCP-05}}
    Secrets policy: {{xref:IXS-04}} | OPTIONAL
    AuthZ rules: {{xref:API-04}} | OPTIONAL
    Admin capabilities: {{xref:ADMIN-01}} | OPTIONAL

Cross-References
Upstream: {{xref:WHCP-01}}, {{xref:WHCP-05}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:WHCP-07}}, {{xref:WHCP-08}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define subscription model + url validation + allowed webhook ids.
intermediate: Required. Define secrets rotation + pause/delete + access control + audit events.
advanced: Required. Add verification handshake, subscription limits, and test delivery and telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, tenant/status fields, handshake, allowed domains, normalization, defaults, rotation process, secret ref policy, resume/delete behavior, admin override, audit fields, optional telemetry metrics, test delivery, limits, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If reg.steps is UNKNOWN → block Completeness Gate.
If url.https_required is UNKNOWN → block Completeness Gate.
If events.allowed_webhook_ids is UNKNOWN → block Completeness Gate.
If audit.required is UNKNOWN → block Completeness Gate.
If telemetry.active_subscriptions_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.WHCP
Pass conditions:
required_fields_present == true
registration_and_url_validation_defined == true
event_selection_defined == true

access_control_and_audit_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

WHCP-07

WHCP-07 — Error Handling (DLQ, quarantine, manual replay)
Header Block
template_id: WHCP-07
title: Error Handling (DLQ, quarantine, manual replay)
type: webhook_error_handling
template_version: 1.0.0
output_path: 10_app/webhooks/WHCP-07_Error_Handling.md
compliance_gate_id: TMP-05.PRIMARY.WHCP
upstream_dependencies: ["WHCP-04", "IXS-06", "JBS-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "WHCP-01", "WHCP-02", "WHCP-03", "WHCP-04", "IXS-06", "JBS-04", "ADMIN-02", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical error handling lifecycle for webhooks (inbound and outbound): how failures are classified, when items go to DLQ/quarantine, how manual replay works, and what operator actions exist. This template must be consistent with delivery semantics and global integration error handling.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
WHCP-01 Webhook Catalog: {{whcp.catalog}}
WHCP-02 Outbound Producer Spec: {{whcp.outbound}} | OPTIONAL
WHCP-03 Inbound Consumer Spec: {{whcp.inbound}} | OPTIONAL
WHCP-04 Delivery Semantics: {{whcp.delivery_semantics}}
IXS-06 Integration Error Handling: {{ixs.error_recovery}} | OPTIONAL
JBS-04 Retry/DLQ Policy: {{jobs.retry_dlq}} | OPTIONAL
ADMIN-02 Support Tools Spec: {{admin.support_tools}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Failure taxonomy (network, 4xx, 5xx, invalid signature, invalid schema)
Retry exhaustion rule (when to stop)
DLQ/quarantine supported (yes/no/UNKNOWN)
DLQ trigger rule
Quarantine contents (what stored, redacted)
Replay supported (yes/no/UNKNOWN)
Replay authorization rules (who can replay)
Replay scope rules (by webhook_id/subscription_id/time)

Backfill/retry safety checks (idempotency)
Operator actions (pause sub, rotate secret, resend test)
Telemetry requirements (dlq depth, replay success)
Audit logging requirements (replay actions, overrides)

Optional Fields
Auto-replay policy | OPTIONAL
Customer notification policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Do not DLQ raw secrets or unredacted PII.
Replay must not violate ordering/duplication rules; idempotency must be enforced.
Operator actions must be permissioned and auditable.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Failure Taxonomy
   classes: {{fail.classes}}
   mapping_notes: {{fail.mapping_notes}} | OPTIONAL
2. Retry Exhaustion
   stop_conditions: {{retry.stop_conditions}}
   max_attempts_ref: {{retry.max_attempts_ref}} (expected: {{xref:WHCP-04}}) | OPTIONAL
3. DLQ / Quarantine
   dlq_supported: {{dlq.supported}}
   dlq_trigger_rule: {{dlq.trigger_rule}}
   redaction_rule: {{dlq.redaction_rule}}
   retention_ms: {{dlq.retention_ms}} | OPTIONAL
4. Replay
   replay_supported: {{replay.supported}}
   who_can_replay: {{replay.who_can_replay}}
   scope_rules: {{replay.scope_rules}}
   safety_checks: {{replay.safety_checks}} | OPTIONAL
5. Operator Actions
   Action
   action_id: {{ops.actions[0].action_id}}
   name: {{ops.actions[0].name}}
   when_allowed: {{ops.actions[0].when_allowed}}
   required_role: {{ops.actions[0].required_role}} | OPTIONAL
   audit_required: {{ops.actions[0].audit_required}}

(Repeat per action.)

6. Auto-Replay (Optional)
   auto_replay_supported: {{auto.supported}} | OPTIONAL
   auto_replay_rules: {{auto.rules}} | OPTIONAL
7. Telemetry
   dlq_depth_metric: {{telemetry.dlq_depth_metric}}
   replay_success_metric: {{telemetry.replay_success_metric}} | OPTIONAL
   quarantine_rate_metric: {{telemetry.quarantine_rate_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
8. Audit Logging
   audit_required: {{audit.required}}
   audit_events: {{audit.events}}
   audit_fields: {{audit.fields}} | OPTIONAL
9. References
   Delivery semantics: {{xref:WHCP-04}}
   Security rules: {{xref:WHCP-05}} | OPTIONAL
   Endpoint management: {{xref:WHCP-06}} | OPTIONAL
   Support tools: {{xref:ADMIN-02}} | OPTIONAL
   Audit trail: {{xref:ADMIN-03}} | OPTIONAL

Cross-References
Upstream: {{xref:WHCP-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:WHCP-08}}, {{xref:ADMIN-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define taxonomy + dlq trigger + replay supported and who can replay.
intermediate: Required. Define redaction/retention, scope rules, operator actions, telemetry.
advanced: Required. Add auto-replay, customer comms policy, and safety checks rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, mapping notes, max attempts ref, retention ms, safety checks, required roles, auto replay, telemetry fields, audit fields, customer notifications, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If dlq.supported is UNKNOWN → block Completeness Gate.
If dlq.trigger_rule is UNKNOWN → block Completeness Gate (when dlq.supported == true).
If dlq.redaction_rule is UNKNOWN → block Completeness Gate (when dlq.supported == true).
If replay.supported is UNKNOWN → block Completeness Gate.
If telemetry.dlq_depth_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.WHCP
Pass conditions:
required_fields_present == true
dlq_and_redaction_defined == true
replay_policy_defined == true
operator_actions_defined == true
telemetry_defined == true
audit_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

WHCP-08

WHCP-08 — Observability (delivery rate, latency, failures)
Header Block
template_id: WHCP-08
title: Observability (delivery rate, latency, failures)
type: webhook_observability
template_version: 1.0.0
output_path: 10_app/webhooks/WHCP-08_Observability.md
compliance_gate_id: TMP-05.PRIMARY.WHCP
upstream_dependencies: ["WHCP-04", "WHCP-07", "IXS-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "WHCP-01", "WHCP-04", "WHCP-07", "IXS-07", "CER-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical observability requirements for webhooks: delivery rate, latency, failures, signature/validation rejects, DLQ depth, dashboards, and alerts. This template must be consistent with integration observability and must not introduce telemetry that violates privacy/redaction rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
WHCP-01 Webhook Catalog: {{whcp.catalog}}
WHCP-04 Delivery Semantics: {{whcp.delivery_semantics}} | OPTIONAL
WHCP-07 Error Handling: {{whcp.error_handling}} | OPTIONAL
IXS-07 Integration Observability: {{ixs.observability}} | OPTIONAL
CER-05 Logging/Redaction: {{cer.logging}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Core metrics set (attempts, success, failure)
Latency metrics (p50/p95 or ms)
Reject metrics (invalid signature, invalid schema)
Per-webhook/per-subscription breakdown policy (avoid high-cardinality)
DLQ/quarantine metrics (depth, age)
Dashboards minimum panels
Alerts (failure spike, latency spike, dlq stuck)
Log field requirements (webhook_id, subscription_id hash, delivery_id)
Redaction/no-PII rule
Runbook references (triage)

Optional Fields
Tracing spans for delivery pipeline | OPTIONAL
Sampling policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Do not log raw payloads unless explicitly allowed and redacted.
Avoid high-cardinality dimensions; hash subscription identifiers where needed.
Alerts must be actionable and tied to runbooks.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Core Metrics
   attempt_metric: {{metrics.attempt_metric}}
   success_metric: {{metrics.success_metric}}
   failure_metric: {{metrics.failure_metric}} | OPTIONAL
2. Latency
   latency_ms_metric: {{metrics.latency_ms_metric}}
   percentiles: {{metrics.percentiles}} | OPTIONAL
3. Reject Metrics
   invalid_signature_metric: {{reject.invalid_signature_metric}}
   invalid_schema_metric: {{reject.invalid_schema_metric}} | OPTIONAL
4. DLQ / Quarantine Metrics
   dlq_depth_metric: {{dlq.depth_metric}}
   dlq_age_metric: {{dlq.age_metric}} | OPTIONAL
5. Breakdown Policy
   allowed_dimensions: {{breakdown.allowed_dimensions}}
   subscription_id_handling: {{breakdown.subscription_id_handling}}
   (hash/omit/UNKNOWN)
6. Logs — Required Fields
   timestamp: {{logs.timestamp}}
   webhook_id: {{logs.webhook_id}}
   direction: {{logs.direction}} | OPTIONAL
   delivery_id: {{logs.delivery_id}} | OPTIONAL
   subscription_id_hash: {{logs.subscription_id_hash}} | OPTIONAL
   status_code: {{logs.status_code}} | OPTIONAL
   duration_ms: {{logs.duration_ms}} | OPTIONAL
   redaction_applied: {{logs.redaction_applied}}
7. Dashboards
   dashboard_name: {{dash.name}}
   minimum_panels: {{dash.minimum_panels}}
   panel_list: {{dash.panel_list}} | OPTIONAL

8. Alerts
   Alert
   alert_id: {{alerts[0].alert_id}}
   condition: {{alerts[0].condition}}
   severity: {{alerts[0].severity}} (low/med/high/UNKNOWN)
   routing: {{alerts[0].routing}}
   runbook_ref: {{alerts[0].runbook_ref}} | OPTIONAL

(Repeat per alert.)

9. Runbooks
   triage_runbook_ref: {{runbooks.triage_ref}} (expected: {{xref:WHCP-07}}) | OPTIONAL
10. References
   Webhook catalog: {{xref:WHCP-01}}
   Delivery semantics: {{xref:WHCP-04}} | OPTIONAL
   Error handling: {{xref:WHCP-07}} | OPTIONAL
   Integration observability baseline: {{xref:IXS-07}} | OPTIONAL

Cross-References
Upstream: {{xref:WHCP-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define core metrics + no-PII rule + minimum panels.
intermediate: Required. Define alerts and log required fields and breakdown policy.
advanced: Required. Add tracing/sampling and refined dimension controls and runbook rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, percentiles, optional reject metrics, dlq age metric, panel list, runbook refs, optional log fields, optional alerts fields, tracing/sampling, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If metrics.attempt_metric is UNKNOWN → block Completeness Gate.
If reject.invalid_signature_metric is UNKNOWN → block Completeness Gate.
If dash.minimum_panels is UNKNOWN → block Completeness Gate.
If logs.webhook_id is UNKNOWN → block Completeness Gate.
If breakdown.subscription_id_handling is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.WHCP
Pass conditions:
required_fields_present == true
core_metrics_defined == true

logging_and_redaction_defined == true
dashboards_and_alerts_defined == true
breakdown_policy_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

WHCP-09

WHCP-09 — Payload Versioning & Compatibility (schema evolution)
Header Block
template_id: WHCP-09
title: Payload Versioning & Compatibility (schema evolution)
type: webhook_payload_versioning_compatibility
template_version: 1.0.0
output_path: 10_app/webhooks/WHCP-09_Payload_Versioning_Compatibility.md
compliance_gate_id: TMP-05.PRIMARY.WHCP
upstream_dependencies: ["WHCP-01", "EVT-02", "IXS-10"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "WHCP-01", "EVT-02", "IXS-10", "WHCP-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical payload versioning and compatibility rules for webhooks: how schemas evolve, how versions are signaled, compatibility windows, breaking change rules, and migration/rollback strategy. This template must be consistent with event schema versioning and integration change management policies.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
WHCP-01 Webhook Catalog: {{whcp.catalog}}
EVT-02 Event Schema Spec: {{evt.schema_spec}} | OPTIONAL
IXS-10 Versioning & Change Management: {{ixs.change_mgmt}} | OPTIONAL
WHCP-04 Delivery Semantics: {{whcp.delivery_semantics}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Versioning model (semver/date/int/UNKNOWN)
Where version is carried (header/payload field/path)
Per-webhook version tracking (webhook_id → current version)
Backward compatibility rules (what is allowed)
Breaking change definition
Deprecation policy (notice period)
Migration strategy (dual schema support)
Rollback strategy (version pin, fallback)
Validation behavior by version (strict/lenient)
Telemetry requirements (version usage distribution)

Optional Fields
Consumer capability negotiation | OPTIONAL

Schema registry links | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
A breaking change requires a new major version (or explicit breaking bump rule) and a migration plan.
Compatibility window must be explicit; do not support “all old versions forever.”
Version parsing must be deterministic and validated.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Versioning Model
   model: {{version.model}} (semver/date/int/UNKNOWN)
   version_location: {{version.location}} (header/payload/path/UNKNOWN)
   version_field_or_header: {{version.field_or_header}} | OPTIONAL
2. Per-Webhook Current Versions
   Entry
   webhook_id: {{current[0].webhook_id}}
   current_version: {{current[0].current_version}}
   supported_versions: {{current[0].supported_versions}} | OPTIONAL
   notes: {{current[0].notes}} | OPTIONAL

(Repeat per webhook.)

3. Compatibility Rules
   backward_compatible_changes: {{compat.backward_compatible_changes}}
   breaking_changes: {{compat.breaking_changes}}
   compat_window_days: {{compat.window_days}}
4. Deprecation Policy
   notice_period_days: {{deprec.notice_period_days}}
   deprec_process: {{deprec.process}}
   sunset_behavior: {{deprec.sunset_behavior}} | OPTIONAL
5. Migration Strategy
   dual_version_support: {{migrate.dual_version_support}}
   migration_steps: {{migrate.steps}} | OPTIONAL
6. Rollback Strategy
   rollback_model: {{rollback.model}} (version_pin/fallback/UNKNOWN)
   rollback_steps: {{rollback.steps}} | OPTIONAL
7. Validation Behavior
   validation_by_version: {{validate.by_version}}
   unknown_version_behavior: {{validate.unknown_version_behavior}}

8. Telemetry
   version_usage_metric: {{telemetry.version_usage_metric}}
   unknown_version_metric: {{telemetry.unknown_version_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
9. References
   Webhook catalog: {{xref:WHCP-01}}
   Event schema versioning: {{xref:EVT-02}} | OPTIONAL
   Integration change mgmt: {{xref:IXS-10}} | OPTIONAL
   Delivery semantics: {{xref:WHCP-04}} | OPTIONAL

Cross-References
Upstream: {{xref:WHCP-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:WHCP-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define version model/location and current versions list.
intermediate: Required. Define compat window, deprecation, and unknown version behavior.
advanced: Required. Add migration/rollback steps, capability negotiation, and telemetry fields rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, version field/header, supported versions, notes, sunset behavior, migration/rollback steps, unknown version metric/fields, capability negotiation, schema registry links, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If version.model is UNKNOWN → block Completeness Gate.
If version.location is UNKNOWN → block Completeness Gate.
If compat.window_days is UNKNOWN → block Completeness Gate.
If validate.unknown_version_behavior is UNKNOWN → block Completeness Gate.
If telemetry.version_usage_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.WHCP
Pass conditions:
required_fields_present == true
version_model_and_location_defined == true
per_webhook_versions_defined == true
compat_and_deprecation_defined == true
unknown_version_handling_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

WHCP-10

WHCP-10 — Testing & Sandbox Strategy (test hooks, simulators)
Header Block
template_id: WHCP-10
title: Testing & Sandbox Strategy (test hooks, simulators)
type: webhook_testing_sandbox_strategy
template_version: 1.0.0
output_path: 10_app/webhooks/WHCP-10_Testing_Sandbox_Strategy.md
compliance_gate_id: TMP-05.PRIMARY.WHCP
upstream_dependencies: ["WHCP-02", "WHCP-03", "IXS-09"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "WHCP-01", "WHCP-02", "WHCP-03", "WHCP-06", "WHCP-09", "IXS-09"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical strategy for testing webhooks safely: test endpoints, simulators, sample payload fixtures, signature generation/verification in tests, CI execution rules, and safety constraints (never deliver to prod endpoints). This template must be consistent with the integration sandbox strategy and webhook security/versioning rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
WHCP-01 Webhook Catalog: {{whcp.catalog}}
WHCP-02 Outbound Producer Spec: {{whcp.outbound}} | OPTIONAL
WHCP-03 Inbound Consumer Spec: {{whcp.inbound}} | OPTIONAL
WHCP-06 Endpoint Registration: {{whcp.registration}} | OPTIONAL
WHCP-09 Versioning/Compatibility: {{whcp.versioning}} | OPTIONAL
IXS-09 Integration Sandbox/Test Strategy: {{ixs.test_strategy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Test surfaces (producer tests, consumer tests)
Fixture payload location and format
Signature test strategy (generate/verify)
Simulators/test hooks (what exists)
Sandbox endpoints for subscriptions (test-only)
CI rules (what runs in CI, safe allowlists)
Safety rules (block prod deliveries)
Coverage expectations (success/failure/invalid sig/version mismatch)
Verification commands (how to run)
Telemetry for tests (optional but recommended)

Optional Fields
Contract testing with consumers | OPTIONAL
Load testing strategy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Tests MUST not use production endpoints or production secrets.
Fixtures MUST include version fields and pass schema validation for targeted versions.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Test Surfaces
   producer_tests: {{tests.producer_tests}}
   consumer_tests: {{tests.consumer_tests}} | OPTIONAL
2. Fixtures
   fixture_location: {{fixtures.location}}
   fixture_format: {{fixtures.format}} (json/yaml/UNKNOWN)
   versions_covered: {{fixtures.versions_covered}} | OPTIONAL
3. Signature Testing
   signing_test_supported: {{sig.test_supported}}
   test_key_policy: {{sig.test_key_policy}}
   verification_test_rule: {{sig.verification_test_rule}} | OPTIONAL
4. Simulators / Test Hooks
   simulators: {{sim.simulators}}
   test_hook_endpoints: {{sim.test_hook_endpoints}} | OPTIONAL
5. Sandbox Subscriptions
   sandbox_endpoints: {{sandbox.endpoints}}
   registration_rules: {{sandbox.registration_rules}} | OPTIONAL
6. CI Rules
   ci_enabled: {{ci.enabled}}
   ci_allowlist_rule: {{ci.allowlist_rule}}
   ci_secrets_policy: {{ci.secrets_policy}} | OPTIONAL
7. Safety Rules
   block_prod_delivery_rule: {{safety.block_prod_delivery_rule}}
   env_guard_rule: {{safety.env_guard_rule}} | OPTIONAL
8. Coverage Expectations
   must_cover:
   {{coverage.must_cover[0]}}
   {{coverage.must_cover[1]}}
   {{coverage.must_cover[2]}} | OPTIONAL

9. Verification Commands
   how_to_run: {{verify.how_to_run}}
   expected_outputs: {{verify.expected_outputs}} | OPTIONAL
10. References
   Webhook catalog: {{xref:WHCP-01}}
   Security rules: {{xref:WHCP-05}} | OPTIONAL
   Versioning: {{xref:WHCP-09}} | OPTIONAL
   Endpoint management: {{xref:WHCP-06}} | OPTIONAL
   Integration test strategy: {{xref:IXS-09}} | OPTIONAL

Cross-References
Upstream: {{xref:WHCP-02}}, {{xref:WHCP-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:WHCP-08}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define fixtures + CI safety + how to run tests.
intermediate: Required. Define signature testing and simulators/test hooks and coverage list.
advanced: Required. Add contract/load testing and multi-version fixture coverage rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, consumer tests, versions covered, verification rules, test hooks, registration rules, ci secrets policy, env guard, extra coverage items, expected outputs, contract/load testing, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If fixtures.location is UNKNOWN → block Completeness Gate.
If ci.allowlist_rule is UNKNOWN → block Completeness Gate.
If safety.block_prod_delivery_rule is UNKNOWN → block Completeness Gate.
If verify.how_to_run is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.WHCP
Pass conditions:
required_fields_present == true
fixtures_defined == true
ci_and_safety_rules_defined == true
coverage_expectations_defined == true
verification_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Payments & Billing (PAY)

Payments & Billing (PAY)
PAY-01 Provider Inventory (by provider_id)
PAY-02 Payment Flow Spec (one-time, recurring, invoices)
PAY-03 Pricing & Plan Mapping (plans, tiers, entitlements)
PAY-04 Webhook Handling for Payments (events, idempotency, retries)
PAY-05 Tax/VAT Rules & Compliance (if applicable)
PAY-06 Refunds, Chargebacks & Disputes Policy
PAY-07 Ledger & Reconciliation Rules (source of truth, audits)
PAY-08 Fraud & Abuse Controls (risk checks, velocity limits)
PAY-09 Security & PCI Boundaries (tokenization, SAQ scope)
PAY-10 Billing Observability & Support Runbooks (alerts, workflows)

PAY-01

PAY-01 — Provider Inventory (by provider_id)
Header Block
template_id: PAY-01
title: Provider Inventory (by provider_id)
type: payments_provider_inventory
template_version: 1.0.0
output_path: 10_app/payments/PAY-01_Provider_Inventory.md
compliance_gate_id: TMP-05.PRIMARY.PAY
upstream_dependencies: ["IXS-01", "PRD-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "PRD-06", "API-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical inventory of payment providers used by the application, indexed by provider_id, including enabled environments, supported payment modes, and compliance scope notes. This document must not invent provider_ids not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}} | OPTIONAL
PRD-06 NFRs/Compliance: {{prd.nfrs}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Provider registry (provider_id list)
provider_id (stable identifier)
provider name/vendor
integration_id binding (IXS-01)
supported modes (one-time/recurring/invoices)
supported payment methods (card/ACH/etc.)
environments enabled (dev/stage/prod)
webhook support (yes/no)
criticality (low/med/high)
owner (team/role)
compliance scope notes (PCI boundary)
references to detailed specs (PAY-02..10)

Optional Fields
Status (planned/active/deprecated) | OPTIONAL

SLA/uptime notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent provider_ids; use only {{spec.pay_providers_by_id}} if present, else mark UNKNOWN and flag.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Inventory Summary
   total_providers: {{inv.total}}
   modes_present: {{inv.modes_present}} | OPTIONAL
   notes: {{inv.notes}} | OPTIONAL
2. Provider Entries (by provider_id)
   Provider
   provider_id: {{providers[0].provider_id}}
   name: {{providers[0].name}}
   integration_id: {{providers[0].integration_id}}
   modes: {{providers[0].modes}}
   payment_methods: {{providers[0].payment_methods}}
   env_enabled: {{providers[0].env_enabled}}
   webhooks_supported: {{providers[0].webhooks_supported}}
   criticality: {{providers[0].criticality}}
   owner: {{providers[0].owner}}
   pci_scope_notes: {{providers[0].pci_scope_notes}}
   spec_refs:
- {{providers[0].spec_refs[0]}} (e.g., {{xref:PAY-02}} / {{xref:PAY-04}})
- {{providers[0].spec_refs[1]}} | OPTIONAL
  status: {{providers[0].status}} | OPTIONAL
  sla_notes: {{providers[0].sla_notes}} | OPTIONAL
  open_questions:
  {{providers[0].open_questions[0]}} | OPTIONAL

(Repeat per provider_id.)

3. References
   Payment flows: {{xref:PAY-02}} | OPTIONAL
   Pricing/plan mapping: {{xref:PAY-03}} | OPTIONAL
   Payment webhooks: {{xref:PAY-04}} | OPTIONAL
   Taxes/compliance: {{xref:PAY-05}} | OPTIONAL
   Refunds/disputes: {{xref:PAY-06}} | OPTIONAL
   Ledger/reconciliation: {{xref:PAY-07}} | OPTIONAL

Fraud controls: {{xref:PAY-08}} | OPTIONAL
Security/PCI boundaries: {{xref:PAY-09}} | OPTIONAL
Observability/runbooks: {{xref:PAY-10}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PAY-02}}, {{xref:PAY-04}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Populate provider registry and core fields; do not invent IDs.
intermediate: Required. Add modes/methods and webhook support and spec refs.
advanced: Required. Add PCI boundary notes, ownership/criticality, and traceability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, inv notes, secondary spec refs, status, sla notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If providers[*].provider_id is UNKNOWN → block Completeness Gate.*
*If providers[*].integration_id is UNKNOWN → block Completeness Gate.
If providers[*].modes is UNKNOWN → block Completeness Gate.*
*If providers[*].pci_scope_notes is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PAY
Pass conditions:
required_fields_present == true
provider_ids_unique == true
integration_ids_exist_in_IXS_01 == true
pci_scope_noted == true
spec_refs_present == true
placeholder_resolution == true
no_unapproved_unknowns == true

PAY-02

PAY-02 — Payment Flow Spec (one-time, recurring, invoices)
Header Block
template_id: PAY-02
title: Payment Flow Spec (one-time, recurring, invoices)
type: payments_flow_spec
template_version: 1.0.0
output_path: 10_app/payments/PAY-02_Payment_Flow_Spec.md
compliance_gate_id: TMP-05.PRIMARY.PAY
upstream_dependencies: ["PAY-01", "API-02", "FE-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PAY-01", "API-01", "API-02", "FE-01", "FE-04", "CER-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical payment flows supported by the product (one-time, recurring, invoices): steps, actors, API endpoints involved, UI touchpoints, provider interactions, and failure/retry handling. This template must be consistent with provider inventory and must not invent payment modes/endpoints not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PAY-01 Provider Inventory: {{pay.providers}}
API-01 Endpoint Catalog: {{api.endpoint_catalog}} | OPTIONAL
API-02 Endpoint Specs: {{api.endpoint_specs}} | OPTIONAL
FE-01 Route Map: {{fe.route_layout}} | OPTIONAL
FE-04 Data Binding Rules: {{fe.data_binding}} | OPTIONAL
CER-02 Retry/Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
provider_id binding
Supported flow types list (one_time/recurring/invoice)
Flow steps (client → server → provider)
Payment intent/checkout object rules (provider primitives)
Idempotency rules for create/confirm
Sensitive data handling (no raw card data)
Failure handling (declines, timeouts)
Retry rules (when allowed vs not)
UI touchpoints (screen_id/route_id)
API endpoints used (endpoint_id)
Telemetry requirements (success/fail, decline reasons)

Optional Fields
Offline/manual invoice flow | OPTIONAL
Proration rules (if subscriptions) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Never handle raw card data unless explicitly in PCI scope; prefer provider tokenization.
All create/confirm operations MUST be idempotent.
Declines must not be retried blindly; use provider reason codes.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Flow Types
   provider_id: {{meta.provider_id}}
   flow_types: {{flows.types}}
2. Flow Definition (repeat per flow type)
   Flow
   flow_type: {{flows.items[0].flow_type}}
   name: {{flows.items[0].name}}
   ui_entry_route: {{flows.items[0].ui_entry_route}} | OPTIONAL
   api_endpoints: {{flows.items[0].api_endpoints}}
   provider_objects: {{flows.items[0].provider_objects}} | OPTIONAL

steps:
{{flows.items[0].steps[0]}}
{{flows.items[0].steps[1]}}
{{flows.items[0].steps[2]}} | OPTIONAL

3. Idempotency
   idempotency_required: {{idem.required}}
   key_rule: {{idem.key_rule}}
   scope: {{idem.scope}} | OPTIONAL
4. Failure Handling
   decline_handling: {{fail.decline_handling}}
   timeout_handling: {{fail.timeout_handling}} | OPTIONAL
   retry_rules: {{fail.retry_rules}}
5. Sensitive Data
   tokenization_rule: {{security.tokenization_rule}}
   no_raw_card_rule: {{security.no_raw_card_rule}}
6. Telemetry
   payment_success_metric: {{telemetry.success_metric}}
   payment_failure_metric: {{telemetry.failure_metric}} | OPTIONAL

decline_reason_metric: {{telemetry.decline_reason_metric}} | OPTIONAL
fields: {{telemetry.fields}} | OPTIONAL

7. References
   Provider inventory: {{xref:PAY-01}}
   Pricing/plan mapping: {{xref:PAY-03}} | OPTIONAL
   Payment webhooks: {{xref:PAY-04}} | OPTIONAL
   Refunds/disputes: {{xref:PAY-06}} | OPTIONAL
   Security/PCI: {{xref:PAY-09}} | OPTIONAL

Cross-References
Upstream: {{xref:PAY-01}}, {{xref:API-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PAY-04}}, {{xref:PAY-07}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define flow steps and endpoints and no-raw-card rule.
intermediate: Required. Define idempotency, failure handling, and telemetry.
advanced: Required. Add proration/invoice variants and detailed provider object mapping.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, ui entry, provider objects, optional steps, scope, timeout handling, telemetry fields, invoice/proration, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If flows.types is UNKNOWN → block Completeness Gate.
If idem.key_rule is UNKNOWN → block Completeness Gate.
If security.no_raw_card_rule is UNKNOWN → block Completeness Gate.
If telemetry.success_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PAY
Pass conditions:
required_fields_present == true
flows_defined == true
idempotency_defined == true
sensitive_data_rules_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PAY-03

PAY-03 — Pricing & Plan Mapping (plans, tiers, entitlements)
Header Block
template_id: PAY-03
title: Pricing & Plan Mapping (plans, tiers, entitlements)
type: payments_pricing_plan_mapping
template_version: 1.0.0
output_path: 10_app/payments/PAY-03_Pricing_Plan_Mapping.md
compliance_gate_id: TMP-05.PRIMARY.PAY
upstream_dependencies: ["PRD-06", "FFCFG-01", "API-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PRD-06", "PAY-01", "FFCFG-01", "API-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical mapping from product pricing/plans to payment-provider products/prices, and how those plans map to internal entitlements/feature access. This template must be consistent with AuthZ/entitlements and must not invent plan IDs beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PRD-06 NFRs / Monetization constraints: {{prd.nfrs}} | OPTIONAL
PAY-01 Provider Inventory: {{pay.providers}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.flags}} | OPTIONAL
API-04 AuthZ Rules (roles/entitlements): {{api.authz_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Plan registry (plan_id list)
plan_id (stable identifier)
plan name
billing cadence (monthly/annual/one-time)
price points (amount/currency)
provider mapping (provider_id + product/price ids)
trial policy (if any)
entitlements granted (role/permission/feature flags)
upgrade/downgrade rules
cancellation policy (end of term vs immediate)
tax behavior flag (if applicable)
telemetry requirements (plan conversions, churn)

Optional Fields
Grandfathering rules | OPTIONAL
Promo codes/coupons policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent plan_ids; use only {{spec.plans_by_id}} if present, else mark UNKNOWN and flag.
Entitlements must bind to existing role/permission/flag IDs.
Price changes must follow change management (IXS-10) if external-facing.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Plan Registry Summary
   total_plans: {{plans.total}}
   notes: {{plans.notes}} | OPTIONAL
2. Plan Entries (by plan_id)
   Plan
   plan_id: {{items[0].plan_id}}
   name: {{items[0].name}}
   cadence: {{items[0].cadence}}
   price: {{items[0].price}}
   provider_map:
   provider_id: {{items[0].provider_map.provider_id}}
   provider_product_id: {{items[0].provider_map.product_id}} | OPTIONAL
   provider_price_id: {{items[0].provider_map.price_id}} | OPTIONAL

trial_policy: {{items[0].trial_policy}} | OPTIONAL
entitlements: {{items[0].entitlements}}
upgrade_rules: {{items[0].upgrade_rules}}
downgrade_rules: {{items[0].downgrade_rules}} | OPTIONAL
cancellation_policy: {{items[0].cancellation_policy}}
tax_flag: {{items[0].tax_flag}} | OPTIONAL
grandfathering: {{items[0].grandfathering}} | OPTIONAL
promo_policy: {{items[0].promo_policy}} | OPTIONAL
open_questions:
{{items[0].open_questions[0]}} | OPTIONAL

(Repeat per plan.)

3. Telemetry
   conversion_metric: {{telemetry.conversion_metric}}

churn_metric: {{telemetry.churn_metric}} | OPTIONAL
fields: {{telemetry.fields}} | OPTIONAL

4. References
   Provider inventory: {{xref:PAY-01}} | OPTIONAL
   Payment flow spec: {{xref:PAY-02}} | OPTIONAL
   Payment webhooks: {{xref:PAY-04}} | OPTIONAL
   Change management: {{xref:IXS-10}} | OPTIONAL
   Feature flags: {{xref:FFCFG-01}} | OPTIONAL
   AuthZ/entitlements: {{xref:API-04}} | OPTIONAL

Cross-References
Upstream: {{xref:PRD-06}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PAY-07}}, {{xref:PAY-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define plan registry + prices + entitlements; do not invent IDs.
intermediate: Required. Define upgrade/downgrade/cancellation rules and provider mappings.
advanced: Required. Add grandfathering/promos and strict telemetry + change management linkage.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, provider product/price ids, trial policy, downgrade rules, tax flag, grandfathering, promo policy, churn metric/fields, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If items[*].plan_id is UNKNOWN → block Completeness Gate.
If items[].provider_map.provider_id is UNKNOWN → block Completeness Gate.
If items[*].entitlements is UNKNOWN → block Completeness Gate.
If telemetry.conversion_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PAY
Pass conditions:
required_fields_present == true
plan_ids_unique == true
provider_mappings_defined == true
entitlements_bind_to_existing_ids == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PAY-04

PAY-04 — Webhook Handling for Payments (events, idempotency, retries)
Header Block
template_id: PAY-04
title: Webhook Handling for Payments (events, idempotency, retries)
type: payments_webhook_handling
template_version: 1.0.0
output_path: 10_app/payments/PAY-04_Webhook_Handling_For_Payments.md
compliance_gate_id: TMP-05.PRIMARY.PAY
upstream_dependencies: ["PAY-01", "WHCP-03", "WHCP-05", "PAY-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PAY-01", "PAY-02", "WHCP-03", "WHCP-05", "WHCP-04", "PAY-07", "API-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical handling of payment provider webhooks: which events are consumed, verification rules, idempotency/deduplication, state transitions (payment/subscription/invoice), retries, and failure recovery. This template must be consistent with webhook consumer/security semantics and internal ledger/reconciliation rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PAY-01 Provider Inventory: {{pay.providers}}
PAY-02 Payment Flow Spec: {{pay.flows}} | OPTIONAL
WHCP-03 Inbound Webhook Consumer Spec: {{whcp.inbound}} | OPTIONAL
WHCP-04 Delivery Semantics: {{whcp.delivery_semantics}} | OPTIONAL
WHCP-05 Webhook Security Rules: {{whcp.security_rules}} | OPTIONAL
PAY-07 Ledger & Reconciliation Rules: {{pay.ledger_rules}} | OPTIONAL
API-03 Error Policy: {{api.error_policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
provider_id binding
Webhook endpoint/path binding (endpoint_id/path)
Consumed event list (provider event types → internal event_id)
Signature verification rule (bind to WHCP-05)
Idempotency key rule (provider event id)
State transition rules (what updates what)
Ordering expectations and handling out-of-order events
Retry/ack rules (ack on accept vs processed)
Failure handling (DLQ/quarantine + replay)
Fraud/abuse handling hooks (if any)

Telemetry requirements (webhook received/processed, failures)
Audit requirements (financial events)

Optional Fields
Test mode handling | OPTIONAL
Multi-account/tenant routing | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Never trust webhook payload without verification.
Webhook processing MUST be idempotent and safe to replay.
State transitions must be consistent with ledger source-of-truth rules.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Endpoint Binding
   provider_id: {{meta.provider_id}}
   endpoint_id_or_path: {{endpoint.binding}}
2. Consumed Events
   Event
   provider_event_type: {{events[0].provider_event_type}}
   internal_event_id: {{events[0].internal_event_id}} | OPTIONAL
   handler_action: {{events[0].handler_action}}
   notes: {{events[0].notes}} | OPTIONAL

(Repeat per event.)

3. Verification
   security_ref: {{verify.security_ref}} (expected: {{xref:WHCP-05}}) | OPTIONAL
   verification_required: {{verify.required}}
   idempotency_key_rule: {{verify.idempotency_key_rule}}
4. State Transitions
   object_updated: {{state.object_updated}}
   transition_rules: {{state.transition_rules}}
   out_of_order_rule: {{state.out_of_order_rule}} | OPTIONAL
5. Ack / Retry
   ack_model: {{ack.model}} (ack_on_accept/ack_on_processed/UNKNOWN)
   retry_semantics_ref: {{ack.retry_semantics_ref}} (expected: {{xref:WHCP-04}}) | OPTIONAL
6. Failure Handling
   dlq_supported: {{fail.dlq_supported}}

dlq_trigger_rule: {{fail.dlq_trigger_rule}}
replay_policy_ref: {{fail.replay_policy_ref}} (expected: {{xref:WHCP-07}}) | OPTIONAL

7. Fraud / Abuse Hooks
   fraud_flags_handling: {{fraud.flags_handling}} | OPTIONAL
   velocity_limit_ref: {{fraud.velocity_limit_ref}} | OPTIONAL
8. Telemetry
   webhook_received_metric: {{telemetry.received_metric}}
   webhook_processed_metric: {{telemetry.processed_metric}} | OPTIONAL
   webhook_failed_metric: {{telemetry.failed_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
9. Audit
   audit_required: {{audit.required}}
   audit_fields: {{audit.fields}} | OPTIONAL
10. References
    Provider inventory: {{xref:PAY-01}}
    Payment flows: {{xref:PAY-02}} | OPTIONAL
    Ledger/reconciliation: {{xref:PAY-07}} | OPTIONAL
    Webhook consumer spec: {{xref:WHCP-03}} | OPTIONAL
    Webhook security: {{xref:WHCP-05}} | OPTIONAL
    Webhook error handling: {{xref:WHCP-07}} | OPTIONAL

Cross-References
Upstream: {{xref:PAY-01}}, {{xref:WHCP-05}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PAY-07}}, {{xref:PAY-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define consumed events + verification required + idempotency key rule.
intermediate: Required. Define state transitions, ack model, DLQ trigger, and telemetry.
advanced: Required. Add out-of-order handling, fraud hooks, and audit rigor for financial events.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, internal event id, notes, security ref, out of order rule, retry semantics ref, replay policy ref, fraud fields, telemetry fields, audit fields, test mode/tenant routing, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If endpoint.binding is UNKNOWN → block Completeness Gate.
If verify.required is UNKNOWN → block Completeness Gate.
If verify.idempotency_key_rule is UNKNOWN → block Completeness Gate.
If ack.model is UNKNOWN → block Completeness Gate.
If telemetry.received_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PAY

Pass conditions:
required_fields_present == true
events_defined == true
verification_and_idempotency_defined == true
ack_and_failure_handling_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PAY-05

PAY-05 — Tax/VAT Rules & Compliance (if applicable)
Header Block
template_id: PAY-05
title: Tax/VAT Rules & Compliance (if applicable)
type: payments_tax_vat_compliance
template_version: 1.0.0
output_path: 10_app/payments/PAY-05_Tax_VAT_Rules_Compliance.md
compliance_gate_id: TMP-05.PRIMARY.PAY
upstream_dependencies: ["PAY-03", "PRD-06", "IXS-08"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PAY-01", "PAY-03", "PRD-06", "IXS-08"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical tax/VAT handling rules for payments: whether taxes apply, how jurisdiction is determined, how taxes are calculated/collected/remitted (or delegated to provider), what user/business data is required, and compliance constraints. This template must not invent tax obligations; unknowns must be explicitly marked UNKNOWN.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PAY-01 Provider Inventory: {{pay.providers}} | OPTIONAL
PAY-03 Pricing & Plan Mapping: {{pay.plans}} | OPTIONAL
PRD-06 NFRs / Legal constraints: {{prd.nfrs}} | OPTIONAL
IXS-08 Integration Security & Compliance: {{ixs.security_compliance}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Tax handling enabled (true/false/UNKNOWN)
Tax model (provider-managed/manual/UNKNOWN)
Supported jurisdictions list (countries/states)
Jurisdiction determination rule (billing address, IP, etc.)
Tax-exempt handling rules (if supported)
Invoice/receipt tax line rules
Stored tax data fields policy (PII minimization)
Compliance notes (VAT IDs, W-9/W-8, etc. if applicable)
Audit/retention policy for tax records
Telemetry requirements (tax calculation failures)

Optional Fields
Product tax category mapping | OPTIONAL

EU OSS/IOSS notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not provide legal conclusions; use upstream/legal inputs.
Minimize stored tax PII; store only what is needed for compliance/audit.
If taxes are enabled, the model MUST be explicit and implementable.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Enablement
   tax_enabled: {{tax.enabled}}
   model: {{tax.model}} (provider_managed/manual/UNKNOWN)
2. Jurisdictions
   supported_jurisdictions: {{jurisdictions.supported}}
   determination_rule: {{jurisdictions.determination_rule}}
3. Tax Exemptions
   exempt_supported: {{exempt.supported}}
   exempt_rules: {{exempt.rules}} | OPTIONAL
4. Tax Lines on Invoices/Receipts
   tax_line_rule: {{lines.tax_line_rule}}
   rounding_rule: {{lines.rounding_rule}} | OPTIONAL
5. Data Handling
   tax_data_fields: {{data.tax_data_fields}}
   pii_minimization_rule: {{data.pii_minimization_rule}}
   retention_policy: {{data.retention_policy}}
6. Compliance Notes
   vat_id_supported: {{compliance.vat_id_supported}} | OPTIONAL
   documentation_required: {{compliance.documentation_required}} | OPTIONAL
   notes: {{compliance.notes}} | OPTIONAL
7. Audit / Telemetry
   audit_required: {{audit.required}}
   audit_fields: {{audit.fields}} | OPTIONAL
   tax_calc_failure_metric: {{telemetry.tax_calc_failure_metric}}
8. References
   Pricing/plans: {{xref:PAY-03}} | OPTIONAL
   Provider inventory: {{xref:PAY-01}} | OPTIONAL
   Security/compliance baseline: {{xref:IXS-08}} | OPTIONAL

Cross-References
Upstream: {{xref:PRD-06}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PAY-07}}, {{xref:PAY-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. State whether tax enabled and which jurisdictions; use UNKNOWN if legal inputs absent.
intermediate: Required. Define determination rule, retention policy, and telemetry metric.
advanced: Required. Add product tax categories and exemption workflows and audit fields rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, exempt rules, rounding rule, compliance fields, audit fields, category mapping, EU notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If tax.enabled is UNKNOWN → allowed, but MUST add to open_questions.
If tax.model is UNKNOWN and tax.enabled == true → block Completeness Gate.
If jurisdictions.determination_rule is UNKNOWN and tax.enabled == true → block Completeness Gate.
If telemetry.tax_calc_failure_metric is UNKNOWN → block Completeness Gate (when tax.enabled == true).

Completeness Gate
Gate ID: TMP-05.PRIMARY.PAY
Pass conditions:
required_fields_present == true
if tax_enabled then model_and_jurisdiction_rules_defined == true
data_handling_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PAY-06

PAY-06 — Refunds, Chargebacks & Disputes Policy
Header Block
template_id: PAY-06
title: Refunds, Chargebacks & Disputes Policy
type: payments_refunds_chargebacks_disputes
template_version: 1.0.0
output_path: 10_app/payments/PAY-06_Refunds_Chargebacks_Disputes_Policy.md
compliance_gate_id: TMP-05.PRIMARY.PAY
upstream_dependencies: ["PAY-02", "PAY-04", "ADMIN-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PAY-01", "PAY-02", "PAY-04", "API-04", "ADMIN-02", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical policy and operational workflows for refunds, chargebacks, and disputes: what is allowed, who can initiate actions, state transitions, provider interactions, evidence handling, timelines, and auditability. This template must be consistent with payment flows and webhook handling.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PAY-01 Provider Inventory: {{pay.providers}}
PAY-02 Payment Flows: {{pay.flows}} | OPTIONAL
PAY-04 Payment Webhook Handling: {{pay.webhooks}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
ADMIN-02 Support/Moderation Tools: {{admin.support_tools}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Refund eligibility rules (time window, product types)
Refund initiation roles (who can refund)
Refund types supported (full/partial)
Dispute/chargeback lifecycle states
Provider event handling mapping (dispute opened/won/lost)
Evidence collection rules (what stored, PII constraints)
Customer communication policy (who notifies)
Fraud escalation hooks (PAY-08)
Audit logging requirements (refund/dispute actions)
Telemetry requirements (refund rate, dispute rate)
SLA/timelines (response deadlines)

Optional Fields
Self-serve refunds | OPTIONAL
Goodwill credit policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
All refund/dispute actions MUST be permissioned and auditable.
Evidence storage must comply with PII rules; do not store more than necessary.
Dispute handling must be consistent with provider webhook events.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Refund Policy
   eligibility_rule: {{refund.eligibility_rule}}
   window_days: {{refund.window_days}} | OPTIONAL
   types_supported: {{refund.types_supported}} (full/partial/UNKNOWN)
   who_can_initiate: {{refund.who_can_initiate}}
2. Refund Workflow
   steps:
   {{refund.steps[0]}}
   {{refund.steps[1]}} | OPTIONAL
   provider_actions: {{refund.provider_actions}} | OPTIONAL
3. Disputes / Chargebacks Lifecycle
   states: {{dispute.states}}
   state_transitions: {{dispute.transitions}} | OPTIONAL
4. Provider Event Mapping
   event_map: {{dispute.provider_event_map}}
5. Evidence Handling
   evidence_required: {{evidence.required}}
   allowed_artifacts: {{evidence.allowed_artifacts}} | OPTIONAL
   pii_constraints: {{evidence.pii_constraints}}
6. Customer Communications
   notify_customer_rule: {{comms.notify_customer_rule}}
   comms_owner: {{comms.owner}} | OPTIONAL
7. Fraud Escalation
   escalate_to_fraud: {{fraud.escalate}}
   fraud_policy_ref: {{fraud.policy_ref}} (expected: {{xref:PAY-08}}) | OPTIONAL
8. Audit / Telemetry
   audit_required: {{audit.required}}
   audit_fields: {{audit.fields}} | OPTIONAL
   refund_rate_metric: {{telemetry.refund_rate_metric}}
   dispute_rate_metric: {{telemetry.dispute_rate_metric}} | OPTIONAL

9. Timelines
   response_deadline_rule: {{sla.response_deadline_rule}}
   refund_processing_time_rule: {{sla.refund_processing_time_rule}} | OPTIONAL
10. References
   Provider inventory: {{xref:PAY-01}}
   Payment flows: {{xref:PAY-02}} | OPTIONAL
   Webhook handling: {{xref:PAY-04}} | OPTIONAL
   Fraud controls: {{xref:PAY-08}} | OPTIONAL
   Admin tools: {{xref:ADMIN-02}} | OPTIONAL
   Audit trail: {{xref:ADMIN-03}} | OPTIONAL

Cross-References
Upstream: {{xref:PAY-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PAY-07}}, {{xref:PAY-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define eligibility + who can refund + basic dispute states; use UNKNOWN for self-serve.
intermediate: Required. Define provider event map, evidence rules, and audit/telemetry.
advanced: Required. Add SLAs/timelines rigor and fraud escalation and comms ownership.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, window days, workflow extra steps, provider actions, transitions, allowed artifacts, comms owner, fraud policy ref, audit fields, dispute rate metric, refund processing time, self-serve/goodwill, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If refund.eligibility_rule is UNKNOWN → block Completeness Gate.
If refund.who_can_initiate is UNKNOWN → block Completeness Gate.
If evidence.pii_constraints is UNKNOWN → block Completeness Gate.
If telemetry.refund_rate_metric is UNKNOWN → block Completeness Gate.
If audit.required is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PAY
Pass conditions:
required_fields_present == true
refund_policy_defined == true
dispute_lifecycle_defined == true
evidence_and_comms_defined == true
audit_and_telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PAY-07

PAY-07 — Ledger & Reconciliation Rules (source of truth, audits)
Header Block
template_id: PAY-07
title: Ledger & Reconciliation Rules (source of truth, audits)
type: payments_ledger_reconciliation_rules
template_version: 1.0.0
output_path: 10_app/payments/PAY-07_Ledger_Reconciliation_Rules.md
compliance_gate_id: TMP-05.PRIMARY.PAY
upstream_dependencies: ["PAY-02", "PAY-04", "DATA-06", "ADMIN-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PAY-01", "PAY-02", "PAY-04", "DATA-06", "DQV-03", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical internal ledger model and reconciliation rules for payments: what the source of truth is for financial state, how provider events map to ledger entries, how reconciliation is performed, what counts as mismatch, and what audits/controls exist. This template must be consistent with payment flows and webhook handling and must not invent financial fields beyond upstream schemas.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PAY-01 Provider Inventory: {{pay.providers}}
PAY-02 Payment Flow Spec: {{pay.flows}} | OPTIONAL
PAY-04 Webhook Handling: {{pay.webhooks}} | OPTIONAL
DATA-06 Canonical Data Schemas (ledger): {{data.schemas}} | OPTIONAL
DQV-03 Data Validation Rules: {{dqv.rules}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Ledger source of truth statement (provider vs internal)
Ledger entities/records (entry types)
Event → ledger mapping rules (provider events)
Idempotency rules for ledger writes
Reconciliation cadence (daily/hourly/UNKNOWN)
Reconciliation inputs (provider reports, API fetches)
Mismatch detection rules (amount, currency, status)
Mismatch handling (hold, alert, manual review)
Audit controls (append-only, approvals)

Retention policy for ledger records
Telemetry requirements (mismatch rate, reconciliation success)

Optional Fields
Multi-currency handling | OPTIONAL
Accounting export rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Ledger writes MUST be idempotent and append-only where possible.
Reconciliation must be repeatable and produce artifacts (reports).
Mismatches must not silently self-resolve; must be surfaced.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Source of Truth
   source_of_truth: {{sot.source_of_truth}} (provider/internal/UNKNOWN)
   rationale: {{sot.rationale}} | OPTIONAL
2. Ledger Model
   entry_types: {{ledger.entry_types}}
   schema_ref: {{ledger.schema_ref}} (expected: {{xref:DATA-06}}) | OPTIONAL
3. Event → Ledger Mapping
   Mapping
   provider_event_type: {{map[0].provider_event_type}}
   ledger_entry_type: {{map[0].ledger_entry_type}}
   idempotency_key: {{map[0].idempotency_key}}
   notes: {{map[0].notes}} | OPTIONAL

(Repeat per mapping.)

4. Idempotency
   idempotency_required: {{idem.required}}
   key_rule: {{idem.key_rule}}
5. Reconciliation Cadence
   cadence: {{recon.cadence}}
   inputs: {{recon.inputs}}
   report_artifact_rule: {{recon.report_artifact_rule}} | OPTIONAL
6. Mismatch Detection
   checks: {{mismatch.checks}}
   tolerance_rules: {{mismatch.tolerance_rules}} | OPTIONAL
7. Mismatch Handling
   on_mismatch_behavior: {{mismatch.on_mismatch_behavior}}
   manual_review_required: {{mismatch.manual_review_required}} | OPTIONAL

8. Audit Controls
   append_only_rule: {{audit.append_only_rule}}
   approval_required_for_adjustments: {{audit.approval_required_for_adjustments}} | OPTIONAL
   audit_ref: {{audit.audit_ref}} (expected: {{xref:ADMIN-03}}) | OPTIONAL
9. Retention
   retention_policy: {{retention.policy}}
10. Telemetry
    recon_success_metric: {{telemetry.recon_success_metric}}
    mismatch_rate_metric: {{telemetry.mismatch_rate_metric}} | OPTIONAL
    fields: {{telemetry.fields}} | OPTIONAL
11. References
    Payment flows: {{xref:PAY-02}} | OPTIONAL
    Webhook handling: {{xref:PAY-04}} | OPTIONAL
    Refunds/disputes: {{xref:PAY-06}} | OPTIONAL
    Audit trail: {{xref:ADMIN-03}} | OPTIONAL

Cross-References
Upstream: {{xref:PAY-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PAY-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define source of truth, entry types, and cadence and mismatch behavior.
intermediate: Required. Define event mapping, idempotency, and audit controls.
advanced: Required. Add exports/multi-currency and strict artifact/report definitions.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, rationale, schema ref, notes, report artifact rule, tolerance rules, manual review, approvals, audit ref, mismatch rate metric/fields, currency/export rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If sot.source_of_truth is UNKNOWN → block Completeness Gate.
If ledger.entry_types is UNKNOWN → block Completeness Gate.
If idem.key_rule is UNKNOWN → block Completeness Gate.
If recon.cadence is UNKNOWN → block Completeness Gate.
If telemetry.recon_success_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PAY
Pass conditions:
required_fields_present == true
source_of_truth_defined == true
ledger_model_defined == true

idempotency_defined == true
reconciliation_defined == true
audit_controls_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PAY-08

PAY-08 — Fraud & Abuse Controls (risk checks, velocity limits)
Header Block
template_id: PAY-08
title: Fraud & Abuse Controls (risk checks, velocity limits)
type: payments_fraud_abuse_controls
template_version: 1.0.0
output_path: 10_app/payments/PAY-08_Fraud_Abuse_Controls.md
compliance_gate_id: TMP-05.PRIMARY.PAY
upstream_dependencies: ["PAY-02", "RLIM-03", "RLIM-04", "ADMIN-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PAY-01", "PAY-02", "PAY-04", "RLIM-03", "RLIM-04", "FFCFG-01", "ADMIN-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical fraud and abuse controls for payments: risk checks before/after charges, velocity limits, blocklists/allowlists, manual review workflows, and enforcement actions. This template must be consistent with general abuse detection/enforcement matrices and must not invent enforcement capabilities not present upstream.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PAY-01 Provider Inventory: {{pay.providers}}
PAY-02 Payment Flow Spec: {{pay.flows}} | OPTIONAL
PAY-04 Webhook Handling: {{pay.webhooks}} | OPTIONAL
RLIM-03 Abuse Signals & Detection: {{rlim.signals}} | OPTIONAL
RLIM-04 Enforcement Actions Matrix: {{rlim.actions}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.flags}} | OPTIONAL
ADMIN-02 Support Tools Spec: {{admin.support_tools}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Risk checks list (pre-charge/post-charge)
Velocity limits (per user/card/ip)
Provider risk signals used (AVS/CVV/3DS) if applicable
Decision outcomes (allow/deny/review)
Manual review workflow (queue, actions)
Enforcement actions (throttle/ban/disable payments)
Kill switch rules (global payments off)
False positive handling (override)
Telemetry requirements (fraud blocks, review rate)
Audit logging requirements (overrides/blocks)

Optional Fields
3DS policy | OPTIONAL
Machine risk scoring notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Fraud decisions must be explainable via reason codes (where possible).
Overrides must be permissioned and auditable.
Kill switch must exist for incident response.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Risk Checks
   pre_charge_checks: {{risk.pre_charge_checks}}
   post_charge_checks: {{risk.post_charge_checks}} | OPTIONAL
2. Velocity Limits
   limits: {{velocity.limits}}
   window_rule: {{velocity.window_rule}} | OPTIONAL
3. Provider Signals
   signals_used: {{signals.used}} | OPTIONAL
   3ds_policy: {{signals.threeds_policy}} | OPTIONAL
4. Decision Outcomes
   outcomes: {{decision.outcomes}} (allow/deny/review/UNKNOWN)
   reason_code_policy: {{decision.reason_code_policy}} | OPTIONAL
5. Manual Review
   review_supported: {{review.supported}}
   review_queue: {{review.queue}} | OPTIONAL
   review_actions: {{review.actions}} | OPTIONAL
6. Enforcement
   enforcement_ref: {{enforce.ref}} (expected: {{xref:RLIM-04}}) | OPTIONAL
   payment_disable_rule: {{enforce.payment_disable_rule}}
   override_allowed: {{enforce.override_allowed}} | OPTIONAL
7. Kill Switch
   kill_switch_flag_ref: {{kill.flag_ref}} (expected: {{xref:FFCFG-01}}) | OPTIONAL
   kill_switch_behavior: {{kill.behavior}}
8. False Positives
   override_process: {{fp.override_process}} | OPTIONAL
   appeals_process: {{fp.appeals_process}} | OPTIONAL
9. Telemetry
   fraud_block_metric: {{telemetry.fraud_block_metric}}
   review_rate_metric: {{telemetry.review_rate_metric}} | OPTIONAL
   override_metric: {{telemetry.override_metric}} | OPTIONAL

10. Audit
    audit_required: {{audit.required}}
    audit_fields: {{audit.fields}} | OPTIONAL
11. References
    Payment flows: {{xref:PAY-02}} | OPTIONAL
    Webhook handling: {{xref:PAY-04}} | OPTIONAL
    Abuse signals: {{xref:RLIM-03}} | OPTIONAL
    Enforcement matrix: {{xref:RLIM-04}} | OPTIONAL
    Feature flags: {{xref:FFCFG-01}} | OPTIONAL
    Support tools: {{xref:ADMIN-02}} | OPTIONAL

Cross-References
Upstream: {{xref:PAY-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PAY-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define velocity limits, decision outcomes, and kill switch behavior.
intermediate: Required. Define manual review workflow and enforcement rules and telemetry.
advanced: Required. Add 3DS policy, risk scoring notes, and override/appeals rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, post-charge checks, window rule, provider signals, 3ds policy, reason code policy, review queue/actions, enforcement ref, override allowed, kill switch flag ref, false positive processes, optional metrics, audit fields, risk scoring, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If velocity.limits is UNKNOWN → block Completeness Gate.
If decision.outcomes is UNKNOWN → block Completeness Gate.
If enforce.payment_disable_rule is UNKNOWN → block Completeness Gate.
If kill.behavior is UNKNOWN → block Completeness Gate.
If telemetry.fraud_block_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PAY
Pass conditions:
required_fields_present == true
risk_and_velocity_defined == true
enforcement_and_kill_switch_defined == true
telemetry_defined == true
audit_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PAY-09

PAY-09 — Security & PCI Boundaries (tokenization, SAQ scope)
Header Block
template_id: PAY-09
title: Security & PCI Boundaries (tokenization, SAQ scope)
type: payments_security_pci_boundaries
template_version: 1.0.0
output_path: 10_app/payments/PAY-09_Security_PCI_Boundaries.md
compliance_gate_id: TMP-05.PRIMARY.PAY
upstream_dependencies: ["PAY-01", "IXS-08", "CSec-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PAY-01", "PAY-02", "IXS-08", "CSec-02", "CSec-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical security boundaries for payments, especially PCI scope: what systems touch payment data, tokenization rules, what is prohibited (raw PAN/CVV), required controls, and how SAQ scope is determined/maintained. This template must be consistent with integration security/compliance and client data protection policies.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PAY-01 Provider Inventory: {{pay.providers}}
PAY-02 Payment Flow Spec: {{pay.flows}} | OPTIONAL
IXS-08 Integration Security & Compliance: {{ixs.security_compliance}} | OPTIONAL
CSec-02 Client Data Protection: {{csec.data_protection}} | OPTIONAL
CSec-01 Token Storage Policy: {{csec.token_storage}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
PCI in scope (true/false/UNKNOWN)
PCI scope boundary statement (what is in/out)
Tokenization rule (provider-hosted fields, tokens only)
Prohibited data rule (no PAN/CVV storage/logging)
Allowed payment identifiers (last4, brand, token ids)
Network and transport requirements (TLS)
Secrets handling reference (IXS-04)
Access control requirements (least privilege)
Logging/redaction requirements (CER-05/CSec-02)
Vulnerability/patch expectations (if in scope)
Telemetry requirements (security events)

Optional Fields
SAQ type notes | OPTIONAL
Pen test / ASV scan notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Never store raw PAN or CVV.
If PCI is in scope, scope boundaries and controls MUST be explicit and enforceable.
Payment identifiers logged must be non-sensitive (token/last4 only).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. PCI Scope
   pci_in_scope: {{pci.in_scope}}
   scope_boundary: {{pci.boundary}}
   saq_type: {{pci.saq_type}} | OPTIONAL
2. Tokenization
   tokenization_rule: {{token.rule}}
   provider_hosted_fields: {{token.provider_hosted_fields}} | OPTIONAL
3. Prohibited Data
   no_pan_rule: {{ban.no_pan_rule}}
   no_cvv_rule: {{ban.no_cvv_rule}}
   no_sensitive_auth_data_logs: {{ban.no_sensitive_auth_data_logs}} | OPTIONAL
4. Allowed Identifiers
   allowed_fields: {{allow.allowed_fields}}
   redaction_rule: {{allow.redaction_rule}} | OPTIONAL
5. Transport / Network
   tls_required: {{net.tls_required}}
   network_allowlist_ref: {{net.allowlist_ref}} (expected: {{xref:IXS-03}}) | OPTIONAL
6. Secrets & Access
   secrets_ref: {{sec.secrets_ref}} (expected: {{xref:IXS-04}}) | OPTIONAL
   least_privilege_rule: {{sec.least_privilege_rule}}
7. Logging / Redaction
   no_payment_data_in_logs: {{logs.no_payment_data_in_logs}}
   client_data_protection_ref: {{logs.client_data_protection_ref}} (expected: {{xref:CSec-02}}) | OPTIONAL
8. Vulnerability Expectations
   patch_policy: {{vuln.patch_policy}} | OPTIONAL
   scan_policy: {{vuln.scan_policy}} | OPTIONAL
9. Telemetry
   security_event_metric: {{telemetry.security_event_metric}}
   fields: {{telemetry.fields}} | OPTIONAL

10. References
    Provider inventory: {{xref:PAY-01}}
    Payment flows: {{xref:PAY-02}} | OPTIONAL
    Integration security baseline: {{xref:IXS-08}} | OPTIONAL
    Token storage: {{xref:CSec-01}} | OPTIONAL
    Client data protection: {{xref:CSec-02}} | OPTIONAL

Cross-References
Upstream: {{xref:PAY-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PAY-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define prohibited data rules + tokenization + pci_in_scope flag.
intermediate: Required. Define scope boundary + logging/redaction + least privilege.
advanced: Required. Add SAQ/pentest/scan notes and strict telemetry fields.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, saq type, hosted fields, optional ban/log rules, redaction rule, allowlist ref, secrets ref, vuln policies, telemetry fields, pen test notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If pci.in_scope is UNKNOWN → allowed, but MUST be flagged in open_questions.
If pci.in_scope == true and pci.boundary is UNKNOWN → block Completeness Gate.
If ban.no_pan_rule is UNKNOWN → block Completeness Gate.
If token.rule is UNKNOWN → block Completeness Gate.
If telemetry.security_event_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PAY
Pass conditions:
required_fields_present == true
tokenization_and_prohibited_data_defined == true
pci_scope_defined_or_not_in_scope == true
least_privilege_and_logging_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PAY-10

PAY-10 — Billing Observability & Support Runbooks (alerts, workflows)
Header Block
template_id: PAY-10
title: Billing Observability & Support Runbooks (alerts, workflows)
type: payments_billing_observability_support_runbooks
template_version: 1.0.0
output_path: 10_app/payments/PAY-10_Billing_Observability_Support_Runbooks.md
compliance_gate_id: TMP-05.PRIMARY.PAY
upstream_dependencies: ["PAY-04", "PAY-06", "PAY-07", "IXS-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PAY-01", "PAY-04", "PAY-06", "PAY-07", "IXS-07", "ADMIN-02", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical observability and support runbooks for billing/payments: dashboards, alerts, triage workflows, operator actions (refunds, dispute handling, entitlement fixes), and incident response procedures. This template must be consistent with payment webhook, ledger, and dispute policies and must not introduce unsafe support actions.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PAY-01 Provider Inventory: {{pay.providers}}
PAY-04 Payment Webhook Handling: {{pay.webhooks}} | OPTIONAL
PAY-06 Refunds/Disputes Policy: {{pay.disputes}} | OPTIONAL
PAY-07 Ledger/Reconciliation Rules: {{pay.ledger}} | OPTIONAL
IXS-07 Integration Observability: {{ixs.observability}} | OPTIONAL
ADMIN-02 Support Tools Spec: {{admin.support_tools}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Dashboards minimum panels (payments health)
Key metrics list (success rate, declines, webhook failures, mismatches)
Alert definitions (payment failure spike, webhook processing lag, mismatch spike)
Alert routing policy (oncall/support)
Runbook location and format
Triage flows (by incident type)
Operator actions list (refund, entitlement restore, replay webhook)
Permissioning requirements for actions (who can do what)
Audit requirements (all financial actions logged)

Customer comms policy (billing incidents)
Post-incident review requirements

Optional Fields
Per-provider dashboards | OPTIONAL
SLOs for billing pipeline | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Operator actions affecting money or entitlements MUST be permissioned and auditable.
Alerts must be actionable and link to runbook steps.
Do not expose sensitive payment details in logs/dashboards.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Dashboards
   dashboard_name: {{dash.name}}
   minimum_panels: {{dash.minimum_panels}}
   panel_list: {{dash.panel_list}} | OPTIONAL
2. Key Metrics
   metrics: {{metrics.list}}
3. Alerts
   Alert
   alert_id: {{alerts[0].alert_id}}
   condition: {{alerts[0].condition}}
   severity: {{alerts[0].severity}} (low/med/high/UNKNOWN)
   routing: {{alerts[0].routing}}
   runbook_ref: {{alerts[0].runbook_ref}} | OPTIONAL

(Repeat per alert.)

4. Runbooks
   runbook_location: {{runbooks.location}}
   format: {{runbooks.format}} (md/wiki/UNKNOWN) | OPTIONAL
5. Triage Flows
   Flow
   incident_type: {{triage[0].incident_type}}
   steps:
   {{triage[0].steps[0]}}
   {{triage[0].steps[1]}} | OPTIONAL
   primary_tools: {{triage[0].primary_tools}} | OPTIONAL

(Repeat per incident type.)

6. Operator Actions
   Action
   action_id: {{ops[0].action_id}}
   name: {{ops[0].name}}
   when_allowed: {{ops[0].when_allowed}}
   required_role: {{ops[0].required_role}}
   audit_required: {{ops[0].audit_required}}

(Repeat per action.)

7. Permissioning
   permissions_ref: {{perm.ref}} (expected: {{xref:API-04}}/{{xref:ADMIN-01}}) | OPTIONAL
8. Audit
   audit_required: {{audit.required}}
   audit_events: {{audit.events}}
   audit_fields: {{audit.fields}} | OPTIONAL
9. Customer Comms
   comms_policy: {{comms.policy}}
   comms_owner: {{comms.owner}} | OPTIONAL
10. Post-Incident
    rca_required: {{post.rca_required}}
    rca_location: {{post.rca_location}} | OPTIONAL
11. References
    Payment webhooks: {{xref:PAY-04}} | OPTIONAL
    Refunds/disputes: {{xref:PAY-06}} | OPTIONAL
    Ledger/reconciliation: {{xref:PAY-07}} | OPTIONAL
    Fraud controls: {{xref:PAY-08}} | OPTIONAL
    Support tools: {{xref:ADMIN-02}} | OPTIONAL
    Audit trail: {{xref:ADMIN-03}} | OPTIONAL

Cross-References
Upstream: {{xref:PAY-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define dashboards + metrics + basic triage steps.
intermediate: Required. Define alerts, operator actions, and audit/comms policies.
advanced: Required. Add per-provider SLOs and refined incident-type triage workflows.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, panel list, optional tools, runbook format, permissions ref, audit fields, comms owner, rca location, per-provider dashboards, SLOs, open_questions
If any Required Field is UNKNOWN, allow only if:

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If dash.minimum_panels is UNKNOWN → block Completeness Gate.
If metrics.list is UNKNOWN → block Completeness Gate.
If alerts list is UNKNOWN → block Completeness Gate.
If ops list is UNKNOWN → block Completeness Gate.
If audit.required is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PAY
Pass conditions:
required_fields_present == true
dashboards_and_alerts_defined == true
triage_and_operator_actions_defined == true
audit_and_comms_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Notifications & Comms (NOTIF)

NOTIF-01

NOTIF-01 — Channel Inventory (email/sms/push/in-app)
Header Block
template_id: NOTIF-01
title: Channel Inventory (email/sms/push/in-app)
type: notifications_channel_inventory
template_version: 1.0.0
output_path: 10_app/notifications/NOTIF-01_Channel_Inventory.md
compliance_gate_id: TMP-05.PRIMARY.NOTIF
upstream_dependencies: ["MPUSH-01", "IXS-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "MPUSH-01", "MPUSH-02", "EVT-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical inventory of notification channels used by the product (email/SMS/push/in-app), including what each channel is used for, constraints, and which downstream templates govern each channel. This document must not invent channel types beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}} | OPTIONAL
MPUSH-01 Notification Types Catalog: {{mpush.types}} | OPTIONAL
MPUSH-02 Push Payload Contract: {{mpush.payload_contract}} | OPTIONAL
EVT-01 Event Catalog: {{evt.catalog}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Channel registry (channel_id list)
channel_id (email/sms/push/in_app)
channel purpose (what it’s for)
supported message types (transactional/marketing/UNKNOWN)
delivery constraints (max length, attachments)
PII/compliance constraints (consent)
quiet hours applicability
opt-out model (global/per-type)
provider dependency (provider_id ref if any)
telemetry requirements (send/deliver/fail)
references to channel-specific specs (NOTIF-03..10 and MPUSH docs for push)

Optional Fields
Fallback channel rules | OPTIONAL
Regional restrictions | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent channel_ids; use the allowed set {email, sms, push, in_app}.
Each channel must define opt-out + consent handling for message types that require it.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Inventory Summary
   total_channels: {{inv.total}}
   notes: {{inv.notes}} | OPTIONAL
2. Channel Entries (by channel_id)
   Channel
   channel_id: {{channels[0].channel_id}}
   purpose: {{channels[0].purpose}}
   message_types: {{channels[0].message_types}}
   constraints: {{channels[0].constraints}}
   compliance_constraints: {{channels[0].compliance_constraints}}
   quiet_hours_applicable: {{channels[0].quiet_hours_applicable}}
   opt_out_model: {{channels[0].opt_out_model}}
   provider_ref: {{channels[0].provider_ref}} | OPTIONAL
   telemetry:
   send_metric: {{channels[0].telemetry.send_metric}}
   deliver_metric: {{channels[0].telemetry.deliver_metric}} | OPTIONAL
   fail_metric: {{channels[0].telemetry.fail_metric}} | OPTIONAL
   spec_refs:
- {{channels[0].spec_refs[0]}} (e.g., {{xref:NOTIF-04}} / {{xref:NOTIF-06}} / {{xref:MPUSH-01}})
- {{channels[0].spec_refs[1]}} | OPTIONAL
  open_questions:
  {{channels[0].open_questions[0]}} | OPTIONAL

(Repeat per channel_id.)

3. References
   Provider inventory: {{xref:NOTIF-02}} | OPTIONAL
   Template/localization mapping: {{xref:NOTIF-03}} | OPTIONAL
   Send policy: {{xref:NOTIF-04}} | OPTIONAL
   Deliverability: {{xref:NOTIF-05}} | OPTIONAL
   Preference center: {{xref:NOTIF-06}} | OPTIONAL

Security/compliance: {{xref:NOTIF-07}} | OPTIONAL
Event→notification mapping: {{xref:NOTIF-08}} | OPTIONAL
Failure handling: {{xref:NOTIF-09}} | OPTIONAL
Observability/runbooks: {{xref:NOTIF-10}} | OPTIONAL
Push catalog: {{xref:MPUSH-01}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:NOTIF-02}}, {{xref:NOTIF-04}}, {{xref:NOTIF-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define channels and purposes and opt-out model; no invention.
intermediate: Required. Add constraints/compliance and telemetry and spec refs.
advanced: Required. Add fallback/regional restrictions and traceability to provider/event IDs.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, inv notes, provider refs, optional telemetry metrics, secondary spec refs, fallback/regional restrictions, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If channels[*].channel_id is UNKNOWN → block Completeness Gate.*
*If channels[*].opt_out_model is UNKNOWN → block Completeness Gate.
If channels[*].telemetry.send_metric is UNKNOWN → block Completeness Gate.*
*If channels[*].spec_refs is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.NOTIF
Pass conditions:
required_fields_present == true
channel_ids_valid == true
opt_out_models_defined == true
telemetry_defined == true
spec_refs_present == true
placeholder_resolution == true
no_unapproved_unknowns == true

NOTIF-02

NOTIF-02 — Provider Inventory (by provider_id)
Header Block
template_id: NOTIF-02
title: Provider Inventory (by provider_id)
type: notifications_provider_inventory
template_version: 1.0.0
output_path: 10_app/notifications/NOTIF-02_Provider_Inventory.md
compliance_gate_id: TMP-05.PRIMARY.NOTIF
upstream_dependencies: ["NOTIF-01", "IXS-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "NOTIF-01", "IXS-01", "IXS-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical inventory of notification/comms providers (email/SMS/push/in-app vendors), indexed by provider_id, including which channels each provider supports, enabled environments, credential references, and operational constraints. This document must not invent provider_ids beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
NOTIF-01 Channel Inventory: {{notif.channels}}
IXS-01 Integration Inventory: {{ixs.inventory}} | OPTIONAL
IXS-04 Secrets/Credential Handling: {{ixs.secrets_policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Provider registry (provider_id list)
provider_id (stable identifier)
provider name/vendor
supported channels (email/sms/push/in_app)
integration_id binding (IXS-01)
credentials reference (IXS-04)
environments enabled (dev/stage/prod)
rate limits/quotas summary
deliverability constraints (email) or carrier constraints (sms) if applicable
criticality (low/med/high)
owner (team/role)
references to detailed specs (NOTIF-03..10 and MPUSH for push)

Optional Fields
Status (planned/active/deprecated) | OPTIONAL
SLA/uptime notes | OPTIONAL
Regional availability notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent provider_ids; use only {{spec.notif_providers_by_id}} if present, else mark UNKNOWN and flag.
Each provider must bind to an integration_id and have a credential reference (or approved UNKNOWN).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Inventory Summary
   total_providers: {{inv.total}}
   channels_covered: {{inv.channels_covered}} | OPTIONAL
   notes: {{inv.notes}} | OPTIONAL
2. Provider Entries (by provider_id)
   Provider
   provider_id: {{providers[0].provider_id}}
   name: {{providers[0].name}}
   supported_channels: {{providers[0].supported_channels}}
   integration_id: {{providers[0].integration_id}}
   credential_ref: {{providers[0].credential_ref}} (expected: {{xref:IXS-04}}) | OPTIONAL
   env_enabled: {{providers[0].env_enabled}}
   rate_limits: {{providers[0].rate_limits}}
   constraints: {{providers[0].constraints}} | OPTIONAL
   criticality: {{providers[0].criticality}}
   owner: {{providers[0].owner}}
   spec_refs:

- {{providers[0].spec_refs[0]}} (e.g., {{xref:NOTIF-04}} / {{xref:NOTIF-05}} / {{xref:MPUSH-04}})
- {{providers[0].spec_refs[1]}} | OPTIONAL
  status: {{providers[0].status}} | OPTIONAL
  sla_notes: {{providers[0].sla_notes}} | OPTIONAL
  region_notes: {{providers[0].region_notes}} | OPTIONAL
  open_questions:
  {{providers[0].open_questions[0]}} | OPTIONAL

(Repeat per provider_id.)

3. References
   Template/localization mapping: {{xref:NOTIF-03}} | OPTIONAL
   Send policy: {{xref:NOTIF-04}} | OPTIONAL
   Deliverability: {{xref:NOTIF-05}} | OPTIONAL
   Preference center: {{xref:NOTIF-06}} | OPTIONAL
   Security/compliance: {{xref:NOTIF-07}} | OPTIONAL
   Event mapping: {{xref:NOTIF-08}} | OPTIONAL
   Failure handling: {{xref:NOTIF-09}} | OPTIONAL
   Observability/runbooks: {{xref:NOTIF-10}} | OPTIONAL

Cross-References
Upstream: {{xref:NOTIF-01}}, {{xref:IXS-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:NOTIF-04}}, {{xref:NOTIF-05}}, {{xref:NOTIF-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define providers and supported channels and env enablement.
intermediate: Required. Add credentials refs, rate limits, and spec refs.
advanced: Required. Add deliverability constraints, regional notes, and criticality/ownership rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, inv notes, credential ref, constraints, secondary spec refs, status/sla/region notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If providers[*].provider_id is UNKNOWN → block Completeness Gate.*
*If providers[*].integration_id is UNKNOWN → block Completeness Gate.
If providers[*].supported_channels is UNKNOWN → block Completeness Gate.*
*If providers[*].rate_limits is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.NOTIF
Pass conditions:
required_fields_present == true
provider_ids_unique == true
integration_ids_exist_in_IXS_01 == true
channels_supported_valid == true
placeholder_resolution == true
no_unapproved_unknowns == true

NOTIF-03

NOTIF-03 — Template & Localization Mapping (template_id, variables)
Header Block
template_id: NOTIF-03
title: Template & Localization Mapping (template_id, variables)
type: notifications_template_localization_mapping
template_version: 1.0.0
output_path: 10_app/notifications/NOTIF-03_Template_Localization_Mapping.md
compliance_gate_id: TMP-05.PRIMARY.NOTIF
upstream_dependencies: ["NOTIF-01", "NOTIF-02", "MPUSH-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "NOTIF-01", "NOTIF-02", "MPUSH-02", "FFCFG-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical mapping from notification types to message templates and localization assets: template identifiers per channel/provider, required variables, default language, fallback behavior, and validation rules. This template must be consistent with payload contracts and must not invent template IDs beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
NOTIF-01 Channel Inventory: {{notif.channels}}
NOTIF-02 Provider Inventory: {{notif.providers}} | OPTIONAL
MPUSH-02 Payload Contract (push copy keys): {{mpush.payload_contract}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.flags}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Template registry (template_id list)
template_id (stable identifier)
notification type binding (notif_id or internal notification name)
channel binding (email/sms/push/in_app)
provider binding (provider_id)
template reference (provider template key/id)
required variables list
localization keys (title/body keys or template keys)
supported locales list
fallback locale rule
missing variable handling rule
template validation rule (pre-send checks)
telemetry requirements (template render failures)

Optional Fields
A/B variant support | OPTIONAL
Per-tenant branding overrides | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent template_ids; use upstream registry if present, else mark UNKNOWN and flag.
Variables must be explicit; no “dynamic” without a list.
Missing required variables MUST fail or use explicit defaults; no silent blanks.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Template Registry Summary
   total_templates: {{templates.total}}
   channels_covered: {{templates.channels_covered}} | OPTIONAL
   notes: {{templates.notes}} | OPTIONAL
2. Template Entries (by template_id)
   Template
   template_id: {{items[0].template_id}}
   notif_binding: {{items[0].notif_binding}}
   channel: {{items[0].channel}}
   provider_id: {{items[0].provider_id}}
   provider_template_ref: {{items[0].provider_template_ref}}
   required_variables: {{items[0].required_variables}}
   localization:
   supported_locales: {{items[0].locales.supported}}
   default_locale: {{items[0].locales.default}}
   fallback_locale_rule: {{items[0].locales.fallback_rule}}
   missing_var_behavior: {{items[0].missing_var_behavior}}
   validation_rule: {{items[0].validation_rule}}
   ab_variant: {{items[0].ab_variant}} | OPTIONAL
   branding_override: {{items[0].branding_override}} | OPTIONAL
   open_questions:
   {{items[0].open_questions[0]}} | OPTIONAL

(Repeat per template_id.)

3. Telemetry
   render_failure_metric: {{telemetry.render_failure_metric}}
   fields: {{telemetry.fields}} | OPTIONAL
4. References
   Channel inventory: {{xref:NOTIF-01}}
   Provider inventory: {{xref:NOTIF-02}} | OPTIONAL

Push payload/copy keys: {{xref:MPUSH-02}} | OPTIONAL
Send policy: {{xref:NOTIF-04}} | OPTIONAL

Cross-References
Upstream: {{xref:NOTIF-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:NOTIF-04}}, {{xref:NOTIF-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define template entries and required variables and fallback locale rule.
intermediate: Required. Define validation rule and missing var behavior and render failure metric.
advanced: Required. Add A/B variants and branding overrides with strict traceability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, channels covered, ab variant, branding override, telemetry fields, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If items[*].template_id is UNKNOWN → block Completeness Gate.
If items[*].required_variables is UNKNOWN → block Completeness Gate.
If items[*].locales.fallback_rule is UNKNOWN → block Completeness Gate.
If telemetry.render_failure_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.NOTIF
Pass conditions:
required_fields_present == true
template_ids_unique == true
channel_and_provider_bindings_valid == true
variable_lists_defined == true
localization_fallback_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

NOTIF-04

NOTIF-04 — Send Policy (throttles, quiet hours, batching)
Header Block
template_id: NOTIF-04
title: Send Policy (throttles, quiet hours, batching)
type: notifications_send_policy
template_version: 1.0.0
output_path: 10_app/notifications/NOTIF-04_Send_Policy.md
compliance_gate_id: TMP-05.PRIMARY.NOTIF
upstream_dependencies: ["NOTIF-01", "RLIM-01", "MPUSH-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "NOTIF-01", "NOTIF-02", "RLIM-01", "RLIM-02", "MPUSH-04", "FFCFG-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical send policy for notifications across channels: throttling rules, batching, quiet hours behavior, priority classes, and safeguards against notification storms. This template must be consistent with rate limit governance and push delivery rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
NOTIF-01 Channel Inventory: {{notif.channels}}
NOTIF-02 Provider Inventory: {{notif.providers}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}} | OPTIONAL
RLIM-02 Rate Limit Catalog: {{rlim.catalog}} | OPTIONAL
MPUSH-04 Push Delivery/Retry Rules: {{mpush.delivery_rules}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.flags}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Priority classes (low/normal/high)
Per-channel throttle rules (limits by time window)
Per-user caps (avoid spam)
Quiet hours supported (yes/no/UNKNOWN)
Quiet hours behavior (delay/drop)
Batching supported (yes/no/UNKNOWN)
Batching rules (grouping window)
Storm prevention rules (global caps, kill switch)
Rate limit enforcement reference (RLIM)
Telemetry requirements (throttled, delayed, dropped)

Optional Fields
Per-notif-type overrides | OPTIONAL
Regional quiet hours | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Storm prevention must exist and be quickly actionable.
Quiet hours behavior must be deterministic and user-respecting.
If batching is enabled, batching must not merge semantically different notifications unless explicitly allowed.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Priority Classes
   priorities: {{priority.classes}}
2. Per-Channel Throttles
   Channel Throttle
   channel_id: {{throttle[0].channel_id}}
   limits: {{throttle[0].limits}}
   per_user_caps: {{throttle[0].per_user_caps}} | OPTIONAL
   notes: {{throttle[0].notes}} | OPTIONAL

(Repeat per channel.)

3. Quiet Hours
   supported: {{quiet.supported}}
   behavior: {{quiet.behavior}} (delay/drop/UNKNOWN)
   window_rule: {{quiet.window_rule}} | OPTIONAL
   regional_support: {{quiet.regional_support}} | OPTIONAL
4. Batching
   supported: {{batch.supported}}
   grouping_window_seconds: {{batch.grouping_window_seconds}} | OPTIONAL
   grouping_key_rule: {{batch.grouping_key_rule}} | OPTIONAL
5. Storm Prevention
   global_caps: {{storm.global_caps}}
   kill_switch_flag_ref: {{storm.kill_switch_flag_ref}} (expected: {{xref:FFCFG-01}}) | OPTIONAL
   incident_mode_behavior: {{storm.incident_mode_behavior}} | OPTIONAL
6. Enforcement
   rate_limit_ref: {{enforce.rate_limit_ref}} (expected: {{xref:RLIM-01}}/{{xref:RLIM-02}}) | OPTIONAL
   enforcement_point: {{enforce.enforcement_point}} (queue/worker/api/UNKNOWN)

7. Telemetry
   throttled_metric: {{telemetry.throttled_metric}}
   delayed_metric: {{telemetry.delayed_metric}} | OPTIONAL
   dropped_metric: {{telemetry.dropped_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
8. References
   Channel inventory: {{xref:NOTIF-01}}
   Provider inventory: {{xref:NOTIF-02}} | OPTIONAL
   Rate limit policy: {{xref:RLIM-01}} | OPTIONAL
   Push delivery rules: {{xref:MPUSH-04}} | OPTIONAL
   Abuse controls: {{xref:MPUSH-06}} | OPTIONAL

Cross-References
Upstream: {{xref:NOTIF-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:NOTIF-05}}, {{xref:NOTIF-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define per-channel throttles and storm caps and throttled metric.
intermediate: Required. Define quiet hours and batching and enforcement point.
advanced: Required. Add overrides/regional quiet hours and kill switch rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, per-user caps, quiet window/regional support, batching details, kill switch ref, incident behavior, rate limit ref, delayed/dropped metrics, telemetry fields, overrides, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If priority.classes is UNKNOWN → block Completeness Gate.
If storm.global_caps is UNKNOWN → block Completeness Gate.
If enforce.enforcement_point is UNKNOWN → block Completeness Gate.
If telemetry.throttled_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.NOTIF
Pass conditions:
required_fields_present == true
throttles_defined == true
storm_prevention_defined == true
enforcement_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

NOTIF-05

NOTIF-05 — Deliverability Rules (bounces, spam, suppression lists)
Header Block
template_id: NOTIF-05
title: Deliverability Rules (bounces, spam, suppression lists)
type: notifications_deliverability_rules
template_version: 1.0.0
output_path: 10_app/notifications/NOTIF-05_Deliverability_Rules.md
compliance_gate_id: TMP-05.PRIMARY.NOTIF
upstream_dependencies: ["NOTIF-02", "WHCP-01", "NOTIF-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "NOTIF-01", "NOTIF-02", "NOTIF-04", "WHCP-01", "WHCP-03", "RLIM-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical deliverability controls for notifications (primarily email/SMS): bounce handling, spam complaints, suppression lists, sender reputation safeguards, and re-enable policies. This template must be consistent with provider setup and send policy, and must not invent deliverability capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
NOTIF-01 Channel Inventory: {{notif.channels}}
NOTIF-02 Provider Inventory: {{notif.providers}} | OPTIONAL
NOTIF-04 Send Policy: {{notif.send_policy}} | OPTIONAL
WHCP-01 Webhook Catalog (bounce/complaint events): {{whcp.catalog}} | OPTIONAL
WHCP-03 Webhook Consumer Spec: {{whcp.inbound}} | OPTIONAL
RLIM-03 Abuse Signals & Detection: {{rlim.signals}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Supported deliverability signals list (hard bounce, soft bounce, complaint)
Signal ingestion method (webhook/polling/UNKNOWN)
Suppression list model (global/per-channel)
Suppression triggers (what adds to suppression)
Suppression duration policy (permanent/ttl/UNKNOWN)
Re-enable policy (how user can be re-enabled)
Sender identity policy (from domains/numbers)
Content policy pointers (avoid spam patterns)
Telemetry requirements (bounce rate, complaint rate)
Alert thresholds (bounce spike)

Optional Fields
Warm-up/ramp policy | OPTIONAL
Dedicated IP policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Hard bounces/complaints must result in suppression per policy.
Deliverability events must not leak PII beyond what is needed.
Alert thresholds must be actionable and tied to runbooks.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Signals
   signals_supported: {{signals.supported}}
   ingestion_method: {{signals.ingestion_method}} (webhook/polling/UNKNOWN)
   webhook_ref: {{signals.webhook_ref}} (expected: {{xref:WHCP-01}}) | OPTIONAL
2. Suppression Lists
   model: {{suppression.model}} (global/per_channel/UNKNOWN)
   list_location: {{suppression.location}} | OPTIONAL
3. Suppression Triggers
   on_hard_bounce: {{suppression.on_hard_bounce}}
   on_soft_bounce: {{suppression.on_soft_bounce}} | OPTIONAL
   on_complaint: {{suppression.on_complaint}}
4. Suppression Duration
   duration_policy: {{suppression.duration_policy}} (permanent/ttl/UNKNOWN)
   ttl_days: {{suppression.ttl_days}} | OPTIONAL
5. Re-Enable Policy
   reenable_supported: {{reenable.supported}}
   reenable_process: {{reenable.process}}
6. Sender Identity
   from_domain_policy: {{sender.from_domain_policy}} | OPTIONAL
   sms_sender_policy: {{sender.sms_sender_policy}} | OPTIONAL
7. Content Policy
   content_rules: {{content.rules}}
   unsubscribe_requirements: {{content.unsubscribe_requirements}} | OPTIONAL
8. Telemetry
   bounce_rate_metric: {{telemetry.bounce_rate_metric}}
   complaint_rate_metric: {{telemetry.complaint_rate_metric}} | OPTIONAL
   suppression_add_metric: {{telemetry.suppression_add_metric}} | OPTIONAL
9. Alerts
   bounce_spike_alert: {{alerts.bounce_spike_alert}}
   complaint_spike_alert: {{alerts.complaint_spike_alert}} | OPTIONAL

10. References
    Provider inventory: {{xref:NOTIF-02}} | OPTIONAL
    Send policy: {{xref:NOTIF-04}} | OPTIONAL
    Preference center: {{xref:NOTIF-06}} | OPTIONAL
    Failure handling: {{xref:NOTIF-09}} | OPTIONAL
    Observability/runbooks: {{xref:NOTIF-10}} | OPTIONAL

Cross-References
Upstream: {{xref:NOTIF-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:NOTIF-06}}, {{xref:NOTIF-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define signals, suppression triggers, duration, and bounce rate metric.
intermediate: Required. Define re-enable process and alert thresholds and sender identity.
advanced: Required. Add warm-up/dedicated IP policy and stricter content rules and metrics.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, webhook ref, list location, soft bounce, ttl days, sender policies, unsubscribe requirements, optional metrics, complaint alert, warm-up/ip policy, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If signals.supported is UNKNOWN → block Completeness Gate.
If suppression.on_hard_bounce is UNKNOWN → block Completeness Gate.
If suppression.on_complaint is UNKNOWN → block Completeness Gate.
If reenable.process is UNKNOWN → block Completeness Gate.
If telemetry.bounce_rate_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.NOTIF
Pass conditions:
required_fields_present == true
signals_and_ingestion_defined == true
suppression_rules_defined == true
reenable_defined == true
telemetry_and_alerts_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

NOTIF-06

NOTIF-06 — Preference Center & Opt-Out Rules (global/per-type)
Header Block
template_id: NOTIF-06
title: Preference Center & Opt-Out Rules (global/per-type)
type: notifications_preference_center_optout_rules
template_version: 1.0.0
output_path: 10_app/notifications/NOTIF-06_Preference_Center_Opt_Out_Rules.md
compliance_gate_id: TMP-05.PRIMARY.NOTIF
upstream_dependencies: ["NOTIF-01", "NOTIF-03", "ROUTE-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "NOTIF-01", "NOTIF-03", "FE-01", "ROUTE-01", "CSec-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical preference center model and opt-out rules: how users control notification settings globally and per-type, how consent is recorded, and how opt-outs are enforced across channels/providers. This template must be consistent with send policy and deliverability suppression rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
NOTIF-01 Channel Inventory: {{notif.channels}}
NOTIF-03 Template/Notification Mapping: {{notif.template_mapping}} | OPTIONAL
FE-01 Route Map/Layout: {{fe.routes}} | OPTIONAL
ROUTE-01 Route Contract: {{route.contract}} | OPTIONAL
CSec-02 Client Data Protection: {{csec.data_protection}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Preference model (global + per-type)
Preference entity fields (user_id, channel, notif_type)
Default opt-in states (per type/channel)
Opt-out enforcement rules (at send time)
Unsubscribe link behavior (email)
Channel-specific opt-out rules (sms STOP, push OS settings)
Transactional vs marketing rules (cannot opt-out transactional?)
Consent capture rules (when/how recorded)
Data retention for preferences
Telemetry requirements (opt-out rate, preference changes)

Optional Fields
Per-tenant preference overrides | OPTIONAL
Double opt-in policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Opt-out must be enforced server-side at send time.
Consent must be explicit where required; do not default opt-in for marketing unless allowed.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Preference Model
   model: {{prefs.model}} (global_plus_per_type/UNKNOWN)
   supported_channels: {{prefs.supported_channels}}
2. Preference Entity
   fields: {{prefs.entity_fields}}
   storage_location: {{prefs.storage_location}} | OPTIONAL
3. Defaults
   default_states: {{prefs.default_states}}
4. Enforcement
   enforce_at_send_time: {{enforce.at_send_time}}
   suppression_interaction_rule: {{enforce.suppression_interaction_rule}} | OPTIONAL
5. Unsubscribe Mechanisms
   email_unsubscribe_rule: {{unsub.email_rule}}
   sms_unsubscribe_rule: {{unsub.sms_rule}} | OPTIONAL
   push_unsubscribe_rule: {{unsub.push_rule}} | OPTIONAL
6. Transactional vs Marketing
   classification_rule: {{class.rule}}
   marketing_opt_in_required: {{class.marketing_opt_in_required}} | OPTIONAL
   transactional_opt_out_allowed: {{class.transactional_opt_out_allowed}}
7. Consent Capture
   capture_points: {{consent.capture_points}}
   consent_fields: {{consent.fields}} | OPTIONAL
8. Retention
   retention_policy: {{retention.policy}}
9. Telemetry
   preference_change_metric: {{telemetry.preference_change_metric}}
   opt_out_metric: {{telemetry.opt_out_metric}} | OPTIONAL
10. References
    Send policy: {{xref:NOTIF-04}} | OPTIONAL
    Deliverability/suppression: {{xref:NOTIF-05}} | OPTIONAL
    Template mapping: {{xref:NOTIF-03}} | OPTIONAL
    Push permission rules: {{xref:MPUSH-03}} | OPTIONAL

Cross-References
Upstream: {{xref:NOTIF-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:NOTIF-07}}, {{xref:NOTIF-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define defaults, unsubscribe mechanisms, and enforce_at_send_time.
intermediate: Required. Define entity fields, consent capture, and transactional vs marketing rules.
advanced: Required. Add double opt-in and suppression interaction and telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, storage location, suppression interaction, sms/push rules, marketing opt-in, consent fields, opt-out metric, overrides, double opt-in, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If prefs.default_states is UNKNOWN → block Completeness Gate.
If enforce.at_send_time is UNKNOWN → block Completeness Gate.
If unsub.email_rule is UNKNOWN → block Completeness Gate.
If retention.policy is UNKNOWN → block Completeness Gate.
If telemetry.preference_change_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.NOTIF
Pass conditions:
required_fields_present == true
defaults_defined == true
unsubscribe_defined == true
enforcement_defined == true
consent_and_retention_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

NOTIF-07

NOTIF-07 — Security & Compliance (PII, CAN-SPAM, consent)
Header Block
template_id: NOTIF-07
title: Security & Compliance (PII, CAN-SPAM, consent)
type: notifications_security_compliance
template_version: 1.0.0
output_path: 10_app/notifications/NOTIF-07_Security_Compliance.md
compliance_gate_id: TMP-05.PRIMARY.NOTIF
upstream_dependencies: ["NOTIF-06", "IXS-08", "CSec-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "NOTIF-01", "NOTIF-06", "IXS-08", "CSec-02", "CER-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical security and compliance requirements for notifications: PII handling, consent and opt-out enforcement, regulatory constraints (e.g., CAN-SPAM / SMS consent), data retention, and safe logging/redaction. This template must be consistent with integration security baseline and preference center rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
NOTIF-01 Channel Inventory: {{notif.channels}}
NOTIF-06 Preference Center/Opt-Out Rules: {{notif.prefs}} | OPTIONAL
IXS-08 Integration Security & Compliance: {{ixs.security_compliance}} | OPTIONAL
CSec-02 Client Data Protection: {{csec.data_protection}} | OPTIONAL
CER-05 Logging/Redaction: {{cer.logging}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Data classification for notification content (PII/no PII)
Consent requirements by channel (email/sms/push/in-app)
Marketing vs transactional enforcement rules
Unsubscribe requirements (email footer, SMS STOP)
Sender identity requirements (from domain/phone)
PII minimization rules (what is allowed in message bodies)
Logging/redaction rules (no message body logs)
Retention rules for notification logs/events
Vendor risk notes (provider compliance)
Telemetry requirements (consent violations, blocked sends)

Optional Fields
Regional compliance differences | OPTIONAL
Age-restricted user rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Opt-out and consent must be enforced at send time.
Do not send sensitive PII in notifications unless explicitly approved and necessary.
Logs must not store message bodies by default; store metadata only.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Data Classification
   allowed_content_classes: {{data.allowed_content_classes}}
   pii_allowed_rule: {{data.pii_allowed_rule}}
2. Consent Requirements
   email_consent_rule: {{consent.email_rule}}
   sms_consent_rule: {{consent.sms_rule}} | OPTIONAL
   push_consent_rule: {{consent.push_rule}} | OPTIONAL
   in_app_consent_rule: {{consent.in_app_rule}} | OPTIONAL
3. Marketing vs Transactional
   classification_rule: {{class.rule}}
   marketing_enforcement: {{class.marketing_enforcement}}
   transactional_enforcement: {{class.transactional_enforcement}} | OPTIONAL
4. Unsubscribe Requirements
   email_unsubscribe_required: {{unsub.email_required}}
   sms_stop_required: {{unsub.sms_stop_required}} | OPTIONAL
5. Sender Identity
   from_domain_policy: {{sender.from_domain_policy}}
   sms_sender_policy: {{sender.sms_sender_policy}} | OPTIONAL
6. PII Minimization
   allowed_fields_in_body: {{pii.allowed_fields_in_body}}
   forbidden_fields_in_body: {{pii.forbidden_fields_in_body}} | OPTIONAL
7. Logging / Redaction
   no_body_logging_rule: {{logs.no_body_logging_rule}}
   metadata_fields_allowed: {{logs.metadata_fields_allowed}} | OPTIONAL
   redaction_rule: {{logs.redaction_rule}} | OPTIONAL
8. Retention
   notification_log_retention: {{retention.log_retention}}
   suppression_retention: {{retention.suppression_retention}} | OPTIONAL
9. Vendor Risk
   provider_compliance_notes: {{vendor.compliance_notes}}
   security_review_required: {{vendor.security_review_required}} | OPTIONAL

10. Telemetry
    blocked_send_metric: {{telemetry.blocked_send_metric}}
    consent_violation_metric: {{telemetry.consent_violation_metric}} | OPTIONAL
11. References
    Preference center: {{xref:NOTIF-06}} | OPTIONAL
    Send policy: {{xref:NOTIF-04}} | OPTIONAL
    Deliverability: {{xref:NOTIF-05}} | OPTIONAL
    Integration security baseline: {{xref:IXS-08}} | OPTIONAL
    Client data protection: {{xref:CSec-02}} | OPTIONAL

Cross-References
Upstream: {{xref:NOTIF-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:NOTIF-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define consent rules and no-body-logging and unsubscribe requirements.
intermediate: Required. Define PII minimization and retention and vendor notes.
advanced: Required. Add regional/age rules and stricter telemetry and auditability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, optional channel consent rules, sms sender policy, forbidden fields list, metadata fields, redaction rule, suppression retention, security review, consent violation metric, regional/age rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If data.pii_allowed_rule is UNKNOWN → block Completeness Gate.
If unsub.email_required is UNKNOWN → block Completeness Gate.
If logs.no_body_logging_rule is UNKNOWN → block Completeness Gate.
If retention.log_retention is UNKNOWN → block Completeness Gate.
If telemetry.blocked_send_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.NOTIF
Pass conditions:
required_fields_present == true
consent_and_unsubscribe_defined == true
pii_and_logging_rules_defined == true
retention_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

NOTIF-08

NOTIF-08 — Event-to-Notification Mapping (events → messages)
Header Block
template_id: NOTIF-08
title: Event-to-Notification Mapping (events → messages)
type: notifications_event_to_message_mapping
template_version: 1.0.0
output_path: 10_app/notifications/NOTIF-08_Event_To_Notification_Mapping.md
compliance_gate_id: TMP-05.PRIMARY.NOTIF
upstream_dependencies: ["EVT-01", "NOTIF-01", "NOTIF-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "EVT-01", "EVT-02", "NOTIF-01", "NOTIF-03", "FFCFG-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical mapping from internal events (event_id) to notification messages (template_id/channel), including eligibility rules, preference enforcement, deduping, and scheduling behavior. This template must be consistent with the event catalog and notification template registry and must not invent event IDs or template IDs beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
EVT-01 Event Catalog: {{evt.catalog}}
EVT-02 Event Schemas: {{evt.schemas}} | OPTIONAL
NOTIF-01 Channel Inventory: {{notif.channels}}
NOTIF-03 Template & Localization Mapping: {{notif.templates}}
FFCFG-01 Feature Flags: {{ffcfg.flags}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Mapping registry (map_id list)
map_id (stable identifier)
event_id binding (must exist)
notification type binding (notif_id/name)
channel(s) used
template_id(s) used
eligibility rules (who receives)
preference enforcement rule (NOTIF-06)
dedupe rule (avoid duplicates per event)
timing rule (immediate/delayed/batched)
failure handling ref (NOTIF-09)
telemetry requirements (enqueued/sent per mapping)

Optional Fields
Feature flag gating | OPTIONAL
Localization override rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
event_id MUST exist in {{xref:EVT-01}}; template_id MUST exist in {{xref:NOTIF-03}}.
Preference enforcement MUST occur before send.
Dedupe must be deterministic and keyed by event_id + subject.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Mapping Summary
   total_mappings: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Mapping Entries (by map_id)
   Mapping
   map_id: {{maps[0].map_id}}
   event_id: {{maps[0].event_id}}
   notif_binding: {{maps[0].notif_binding}}
   channels: {{maps[0].channels}}
   template_ids: {{maps[0].template_ids}}
   eligibility_rule: {{maps[0].eligibility_rule}}
   preference_rule_ref: {{maps[0].preference_rule_ref}} (expected: {{xref:NOTIF-06}}) | OPTIONAL
   dedupe_rule: {{maps[0].dedupe_rule}}
   timing_rule: {{maps[0].timing_rule}} (immediate/delayed/batched/UNKNOWN)
   failure_ref: {{maps[0].failure_ref}} (expected: {{xref:NOTIF-09}}) | OPTIONAL
   flag_gate: {{maps[0].flag_gate}} | OPTIONAL
   locale_override: {{maps[0].locale_override}} | OPTIONAL
   open_questions:
   {{maps[0].open_questions[0]}} | OPTIONAL

(Repeat per mapping.)

3. Telemetry
   enqueued_metric: {{telemetry.enqueued_metric}}
   sent_metric: {{telemetry.sent_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
4. References
   Event catalog: {{xref:EVT-01}}
   Template mapping: {{xref:NOTIF-03}}
   Send policy: {{xref:NOTIF-04}} | OPTIONAL

Preference center: {{xref:NOTIF-06}} | OPTIONAL
Failure handling: {{xref:NOTIF-09}} | OPTIONAL

Cross-References
Upstream: {{xref:EVT-01}}, {{xref:NOTIF-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:NOTIF-09}}, {{xref:NOTIF-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define event_id → template/channel mappings and dedupe rule.
intermediate: Required. Add eligibility and preference enforcement and timing rules.
advanced: Required. Add feature flag gates, localization overrides, and telemetry field rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, meta notes, preference ref, failure ref, flag gate, locale override, sent metric/fields, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If maps[*].event_id is UNKNOWN → block Completeness Gate.
If maps[*].template_ids is UNKNOWN → block Completeness Gate.
If maps[*].dedupe_rule is UNKNOWN → block Completeness Gate.
If telemetry.enqueued_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.NOTIF
Pass conditions:
required_fields_present == true
event_ids_valid == true
template_ids_valid == true
dedupe_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

NOTIF-09

NOTIF-09 — Failure Handling (retries, fallback channels, DLQ)
Header Block
template_id: NOTIF-09
title: Failure Handling (retries, fallback channels, DLQ)
type: notifications_failure_handling
template_version: 1.0.0
output_path: 10_app/notifications/NOTIF-09_Failure_Handling.md
compliance_gate_id: TMP-05.PRIMARY.NOTIF
upstream_dependencies: ["NOTIF-04", "IXS-06", "JBS-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "NOTIF-01", "NOTIF-02", "NOTIF-04", "IXS-06", "JBS-04", "RLIM-04", "ADMIN-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical failure handling for notification delivery across channels/providers: retry eligibility, backoff, DLQ/quarantine, fallback channel rules, and operator recovery actions. This template must be consistent with send policy, integration error handling, and job DLQ policies.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
NOTIF-01 Channel Inventory: {{notif.channels}}
NOTIF-02 Provider Inventory: {{notif.providers}} | OPTIONAL
NOTIF-04 Send Policy: {{notif.send_policy}} | OPTIONAL
IXS-06 Integration Error Handling: {{ixs.error_recovery}} | OPTIONAL
JBS-04 Retry/DLQ Policy: {{jobs.retry_dlq}} | OPTIONAL
RLIM-04 Enforcement Actions Matrix: {{rlim.actions}} | OPTIONAL
ADMIN-02 Support Tools Spec: {{admin.support_tools}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Failure taxonomy (provider reject, bounce, throttled, network)
Retry eligibility rules (by channel/failure)
Backoff policy + max attempts
DLQ/quarantine supported (yes/no/UNKNOWN)
DLQ trigger rules
Fallback channel rules (if primary fails)
Suppression interaction rules (do not retry suppressed)
Operator actions (resend, disable provider, rotate creds)
User impact rules (silent vs surfaced)
Telemetry requirements (retry count, DLQ depth)

Optional Fields
Per-notif-type overrides | OPTIONAL
Auto-disable provider thresholds | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Retries must be bounded and must respect suppression/opt-out.
Fallback channels must honor consent/opt-out for that channel.
Operator actions must be permissioned and auditable.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Failure Taxonomy
   classes: {{fail.classes}}
   channel_notes: {{fail.channel_notes}} | OPTIONAL
2. Retry Eligibility
   retryable: {{retry.retryable}}
   non_retryable: {{retry.non_retryable}} | OPTIONAL
3. Backoff
   policy: {{backoff.policy}} (exponential/jitter/UNKNOWN)
   max_attempts: {{backoff.max_attempts}}
   max_age_ms: {{backoff.max_age_ms}} | OPTIONAL
4. DLQ / Quarantine
   dlq_supported: {{dlq.supported}}
   dlq_trigger_rule: {{dlq.trigger_rule}}
   redaction_rule: {{dlq.redaction_rule}} | OPTIONAL
   drain_policy: {{dlq.drain_policy}} | OPTIONAL
5. Fallback Channels
   fallback_supported: {{fallback.supported}}
   fallback_rules: {{fallback.rules}}
   consent_check_rule: {{fallback.consent_check_rule}} | OPTIONAL
6. Suppression Interaction
   no_retry_when_suppressed: {{suppress.no_retry_when_suppressed}}
   suppression_ref: {{suppress.ref}} (expected: {{xref:NOTIF-05}}) | OPTIONAL
7. Operator Actions
   actions: {{ops.actions}}
   permissions_ref: {{ops.permissions_ref}} (expected: {{xref:API-04}}/{{xref:ADMIN-01}}) | OPTIONAL
8. User Impact
   surface_failures: {{impact.surface_failures}}
   user_copy_policy: {{impact.user_copy_policy}} | OPTIONAL
9. Telemetry
   retry_count_metric: {{telemetry.retry_count_metric}}

dlq_depth_metric: {{telemetry.dlq_depth_metric}} | OPTIONAL
fallback_used_metric: {{telemetry.fallback_used_metric}} | OPTIONAL
fields: {{telemetry.fields}} | OPTIONAL

10. References
    Send policy: {{xref:NOTIF-04}} | OPTIONAL
    Deliverability/suppression: {{xref:NOTIF-05}} | OPTIONAL
    Preference center: {{xref:NOTIF-06}} | OPTIONAL
    Observability/runbooks: {{xref:NOTIF-10}} | OPTIONAL

Cross-References
Upstream: {{xref:NOTIF-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:NOTIF-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define retry and max attempts and fallback rules.
intermediate: Required. Define DLQ triggers, suppression interaction, and telemetry.
advanced: Required. Add auto-disable thresholds and per-type overrides and operator permissions rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, channel notes, non-retryable list, max age, dlq redaction/drain, consent check rule, suppression ref, permissions ref, user copy policy, optional metrics/fields, overrides/thresholds, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If backoff.max_attempts is UNKNOWN → block Completeness Gate.
If dlq.supported is UNKNOWN → block Completeness Gate.
If fallback.rules is UNKNOWN → block Completeness Gate (when fallback.supported == true).
If suppress.no_retry_when_suppressed is UNKNOWN → block Completeness Gate.
If telemetry.retry_count_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.NOTIF
Pass conditions:
required_fields_present == true
bounded_retry_defined == true
fallback_and_suppression_defined == true
dlq_policy_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

NOTIF-10

NOTIF-10 — Observability & Support Runbooks (delivery metrics, alerts)
Header Block
template_id: NOTIF-10
title: Observability & Support Runbooks (delivery metrics, alerts)
type: notifications_observability_support_runbooks
template_version: 1.0.0
output_path: 10_app/notifications/NOTIF-10_Observability_Support_Runbooks.md
compliance_gate_id: TMP-05.PRIMARY.NOTIF
upstream_dependencies: ["NOTIF-04", "NOTIF-05", "NOTIF-09", "IXS-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "NOTIF-01", "NOTIF-02", "NOTIF-04", "NOTIF-05", "NOTIF-09", "IXS-07", "ADMIN-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical observability and support runbooks for notifications: dashboards, alerts, triage flows, and operator actions for delivery issues (throttling, bounces, provider outages, DLQ backlogs). This template must be consistent with send/deliverability/failure policies and integration observability.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
NOTIF-01 Channel Inventory: {{notif.channels}}
NOTIF-02 Provider Inventory: {{notif.providers}} | OPTIONAL
NOTIF-04 Send Policy: {{notif.send_policy}} | OPTIONAL
NOTIF-05 Deliverability Rules: {{notif.deliverability}} | OPTIONAL
NOTIF-09 Failure Handling: {{notif.failures}} | OPTIONAL
IXS-07 Integration Observability: {{ixs.observability}} | OPTIONAL
ADMIN-02 Support Tools: {{admin.support_tools}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Dashboards minimum panels (by channel/provider)
Key metrics list (send, delivered, bounced, suppressed, dlq depth)
Alert definitions (delivery drop, bounce spike, dlq stuck, provider outage)
Alert routing policy (oncall/support)
Runbook location and format
Triage flows (by incident type)
Operator actions (pause sends, rotate creds, clear suppression)
Permissioning requirements for actions
Customer comms triggers (when to notify users)
Post-incident review requirements

Optional Fields
SLOs per channel | OPTIONAL
Auto-mitigation rules (switch provider) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Alerts must be actionable and reference triage steps.
Operator actions must be permissioned and auditable.
Dashboards must avoid PII; use aggregates/hashes only.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Dashboards
   dashboard_name: {{dash.name}}
   minimum_panels: {{dash.minimum_panels}}
   panel_list: {{dash.panel_list}} | OPTIONAL
2. Key Metrics
   metrics: {{metrics.list}}
3. Alerts
   Alert
   alert_id: {{alerts[0].alert_id}}
   condition: {{alerts[0].condition}}
   severity: {{alerts[0].severity}} (low/med/high/UNKNOWN)
   routing: {{alerts[0].routing}}
   runbook_ref: {{alerts[0].runbook_ref}} | OPTIONAL

(Repeat per alert.)

4. Runbooks
   runbook_location: {{runbooks.location}}
   format: {{runbooks.format}} (md/wiki/UNKNOWN) | OPTIONAL
5. Triage Flows
   Flow
   incident_type: {{triage[0].incident_type}}
   steps:
   {{triage[0].steps[0]}}
   {{triage[0].steps[1]}} | OPTIONAL
   primary_tools: {{triage[0].primary_tools}} | OPTIONAL

(Repeat per incident type.)

6. Operator Actions
   actions: {{ops.actions}}

permissions_ref: {{ops.permissions_ref}} (expected: {{xref:API-04}}/{{xref:ADMIN-01}}) | OPTIONAL

7. Customer Comms
   notify_users_when: {{comms.notify_users_when}}
   comms_owner: {{comms.owner}} | OPTIONAL
8. Post-Incident
   rca_required: {{post.rca_required}}
   rca_location: {{post.rca_location}} | OPTIONAL
9. References
   Send policy: {{xref:NOTIF-04}} | OPTIONAL
   Deliverability: {{xref:NOTIF-05}} | OPTIONAL
   Failure handling: {{xref:NOTIF-09}} | OPTIONAL
   Preference center: {{xref:NOTIF-06}} | OPTIONAL
   Provider inventory: {{xref:NOTIF-02}} | OPTIONAL
   Support tools: {{xref:ADMIN-02}} | OPTIONAL

Cross-References
Upstream: {{xref:NOTIF-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define dashboards + key metrics + one triage flow.
intermediate: Required. Define alerts, routing, operator actions, and comms triggers.
advanced: Required. Add SLOs and auto-mitigation and incident-type triage rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, panel list, optional tools, runbook format, permissions ref, comms owner, rca location, slos, auto-mitigation, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If dash.minimum_panels is UNKNOWN → block Completeness Gate.
If metrics.list is UNKNOWN → block Completeness Gate.
If alerts list is UNKNOWN → block Completeness Gate.
If triage[0].steps[0] is UNKNOWN → block Completeness Gate.
If post.rca_required is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.NOTIF
Pass conditions:
required_fields_present == true
dashboards_and_alerts_defined == true
triage_and_actions_defined == true
comms_and_post_incident_defined == true

placeholder_resolution == true
no_unapproved_unknowns == true

Files/Media & Storage (FMS)

FMS-01

FMS-01 — Storage Provider Inventory (by provider_id/bucket_id)
Header Block
template_id: FMS-01
title: Storage Provider Inventory (by provider_id/bucket_id)
type: files_media_storage_provider_inventory
template_version: 1.0.0
output_path: 10_app/files_media/FMS-01_Storage_Provider_Inventory.md
compliance_gate_id: TMP-05.PRIMARY.FMS
upstream_dependencies: ["IXS-01", "FPMP-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IXS-01", "FPMP-02", "FPMP-01", "IXS-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical inventory of file/media storage providers and storage targets (buckets/containers), indexed by provider_id and bucket_id, including environment scoping, access modes, and security/compliance flags. This document must not invent provider/bucket IDs beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IXS-01 Integration Inventory: {{ixs.inventory}} | OPTIONAL
FPMP-02 Storage Strategy: {{fpmp.storage_strategy}} | OPTIONAL
FPMP-01 Upload Contract: {{fpmp.upload_contract}} | OPTIONAL
IXS-04 Secrets/Credential Handling: {{ixs.secrets_policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Provider registry (provider_id list)
Bucket/target registry (bucket_id list)
provider_id (stable identifier)
bucket_id (stable identifier)
provider/vendor name
integration_id binding (IXS-01)
env scope (dev/stage/prod)
region/location (if applicable)
access mode (public/private/signed/UNKNOWN)
path prefix rules (naming convention)
encryption flags (at rest)
retention/lifecycle policy ref (FMS-05)

observability/cost tracking flags (FMS-10)
credential reference (IXS-04)

Optional Fields
CDN binding (FMS-04) | OPTIONAL
Replication/DR notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Buckets/targets must be environment-scoped and least-privilege.
Do not list secret values; reference credential policies only.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Inventory Summary
   total_providers: {{inv.total_providers}}
   total_buckets: {{inv.total_buckets}}
   notes: {{inv.notes}} | OPTIONAL
2. Provider Entries
   Provider
   provider_id: {{providers[0].provider_id}}
   name: {{providers[0].name}}
   integration_id: {{providers[0].integration_id}}
   env_enabled: {{providers[0].env_enabled}}
   credential_ref: {{providers[0].credential_ref}} (expected: {{xref:IXS-04}}) | OPTIONAL
   notes: {{providers[0].notes}} | OPTIONAL

(Repeat per provider.)

3. Bucket/Target Entries
   Bucket
   bucket_id: {{buckets[0].bucket_id}}
   provider_id: {{buckets[0].provider_id}}
   env_scope: {{buckets[0].env_scope}}
   region: {{buckets[0].region}} | OPTIONAL
   access_mode: {{buckets[0].access_mode}}
   path_prefix_rule: {{buckets[0].path_prefix_rule}}
   encryption_at_rest: {{buckets[0].encryption_at_rest}}
   retention_ref: {{buckets[0].retention_ref}} (expected: {{xref:FMS-05}}) | OPTIONAL
   cdn_ref: {{buckets[0].cdn_ref}} | OPTIONAL
   dr_notes: {{buckets[0].dr_notes}} | OPTIONAL
   open_questions:
   {{buckets[0].open_questions[0]}} | OPTIONAL

(Repeat per bucket.)

4. References
   Upload contract: {{xref:FPMP-01}} | OPTIONAL
   Storage strategy: {{xref:FPMP-02}} | OPTIONAL
   CDN rules: {{xref:FMS-04}} | OPTIONAL
   Retention/lifecycle: {{xref:FMS-05}} | OPTIONAL
   Security/compliance: {{xref:FMS-06}} | OPTIONAL
   Observability/cost: {{xref:FMS-10}} | OPTIONAL

Cross-References
Upstream: {{xref:IXS-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FMS-02}}, {{xref:FMS-05}}, {{xref:FMS-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. List providers/buckets and access modes and env scopes; do not invent IDs.
intermediate: Required. Add path prefix rules, encryption flags, and retention refs.
advanced: Required. Add CDN/DR bindings and cost tracking flags and rigorous traceability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, inv notes, provider notes, region, retention/cdn refs, dr notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If providers[*].provider_id is UNKNOWN → block Completeness Gate.*
*If buckets[*].bucket_id is UNKNOWN → block Completeness Gate.
If buckets[*].access_mode is UNKNOWN → block Completeness Gate.*
*If buckets[*].path_prefix_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FMS
Pass conditions:
required_fields_present == true
provider_ids_unique == true
bucket_ids_unique == true
env_scoping_defined == true
access_modes_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FMS-02

FMS-02 — Upload/Download Integration Spec (signed URLs, auth, limits)
Header Block
template_id: FMS-02
title: Upload/Download Integration Spec (signed URLs, auth, limits)
type: files_media_upload_download_integration_spec
template_version: 1.0.0
output_path: 10_app/files_media/FMS-02_Upload_Download_Integration_Spec.md
compliance_gate_id: TMP-05.PRIMARY.FMS
upstream_dependencies: ["FMS-01", "FPMP-01", "API-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FMS-01", "FPMP-01", "FPMP-02", "API-01", "API-02", "CSec-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical integration spec for uploading and downloading files/media: signed URL issuance, auth requirements, size/type limits, storage paths, and download access controls. This template must be consistent with upload contract and storage strategy and must not invent API endpoints or access modes beyond upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FMS-01 Storage Provider Inventory: {{fms.storage_inventory}}
FPMP-01 Upload Contract: {{fpmp.upload_contract}}
FPMP-02 Storage Strategy: {{fpmp.storage_strategy}} | OPTIONAL
API-01 Endpoint Catalog: {{api.endpoint_catalog}} | OPTIONAL
API-02 Endpoint Specs: {{api.endpoint_specs}} | OPTIONAL
CSec-01 Token Storage Policy (client): {{csec.token_storage}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Upload flow model (direct-to-storage vs proxy)
Signed URL supported (yes/no/UNKNOWN)
Signed URL issuance endpoint (endpoint_id/path)
Signed URL TTL rule
Auth required to request upload/download
Upload constraints (max size, allowed types)
Path/key construction rule (bucket + prefix)
Download access model (public/private/signed/UNKNOWN)
Authorization checks for download (API-04 ref)
Virus/malware scanning binding (FPMP-06/FPMP-03)
Telemetry requirements (upload success/fail, bytes)

Optional Fields
Multipart upload rules | OPTIONAL
Resume/retry guidance | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
If signed URLs are used, TTL must be bounded and least-privilege (method/path).
Download access must enforce AuthZ for private objects.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Upload Flow
   model: {{upload.model}} (direct_to_storage/proxy/UNKNOWN)
   signed_url_supported: {{upload.signed_url_supported}}
   issuance_endpoint: {{upload.issuance_endpoint}}
   signed_url_ttl_seconds: {{upload.signed_url_ttl_seconds}}
2. Auth
   auth_required: {{auth.required}}
   authz_ref: {{auth.authz_ref}} (expected: {{xref:API-04}}) | OPTIONAL
3. Constraints
   max_size_bytes: {{constraints.max_size_bytes}}
   allowed_types: {{constraints.allowed_types}}
   rate_limit_ref: {{constraints.rate_limit_ref}} | OPTIONAL
4. Storage Paths
   bucket_id: {{paths.bucket_id}}
   key_prefix_rule: {{paths.key_prefix_rule}}
   naming_rule: {{paths.naming_rule}} | OPTIONAL
5. Download
   access_model: {{download.access_model}} (public/private/signed/UNKNOWN)
   download_endpoint: {{download.endpoint}} | OPTIONAL
   download_auth_rule: {{download.auth_rule}} | OPTIONAL
6. Scanning
   scan_required: {{scan.required}}
   scan_ref: {{scan.ref}} (expected: {{xref:FPMP-06}}) | OPTIONAL
7. Telemetry
   upload_success_metric: {{telemetry.upload_success_metric}}
   upload_fail_metric: {{telemetry.upload_fail_metric}} | OPTIONAL
   bytes_uploaded_metric: {{telemetry.bytes_uploaded_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
8. References
   Storage inventory: {{xref:FMS-01}}
   Upload contract: {{xref:FPMP-01}}
   Storage strategy: {{xref:FPMP-02}} | OPTIONAL

File security: {{xref:FPMP-06}} | OPTIONAL
CDN rules: {{xref:FMS-04}} | OPTIONAL

Cross-References
Upstream: {{xref:FMS-01}}, {{xref:FPMP-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FMS-03}}, {{xref:FMS-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define upload model, constraints, and download access model.
intermediate: Required. Define signed URL TTL and auth/authz ref and path rules.
advanced: Required. Add multipart/resume rules and strict telemetry fields and scanning rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, authz ref, rate limit ref, naming rule, download endpoint/auth rule, scan ref, optional telemetry metrics/fields, multipart/resume, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If upload.model is UNKNOWN → block Completeness Gate.
If constraints.max_size_bytes is UNKNOWN → block Completeness Gate.
If download.access_model is UNKNOWN → block Completeness Gate.
If telemetry.upload_success_metric is UNKNOWN → block Completeness Gate.
If scan.required is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FMS
Pass conditions:
required_fields_present == true
upload_and_download_models_defined == true
constraints_defined == true
path_rules_defined == true
scanning_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FMS-03

FMS-03 — Media Processing Integration (transcode/resize/scan vendors)
Header Block
template_id: FMS-03
title: Media Processing Integration (transcode/resize/scan vendors)
type: files_media_processing_integration
template_version: 1.0.0
output_path: 10_app/files_media/FMS-03_Media_Processing_Integration.md
compliance_gate_id: TMP-05.PRIMARY.FMS
upstream_dependencies: ["FPMP-03", "FPMP-04", "FMS-01", "JBS-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FMS-01", "FPMP-03", "FPMP-04", "FPMP-06", "JBS-01", "JBS-02", "EVT-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical integration spec for media/file processing: which processing stages exist (transcode/resize/scan/parse), which vendors or internal services perform them, job triggers, inputs/outputs, retries, and security constraints. This template must be consistent with file processing pipeline stages and async status models.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FMS-01 Storage Provider Inventory: {{fms.storage_inventory}}
FPMP-03 Processing Pipeline Stages: {{fpmp.stages}}
FPMP-04 Async Processing & Status Model: {{fpmp.async_status}} | OPTIONAL
FPMP-06 File Security/Compliance: {{fpmp.security}} | OPTIONAL
JBS-01 Jobs Inventory: {{jobs.inventory}} | OPTIONAL
JBS-02 Job Specs: {{jobs.specs}} | OPTIONAL
EVT-01 Event Catalog (processing events): {{evt.catalog}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Processing stage registry (stage_id list)
Stage definitions (input → output)
Which stages use vendor vs internal service
Job trigger rules (upload complete, manual, cron)
Job IDs involved (JBS binding)
Storage IO paths (source bucket, output bucket)
Retry/DLQ binding (JBS-04)
Idempotency rules (per file hash/version)
Security constraints (scan before publish)
Telemetry requirements (latency, failure rate per stage)

Optional Fields
Parallelization rules | OPTIONAL
Resource limits (CPU/memory) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Files must not be publicly served until required security scanning passes.
Stage outputs must be deterministic and versioned (input hash → output variant).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Stage Registry
   total_stages: {{stages.total}}
   stage_ids: {{stages.ids}}
2. Stage Definitions (repeat per stage_id)
   Stage
   stage_id: {{items[0].stage_id}}
   name: {{items[0].name}}
   processor_type: {{items[0].processor_type}} (vendor/internal/UNKNOWN)
   job_id_ref: {{items[0].job_id_ref}} (expected: {{xref:JBS-01}}) | OPTIONAL
   trigger: {{items[0].trigger}}
   inputs: {{items[0].inputs}}
   outputs: {{items[0].outputs}}
   source_path_rule: {{items[0].source_path_rule}} | OPTIONAL
   output_path_rule: {{items[0].output_path_rule}} | OPTIONAL
   retry_ref: {{items[0].retry_ref}} (expected: {{xref:JBS-04}}) | OPTIONAL
   idempotency_rule: {{items[0].idempotency_rule}}
   security_gate: {{items[0].security_gate}} | OPTIONAL
   telemetry:
   latency_metric: {{items[0].telemetry.latency_metric}}
   failure_metric: {{items[0].telemetry.failure_metric}} | OPTIONAL

(Repeat per stage.)

3. Storage IO
   source_bucket_id: {{io.source_bucket_id}}
   output_bucket_id: {{io.output_bucket_id}} | OPTIONAL
4. Security
   scan_before_publish_rule: {{security.scan_before_publish_rule}}
   required_scans: {{security.required_scans}} | OPTIONAL
5. Telemetry (Global)
   pipeline_success_metric: {{telemetry.pipeline_success_metric}}
   pipeline_failure_metric: {{telemetry.pipeline_failure_metric}} | OPTIONAL

6. References
   Storage inventory: {{xref:FMS-01}}
   Pipeline stages: {{xref:FPMP-03}}
   Async status model: {{xref:FPMP-04}} | OPTIONAL
   File security: {{xref:FPMP-06}} | OPTIONAL
   Jobs specs: {{xref:JBS-02}} | OPTIONAL

Cross-References
Upstream: {{xref:FPMP-03}}, {{xref:FMS-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FMS-06}}, {{xref:FMS-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define stage registry and inputs/outputs and idempotency rule.
intermediate: Required. Bind job refs, retry refs, and scan-before-publish security gate.
advanced: Required. Add parallelization/resource limits and strict output versioning rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, job refs, path rules, retry refs, security gate, optional telemetry metrics, output bucket id, required scans, global failure metric, parallel/resource limits, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If stages.ids is UNKNOWN → block Completeness Gate.
If items[*].idempotency_rule is UNKNOWN → block Completeness Gate.
If security.scan_before_publish_rule is UNKNOWN → block Completeness Gate.
If telemetry.pipeline_success_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FMS
Pass conditions:
required_fields_present == true
stages_defined == true
idempotency_defined == true
security_gates_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FMS-04

FMS-04 — CDN & Delivery Rules (cache headers, variants)
Header Block
template_id: FMS-04
title: CDN & Delivery Rules (cache headers, variants)
type: files_media_cdn_delivery_rules
template_version: 1.0.0
output_path: 10_app/files_media/FMS-04_CDN_Delivery_Rules.md
compliance_gate_id: TMP-05.PRIMARY.FMS
upstream_dependencies: ["FMS-01", "FPMP-05", "FMS-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FMS-01", "FPMP-05", "FMS-02", "FMS-05", "FMS-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical CDN and delivery policy for files/media: cache headers, access models, variants (sizes/bitrates), purge/invalidation rules, and how signed access is enforced at the edge. This template must be consistent with storage access modes and security/retention constraints.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FMS-01 Storage Inventory: {{fms.storage_inventory}}
FPMP-05 CDN/Delivery Rules: {{fpmp.cdn_rules}} | OPTIONAL
FMS-02 Upload/Download Spec: {{fms.upload_download}} | OPTIONAL
FMS-05 Retention/Lifecycle: {{fms.retention}} | OPTIONAL
FMS-06 Security/Compliance: {{fms.security}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
CDN enabled (true/false/UNKNOWN)
CDN provider/edge binding (provider_id)
Delivery domains/URLs policy
Cache policy (public/private, max-age)
Signed delivery supported (yes/no/UNKNOWN)
Signed URL/token TTL rule
Variant strategy (image sizes, video bitrates)
Cache key strategy (include params?)
Purge/invalidation rules (when, how)
Access control enforcement at edge (auth headers/signed tokens)
Telemetry requirements (edge hit rate, errors)

Optional Fields
Regional edge strategy | OPTIONAL
Origin shielding rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Private media must not be cacheable publicly.
If signed delivery is used, TTL must be bounded and tied to object/variant.
Purge rules must not leak private URLs or identifiers.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Enablement
   cdn_enabled: {{cdn.enabled}}
   provider_id: {{cdn.provider_id}} | OPTIONAL
2. Delivery Domains
   domains: {{delivery.domains}}
   url_pattern_rule: {{delivery.url_pattern_rule}} | OPTIONAL
3. Cache Policy
   public_cache_rule: {{cache.public_cache_rule}}
   private_cache_rule: {{cache.private_cache_rule}}
   default_max_age_seconds: {{cache.default_max_age_seconds}} | OPTIONAL
4. Signed Delivery
   signed_supported: {{signed.supported}}
   ttl_seconds: {{signed.ttl_seconds}} | OPTIONAL
   token_location: {{signed.token_location}} (query/header/cookie/UNKNOWN) | OPTIONAL
5. Variants
   variant_strategy: {{variants.strategy}}
   variants_list: {{variants.list}} | OPTIONAL
6. Cache Keys
   cache_key_rule: {{cachekey.rule}}
   include_query_params: {{cachekey.include_query_params}} | OPTIONAL
7. Purge / Invalidation
   purge_supported: {{purge.supported}}
   purge_triggers: {{purge.triggers}}
   purge_method: {{purge.method}} | OPTIONAL
8. Edge Access Control
   edge_enforcement: {{edge.enforcement}}
   auth_model: {{edge.auth_model}} (signed_only/header_pass_through/UNKNOWN) | OPTIONAL

9. Telemetry
   cdn_hit_rate_metric: {{telemetry.hit_rate_metric}}
   cdn_error_metric: {{telemetry.error_metric}} | OPTIONAL
10. References
   Storage inventory: {{xref:FMS-01}}
   Upload/download spec: {{xref:FMS-02}} | OPTIONAL
   Retention/lifecycle: {{xref:FMS-05}} | OPTIONAL
   Security/compliance: {{xref:FMS-06}} | OPTIONAL

Cross-References
Upstream: {{xref:FMS-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FMS-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define cache policy and whether CDN enabled and signed supported.
intermediate: Required. Define variants, cache key rule, and purge triggers.
advanced: Required. Add edge auth model, origin shielding/regional strategy, and telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, provider id, url pattern, max age, ttl, token location, variants list, query params, purge method, auth model, error metric, regional/shielding, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If cdn.enabled is UNKNOWN → block Completeness Gate.
If cache.public_cache_rule is UNKNOWN → block Completeness Gate.
If cache.private_cache_rule is UNKNOWN → block Completeness Gate.
If cachekey.rule is UNKNOWN → block Completeness Gate.
If telemetry.hit_rate_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FMS
Pass conditions:
required_fields_present == true
cache_policy_defined == true
cdn_and_signed_delivery_defined == true
purge_and_cachekey_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FMS-05

FMS-05 — Retention & Lifecycle Rules (TTL, archival, deletion)
Header Block
template_id: FMS-05
title: Retention & Lifecycle Rules (TTL, archival, deletion)
type: files_media_retention_lifecycle_rules
template_version: 1.0.0
output_path: 10_app/files_media/FMS-05_Retention_Lifecycle_Rules.md
compliance_gate_id: TMP-05.PRIMARY.FMS
upstream_dependencies: ["FMS-01", "FMS-06", "FPMP-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FMS-01", "FMS-02", "FMS-06", "FPMP-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical retention, lifecycle, and deletion rules for stored files/media: TTLs by class, archival policies, deletion propagation, and legal/PII constraints. This template must be consistent with file security/compliance rules and must not invent retention obligations without upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FMS-01 Storage Provider Inventory: {{fms.storage_inventory}}
FMS-02 Upload/Download Spec: {{fms.upload_download}} | OPTIONAL
FMS-06 Security/Compliance for Files: {{fms.security}} | OPTIONAL
FPMP-06 Security & Compliance for Files: {{fpmp.file_security}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
File classification model (public/private/pii/UNKNOWN)
Retention policy per class (ttl_days or permanent)
Archival policy (when/how)
Deletion policy (who can delete, how)
Deletion propagation rule (variants, CDN caches)
Legal hold rule (if applicable)
Secure delete expectations (if applicable)
Telemetry requirements (storage growth, deletions)
Audit requirements (deletion events)

Optional Fields
Per-bucket overrides | OPTIONAL

User export/download policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Retention must be explicit; avoid “forever” unless justified.
Deletion must propagate to derived variants and caches per rule.
Legal holds must override deletion/TTL when enabled.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Classification Model
   classes: {{class.model}}
   definitions: {{class.definitions}} | OPTIONAL
2. Retention by Class
   Policy
   class: {{retention[0].class}}
   ttl_days: {{retention[0].ttl_days}}
   archive_after_days: {{retention[0].archive_after_days}} | OPTIONAL
   notes: {{retention[0].notes}} | OPTIONAL

(Repeat per class.)

3. Deletion Policy
   who_can_delete: {{delete.who_can_delete}}
   delete_method: {{delete.method}} (soft/hard/UNKNOWN)
   delete_propagation_rule: {{delete.propagation_rule}}
4. Legal Hold
   legal_hold_supported: {{hold.supported}}
   hold_rule: {{hold.rule}} | OPTIONAL
5. Secure Delete
   secure_delete_required: {{secure.required}} | OPTIONAL
   secure_delete_rule: {{secure.rule}} | OPTIONAL
6. Telemetry
   storage_growth_metric: {{telemetry.storage_growth_metric}}
   deletions_metric: {{telemetry.deletions_metric}} | OPTIONAL
7. Audit
   audit_required: {{audit.required}}
   audit_fields: {{audit.fields}} | OPTIONAL
8. References
   Storage inventory: {{xref:FMS-01}}
   Upload/download: {{xref:FMS-02}} | OPTIONAL
   CDN purge rules: {{xref:FMS-04}} | OPTIONAL

Security/compliance: {{xref:FMS-06}} | OPTIONAL
File security baseline: {{xref:FPMP-06}} | OPTIONAL

Cross-References
Upstream: {{xref:FMS-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FMS-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define retention TTLs and deletion propagation rule.
intermediate: Required. Define archival and audit/telemetry rules.
advanced: Required. Add legal hold and secure delete and per-bucket overrides.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, definitions, archive after days, notes, hold/secure rules, deletions metric, audit fields, overrides/export policy, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If class.model is UNKNOWN → block Completeness Gate.
If delete.propagation_rule is UNKNOWN → block Completeness Gate.
If telemetry.storage_growth_metric is UNKNOWN → block Completeness Gate.
If audit.required is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FMS
Pass conditions:
required_fields_present == true
classification_and_ttls_defined == true
deletion_and_propagation_defined == true
telemetry_defined == true
audit_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FMS-06

FMS-06 — Security & Compliance (PII, malware scanning, encryption)
Header Block
template_id: FMS-06
title: Security & Compliance (PII, malware scanning, encryption)
type: files_media_security_compliance
template_version: 1.0.0
output_path: 10_app/files_media/FMS-06_Security_Compliance.md
compliance_gate_id: TMP-05.PRIMARY.FMS
upstream_dependencies: ["FPMP-06", "IXS-08", "FMS-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FMS-01", "FMS-02", "FPMP-06", "IXS-08", "CSec-02", "CER-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical security and compliance requirements for files/media storage and delivery: encryption, malware scanning, PII handling, access controls, safe URL issuance, and auditability. This template must be consistent with file security baseline and integration compliance rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FMS-01 Storage Inventory: {{fms.storage_inventory}}
FMS-02 Upload/Download Spec: {{fms.upload_download}} | OPTIONAL
FPMP-06 File Security Baseline: {{fpmp.file_security}} | OPTIONAL
IXS-08 Integration Security Baseline: {{ixs.security_compliance}} | OPTIONAL
CSec-02 Client Data Protection: {{csec.data_protection}} | OPTIONAL
CER-05 Logging/Redaction: {{cer.logging}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Encryption in transit rule (TLS)
Encryption at rest rule (bucket encryption)
Malware scanning required (yes/no/UNKNOWN)
Scanning stages/binding (FPMP-06/FPMP-03)
Access control model (public/private/signed)
Signed URL/token rules (TTL, scope)
PII classification for files (what counts)
Retention/deletion binding (FMS-05)
Audit logging requirements (access/download/delete)
Logging/redaction rules (no file contents)
Telemetry requirements (scan failures, access denials)

Optional Fields
Content moderation scanning | OPTIONAL
DLP rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Files must not be served publicly until required scans pass.
Signed access must be least-privilege (method, path) and short-lived.
Do not log file contents; store metadata only.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Encryption
   tls_required: {{crypto.tls_required}}
   at_rest_encryption_required: {{crypto.at_rest_required}}
   kms_notes: {{crypto.kms_notes}} | OPTIONAL
2. Malware Scanning
   scan_required: {{scan.required}}
   scan_binding_ref: {{scan.binding_ref}} (expected: {{xref:FPMP-06}}) | OPTIONAL
   fail_closed_rule: {{scan.fail_closed_rule}} | OPTIONAL
3. Access Control
   access_model: {{access.model}} (public/private/signed/UNKNOWN)
   default_private_rule: {{access.default_private_rule}}
   public_exceptions: {{access.public_exceptions}} | OPTIONAL
4. Signed Access
   signed_supported: {{signed.supported}}
   ttl_seconds: {{signed.ttl_seconds}} | OPTIONAL
   scope_rule: {{signed.scope_rule}} | OPTIONAL
5. PII Classification
   pii_file_types: {{pii.file_types}}
   pii_detection_rule: {{pii.detection_rule}} | OPTIONAL
6. Retention / Deletion Binding
   retention_ref: {{retention.ref}} (expected: {{xref:FMS-05}}) | OPTIONAL
7. Audit
   audit_required: {{audit.required}}
   audit_events: {{audit.events}}
   audit_fields: {{audit.fields}} | OPTIONAL
8. Logging / Redaction
   no_content_logging_rule: {{logs.no_content_logging_rule}}
   metadata_allowlist: {{logs.metadata_allowlist}} | OPTIONAL
9. Telemetry
   scan_failure_metric: {{telemetry.scan_failure_metric}}
   access_denied_metric: {{telemetry.access_denied_metric}} | OPTIONAL

10. References
    Upload/download: {{xref:FMS-02}} | OPTIONAL
    Retention/lifecycle: {{xref:FMS-05}} | OPTIONAL
    File security baseline: {{xref:FPMP-06}} | OPTIONAL
    Integration security baseline: {{xref:IXS-08}} | OPTIONAL

Cross-References
Upstream: {{xref:FMS-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FMS-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define access model and tls/at-rest encryption and scan required.
intermediate: Required. Define signed access TTL/scope and audit events and telemetry.
advanced: Required. Add DLP/moderation rules and fail-closed rigor and KMS details.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, kms notes, scan binding/fail-closed, public exceptions, signed ttl/scope, pii detection, retention ref, audit fields, metadata allowlist, optional metrics, moderation/dlp, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If crypto.tls_required is UNKNOWN → block Completeness Gate.
If crypto.at_rest_required is UNKNOWN → block Completeness Gate.
If scan.required is UNKNOWN → block Completeness Gate.
If access.default_private_rule is UNKNOWN → block Completeness Gate.
If telemetry.scan_failure_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FMS
Pass conditions:
required_fields_present == true
encryption_defined == true
scan_and_access_controls_defined == true
audit_and_logging_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FMS-07

FMS-07 — Access Control Model (ACLs, signed access, tenancy)
Header Block
template_id: FMS-07
title: Access Control Model (ACLs, signed access, tenancy)
type: files_media_access_control_model
template_version: 1.0.0
output_path: 10_app/files_media/FMS-07_Access_Control_Model.md
compliance_gate_id: TMP-05.PRIMARY.FMS
upstream_dependencies: ["FMS-02", "API-04", "CSec-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FMS-01", "FMS-02", "FMS-06", "API-04", "ROUTE-04", "CSec-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical access control model for files/media: tenancy boundaries, ACL semantics, signed access rules, who can read/write/delete, and how authorization is enforced across API, storage, and CDN. This template must be consistent with AuthZ rules and secure deep link handling.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FMS-01 Storage Inventory: {{fms.storage_inventory}} | OPTIONAL
FMS-02 Upload/Download Spec: {{fms.upload_download}}
FMS-06 Security/Compliance: {{fms.security}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
ROUTE-04 Guard Rules: {{route.guard_rules}} | OPTIONAL
CSec-05 Secure Link Handling: {{csec.deep_link_security}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Tenancy model (single/tenant_id/UNKNOWN)
Object ownership rule (who owns a file)
Access modes (private/public/signed) allowed
ACL model (owner, shared, role-based)
Read permission rule (who can read)
Write permission rule (who can upload/replace)
Delete permission rule (who can delete)
Signed access issuance rule (who can request)
Signed token TTL and scope rule
Cross-tenant access prevention rule

Audit requirements (access, delete)
Telemetry requirements (access denied, signed url issued)

Optional Fields
Share links (time-bound) | OPTIONAL
Admin override rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Cross-tenant access must be impossible by default.
Signed access must be scoped and short-lived; never grant blanket bucket access.
ACL changes must be auditable.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Tenancy & Ownership
   tenancy_model: {{tenant.model}} (single/tenant_id/UNKNOWN)
   ownership_rule: {{owner.rule}}
2. Access Modes
   allowed_modes: {{modes.allowed}}
   default_mode: {{modes.default}}
3. ACL Model
   acl_model: {{acl.model}} (owner_only/shared/role_based/UNKNOWN)
   acl_fields: {{acl.fields}} | OPTIONAL
4. Permissions
   read_rule: {{perm.read_rule}}
   write_rule: {{perm.write_rule}}
   delete_rule: {{perm.delete_rule}}
   admin_override: {{perm.admin_override}} | OPTIONAL
5. Signed Access
   who_can_request_signed: {{signed.who_can_request}}
   ttl_seconds: {{signed.ttl_seconds}}
   scope_rule: {{signed.scope_rule}}
   token_validation_rule: {{signed.token_validation_rule}} | OPTIONAL
6. Cross-Tenant Prevention
   cross_tenant_block_rule: {{tenant.cross_tenant_block_rule}}
   resource_id_validation_rule: {{tenant.resource_id_validation_rule}} | OPTIONAL
7. Audit
   audit_required: {{audit.required}}
   audit_events: {{audit.events}}
   audit_fields: {{audit.fields}} | OPTIONAL

8. Telemetry
   access_denied_metric: {{telemetry.access_denied_metric}}
   signed_url_issued_metric: {{telemetry.signed_url_issued_metric}} | OPTIONAL
9. References
   Upload/download spec: {{xref:FMS-02}}
   Security/compliance: {{xref:FMS-06}} | OPTIONAL
   AuthZ rules: {{xref:API-04}} | OPTIONAL
   Secure links: {{xref:CSec-05}} | OPTIONAL
   Retention/lifecycle: {{xref:FMS-05}} | OPTIONAL

Cross-References
Upstream: {{xref:FMS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FMS-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define tenancy model, read/write/delete rules, and signed TTL.
intermediate: Required. Define ACL model, cross-tenant block rule, and audit events.
advanced: Required. Add share links and admin overrides and token validation rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, acl fields, admin override, token validation, resource id validation, audit fields, signed url issued metric, share links, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If tenant.cross_tenant_block_rule is UNKNOWN → block Completeness Gate.
If perm.read_rule is UNKNOWN → block Completeness Gate.
If perm.write_rule is UNKNOWN → block Completeness Gate.
If signed.ttl_seconds is UNKNOWN → block Completeness Gate.
If telemetry.access_denied_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FMS
Pass conditions:
required_fields_present == true
tenancy_and_permissions_defined == true
signed_access_defined == true
cross_tenant_prevention_defined == true
audit_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FMS-08

FMS-08 — Metadata & Indexing (search, tags, EXIF policy)
Header Block
template_id: FMS-08
title: Metadata & Indexing (search, tags, EXIF policy)
type: files_media_metadata_indexing
template_version: 1.0.0
output_path: 10_app/files_media/FMS-08_Metadata_Indexing.md
compliance_gate_id: TMP-05.PRIMARY.FMS
upstream_dependencies: ["FMS-02", "DATA-06", "PFS-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FMS-02", "DATA-06", "PFS-01", "FPMP-03", "FMS-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical metadata model for files/media and how that metadata is indexed/searchable: tag fields, searchable facets, EXIF extraction/stripping policy, and privacy constraints. This template must be consistent with canonical schemas and query contracts.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FMS-02 Upload/Download Spec: {{fms.upload_download}}
DATA-06 Canonical Schemas (file metadata): {{data.schemas}} | OPTIONAL
PFS-01 Query Contract (filters/sorts): {{pfs.query_contract}} | OPTIONAL
FPMP-03 Processing Stages (metadata extraction): {{fpmp.stages}} | OPTIONAL
FMS-06 Security/Compliance: {{fms.security}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Metadata entity/schema ref (file_metadata schema)
Core metadata fields list (size, type, hash, owner, created_at)
Tagging model (user tags/system tags)
Search/index fields list (allowed filter/sort fields)
EXIF policy (strip/retain/UNKNOWN)
Derived metadata extraction rules (dimensions, duration)
PII policy for metadata (no GPS unless allowed)
Index update rules (on upload/process/delete)
Telemetry requirements (index lag, extraction failures)

Optional Fields
Full-text search support | OPTIONAL

Versioning of metadata schema | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
EXIF and other embedded metadata must be handled explicitly; default to stripping sensitive fields unless approved.
Only allow filtering/sorting on indexed fields per PFS contract; reject unsupported queries.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Schema Binding
   metadata_schema_ref: {{schema.ref}} (expected: {{xref:DATA-06}}) | OPTIONAL
2. Core Fields
   core_fields: {{fields.core}}
3. Tags
   tag_model: {{tags.model}} (user_tags/system_tags/UNKNOWN)
   allowed_tag_fields: {{tags.allowed_fields}} | OPTIONAL
4. Search / Index
   indexed_fields: {{index.fields}}
   filterable_fields: {{index.filterable_fields}}
   sortable_fields: {{index.sortable_fields}} | OPTIONAL
5. EXIF & Embedded Metadata
   exif_policy: {{exif.policy}} (strip/retain/UNKNOWN)
   fields_stripped: {{exif.fields_stripped}} | OPTIONAL
   fields_retained: {{exif.fields_retained}} | OPTIONAL
6. Derived Metadata Extraction
   extraction_rules: {{derive.rules}}
   processor_ref: {{derive.processor_ref}} (expected: {{xref:FPMP-03}}) | OPTIONAL
7. Index Update Rules
   on_upload: {{update.on_upload}}
   on_process_complete: {{update.on_process_complete}} | OPTIONAL
   on_delete: {{update.on_delete}}
8. PII Policy
   pii_forbidden_fields: {{pii.forbidden_fields}}
   gps_policy: {{pii.gps_policy}} | OPTIONAL
9. Telemetry
   index_lag_metric: {{telemetry.index_lag_metric}}
   extraction_failure_metric: {{telemetry.extraction_failure_metric}} | OPTIONAL
10. References
    Query contract: {{xref:PFS-01}} | OPTIONAL
    Upload/download: {{xref:FMS-02}}
    Processing stages: {{xref:FPMP-03}} | OPTIONAL
    Security/compliance: {{xref:FMS-06}} | OPTIONAL

Cross-References
Upstream: {{xref:FMS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FMS-09}}, {{xref:FMS-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define core fields and EXIF policy and index fields list.
intermediate: Required. Define update rules and derived extraction and PII forbidden fields.
advanced: Required. Add full-text/versioning and strict filter/sort enforcement and telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, schema ref, allowed tag fields, sortable fields, exif strip/retain lists, processor ref, optional update step, gps policy, extraction failure metric, full-text/versioning, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If fields.core is UNKNOWN → block Completeness Gate.
If index.fields is UNKNOWN → block Completeness Gate.
If exif.policy is UNKNOWN → block Completeness Gate.
If update.on_delete is UNKNOWN → block Completeness Gate.
If telemetry.index_lag_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FMS
Pass conditions:
required_fields_present == true
core_fields_defined == true
index_fields_defined == true
exif_and_pii_policies_defined == true
update_rules_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FMS-09

FMS-09 — Migration & Backfill Plan (moving buckets, rehash)
Header Block
template_id: FMS-09
title: Migration & Backfill Plan (moving buckets, rehash)
type: files_media_migration_backfill_plan
template_version: 1.0.0
output_path: 10_app/files_media/FMS-09_Migration_Backfill_Plan.md
compliance_gate_id: TMP-05.PRIMARY.FMS
upstream_dependencies: ["FMS-01", "FMS-05", "IXS-10"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FMS-01", "FMS-05", "FMS-07", "IXS-10", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical plan for migrating stored files/media (bucket moves, key renames, encryption changes, re-hashing) and executing backfills safely, with auditability and rollback. This template must be consistent with retention/access rules and change management policies.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FMS-01 Storage Inventory: {{fms.storage_inventory}}
FMS-05 Retention/Lifecycle Rules: {{fms.retention}} | OPTIONAL
FMS-07 Access Control Model: {{fms.access_control}} | OPTIONAL
IXS-10 Change Management: {{ixs.change_mgmt}} | OPTIONAL
ADMIN-03 Audit Trail: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Migration scope (what is moving/changing)
Source/target bucket_ids
Key rename rules (if any)
Re-encryption / KMS change rules (if any)
Re-hash rules (if any)
Backfill strategy (copy, verify, switch)
Verification strategy (checksums, counts)
Cutover strategy (dual read, version pin)
Rollback strategy
Safety limits (time windows, max objects)
Audit logging requirements (who ran migration)
Telemetry requirements (progress, errors)

Optional Fields
User-visible impact notes | OPTIONAL
Cost estimation notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Migrations must not violate retention or access control rules.
Verification must be explicit (checksums/counts), not “looks good.”
Rollback must be possible or explicitly documented as not possible.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Scope
   scope: {{scope.statement}}
   objects_in_scope: {{scope.objects_in_scope}} | OPTIONAL
2. Source/Target
   source_bucket_id: {{buckets.source}}
   target_bucket_id: {{buckets.target}}
3. Key/Path Changes
   key_rename_rule: {{keys.rename_rule}} | OPTIONAL
4. Crypto Changes
   reencrypt_supported: {{crypto.reencrypt_supported}} | OPTIONAL
   kms_change_rule: {{crypto.kms_change_rule}} | OPTIONAL
5. Re-Hash
   rehash_supported: {{hash.rehash_supported}} | OPTIONAL
   rehash_rule: {{hash.rehash_rule}} | OPTIONAL
6. Backfill Strategy
   method: {{backfill.method}} (copy_verify_switch/UNKNOWN)
   steps: {{backfill.steps}}
   dual_read_supported: {{backfill.dual_read_supported}} | OPTIONAL
7. Verification
   verification_checks: {{verify.checks}}
   acceptance_thresholds: {{verify.thresholds}} | OPTIONAL
8. Cutover / Rollback
   cutover_rule: {{cutover.rule}}
   rollback_rule: {{rollback.rule}}
9. Safety Limits
   time_window_rule: {{safety.time_window_rule}}
   max_objects_rule: {{safety.max_objects_rule}} | OPTIONAL
10. Audit / Telemetry
    audit_required: {{audit.required}}
    audit_fields: {{audit.fields}} | OPTIONAL

progress_metric: {{telemetry.progress_metric}}
error_metric: {{telemetry.error_metric}} | OPTIONAL

11. References
    Storage inventory: {{xref:FMS-01}}
    Access control: {{xref:FMS-07}} | OPTIONAL
    Retention/lifecycle: {{xref:FMS-05}} | OPTIONAL
    Change management: {{xref:IXS-10}} | OPTIONAL
    Observability/runbooks: {{xref:FMS-10}} | OPTIONAL

Cross-References
Upstream: {{xref:FMS-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FMS-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define source/target buckets, steps, verification checks, rollback rule.
intermediate: Required. Define cutover/dual read and safety limits and audit fields.
advanced: Required. Add crypto/rehash details, cost/impact notes, and telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, objects in scope, rename/crypto/hash sections, thresholds, max objects, audit fields, error metric, impact/cost notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If buckets.source is UNKNOWN → block Completeness Gate.
If buckets.target is UNKNOWN → block Completeness Gate.
If verify.checks is UNKNOWN → block Completeness Gate.
If rollback.rule is UNKNOWN → block Completeness Gate.
If telemetry.progress_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FMS
Pass conditions:
required_fields_present == true
source_and_target_defined == true
backfill_and_verification_defined == true
cutover_and_rollback_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FMS-10

FMS-10 — Observability & Cost Controls (storage growth, egress)
Header Block
template_id: FMS-10
title: Observability & Cost Controls (storage growth, egress)
type: files_media_observability_cost_controls
template_version: 1.0.0
output_path: 10_app/files_media/FMS-10_Observability_Cost_Controls.md
compliance_gate_id: TMP-05.PRIMARY.FMS
upstream_dependencies: ["FMS-01", "FMS-04", "FMS-05", "IXS-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FMS-01", "FMS-04", "FMS-05", "FMS-06", "IXS-07", "RLIM-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical observability and cost controls for files/media: storage growth tracking, egress/CDN costs, alerts for abnormal patterns, lifecycle enforcement verification, and operator runbooks to control spend. This template must be consistent with retention/CDN rules and integration observability baselines.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FMS-01 Storage Inventory: {{fms.storage_inventory}}
FMS-04 CDN/Delivery Rules: {{fms.cdn_delivery}} | OPTIONAL
FMS-05 Retention/Lifecycle Rules: {{fms.retention}} | OPTIONAL
FMS-06 Security/Compliance: {{fms.security}} | OPTIONAL
IXS-07 Integration Observability: {{ixs.observability}} | OPTIONAL
RLIM-01 Rate Limit Policy (egress throttles if any): {{rlim.policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Core metrics (total bytes, object count)
Growth metrics (daily/weekly delta)
Egress/CDN metrics (bytes out, hit rate)
Per-bucket breakdown policy (allowed dimensions)
Cost alert thresholds (growth spike, egress spike)
Lifecycle enforcement checks (TTL actually applied)
Abuse signals (hotlinking, download storms)
Operator actions (tighten TTL, disable public access, purge)
Runbook location and format
Telemetry requirements (cost anomalies surfaced)

Optional Fields
Budget caps by env/tenant | OPTIONAL
Auto-mitigation rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Cost controls must not weaken security (no opening access to reduce costs).
Alerts must be actionable and mapped to operator actions.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Core Metrics
   total_bytes_metric: {{metrics.total_bytes_metric}}
   object_count_metric: {{metrics.object_count_metric}}
2. Growth
   daily_growth_metric: {{growth.daily_metric}}
   weekly_growth_metric: {{growth.weekly_metric}} | OPTIONAL
3. Egress / CDN
   egress_bytes_metric: {{egress.bytes_metric}}
   cdn_hit_rate_metric: {{egress.hit_rate_metric}} | OPTIONAL
4. Breakdown Policy
   allowed_dimensions: {{breakdown.allowed_dimensions}}
   per_bucket_allowed: {{breakdown.per_bucket_allowed}}
5. Cost Alerts
   growth_spike_alert: {{alerts.growth_spike_alert}}
   egress_spike_alert: {{alerts.egress_spike_alert}} | OPTIONAL
   routing: {{alerts.routing}} | OPTIONAL
6. Lifecycle Enforcement Checks
   ttl_enforcement_check: {{lifecycle.ttl_enforcement_check}}
   archive_check: {{lifecycle.archive_check}} | OPTIONAL
7. Abuse Signals
   hotlinking_signal: {{abuse.hotlinking_signal}} | OPTIONAL
   download_storm_signal: {{abuse.download_storm_signal}} | OPTIONAL
8. Operator Actions
   actions: {{ops.actions}}
   permissions_ref: {{ops.permissions_ref}} (expected: {{xref:API-04}}/{{xref:ADMIN-01}}) | OPTIONAL
9. Runbooks
   runbook_location: {{runbooks.location}}
   format: {{runbooks.format}} (md/wiki/UNKNOWN) | OPTIONAL
10. References
    Storage inventory: {{xref:FMS-01}}
    CDN/delivery: {{xref:FMS-04}} | OPTIONAL

Retention/lifecycle: {{xref:FMS-05}} | OPTIONAL
Security/compliance: {{xref:FMS-06}} | OPTIONAL

Cross-References
Upstream: {{xref:FMS-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define core metrics, one alert, and runbook location.
intermediate: Required. Define egress metrics and lifecycle checks and operator actions.
advanced: Required. Add budgets/auto-mitigation and abuse signals with strict dimension policy.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, weekly growth, hit rate, routing, archive check, abuse signals, permissions ref, runbook format, budgets/auto-mitigation, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If metrics.total_bytes_metric is UNKNOWN → block Completeness Gate.
If growth.daily_metric is UNKNOWN → block Completeness Gate.
If egress.bytes_metric is UNKNOWN → block Completeness Gate.
If lifecycle.ttl_enforcement_check is UNKNOWN → block Completeness Gate.
If runbooks.location is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FMS
Pass conditions:
required_fields_present == true
core_and_egress_metrics_defined == true
alerts_defined == true
lifecycle_checks_defined == true
operator_actions_and_runbooks_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true